

The Minimum Wage in Firms' Organizations: Productivity Implications*

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Abstract

We exploit a unique policy setting in France (2003–2006), where multiple minimum wages coexisted, and show that higher minimum wages reduce firm-level employment, flatten hierarchies, increase training and raise revenue productivity per worker. We construct and calibrate a general equilibrium model of optimal hierarchies that rationalizes these findings: higher minimum wages lead firms to rely on more skilled workers who become less dependent on supervision, which results in flatter organizations and endogenous productivity gains that partly offset aggregate output losses. These losses are amplified in economies with advanced communication technologies but mitigated with superior information-processing or problem-solving technologies.

Keywords: Minimum Wage, Firms' Organizations, New Technologies (ICT, AI), Productivity

JEL Classification: D21, D22, D24, J23, J30

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1 Introduction

For decades, labor economists have been studying the effects of the minimum wage on the economy. To date, however, little is known about how minimum wages impact the overall internal organization of labor within firms, even though they unambiguously alter the relative prices of factors (labor) and hence the relative cost of different work organization strategies. These in turn determine the roles assigned to workers with different skills, the structure of wages and, ultimately, the productivity of firms (Garicano, 2000; Caliendo and Rossi-Hansberg, 2012). In this paper, we propose a unified framework to examine these questions and investigate how the ongoing wave of communication and AI-based technologies might affect the relevant trade-offs.

Our contributions are threefold. First, we leverage a unique empirical setting in recent French economic history—namely, the coexistence of different minimum wage schedules between 2003 and 2006—combined with exceptionally detailed matched worker-firm data. This allows us to investigate how the intensity of the minimum wage constraint relates to firms’ organizational strategies and productivity. Second, we propose a structural model of optimal hierarchies that delivers both qualitative and quantitative predictions regarding how firms adjust their internal organization in response to rising minimum wages, with implications for worker skills, wages, and firm productivity. We calibrate the model to key static empirical moments from the French data and evaluate its ability to quantitatively predict the treatment responses estimated in our regression analysis. Finally, we use the model to conduct counterfactual simulations and show that endogenous organizational strategies—and their impact on aggregate productivity—play a central role in mitigating the effects of moderate minimum wage constraints. Importantly, we extend the analysis to different technological environments, exploring whether major shifts—such as the adoption of new AI-based tools or the widespread investment in communication technologies prompted by the Covid-19 pandemic—reinforce or invalidate these findings.

Our empirical analysis first focuses on a specific period of the French economy, which was characterized by the coexistence of several legal minimum wages, each applying to different sets of firms. This situation emerged as a consequence of the 1998 and 2000 “Aubry” laws (named after the Minister of Labor at the time), which mandated a reduction of the legal workweek from 39 to 35 hours from January 1st, 2000 in firms with more than 20 employees, and on January 1st, 2002 for all other firms. Different regimes of GMR (“Garantie Mensuelle de Rémunération”, Aeberhardt et al., 2016) were created, tied to the timing of each firm’s agreement to implement the workweek reduction, in order to preserve workers’ purchasing power. Our period of interest is the subsequent period between 2003 and 2006, during which the government implemented a process of reconvergence of the various GMRs towards a single national minimum wage. Over this period, legal working hours had stabilized at 35 hours in a large sample of companies, which were however – and this is more interesting for us – differentially exposed to increases in the minimum wage (GMR) applicable to them. This setting opens the door to powerful double- and triple-difference strategies, whereby our mechanisms of interest are identified from firms’ differential exposure to changes in the minimum wage, while addressing their endogenous sorting across GMRs.

The French data allow us to accurately capture the evolution of French production hierarchies (Caliendo, Monte, et al., 2015), their worker-level training decisions as well as their performance in terms of sales and productivity. We obtain the following results. First, the increase in labor costs is mainly concentrated at the bottom layer of firms’ hierarchies, while wages at upper layers tend to adjust downwards to mitigate the cost shock. Second, firms that are more exposed to minimum-wage increases significantly reduce their size by flattening their hierarchical organization. For the “medium” minimum-wage increase considered in our setting (3.7% in real terms), evaluated at the sample mean of the exposure measure, our estimates imply a reduction in firm size of 0.3% to 0.9%, depending on whether firm size is measured in jobs or hours, and whether the

effect is estimated using double- or triple-difference specifications.¹ In addition, their workers enroll more often in training programs, suggesting that the skill content of jobs increases. Lastly, these adjustments are associated with significant increases in “revenue” productivity (Foster et al., 2008; Hsieh and Klenow, 2009).

In a second step, we construct a quantitative model of firms’ organization to assess whether the organizational responses observed in the data are likely to qualitatively and quantitatively rationalize the observed firm-level productivity increases, and to investigate further general equilibrium implications of minimum wages. Our framework builds on the knowledge-based management hierarchy framework initially introduced by Garicano (2000) and extended by Caliendo and Rossi-Hansberg (2012) to an economy with heterogeneous firms.² In this framework, production requires labor to be combined with knowledge. The optimal firm organization has a pyramidal structure composed of one layer of production workers, and one or many layers of managers with increasing levels of knowledge (Garicano, 2000). The benefit of this structure is that it prevents knowledgeable managers from dealing with the most frequent problems and allows them to devote their time to less common problems, which they have a comparative advantage in addressing (Garicano, 2000; Garicano and Van Zandt, 2013; Garicano and Hubbard, 2007). The optimal number of layers to use in production is endogenously determined by firms. Their decisions are based on the costs and benefits of introducing an additional layer of management to the firm (Caliendo and Rossi-Hansberg, 2012): from the perspective of the firm, adding a new layer of management is similar to incurring an additional fixed cost, because the firm has to hire an additional knowledgeable manager. However, provided that the firm is producing at a sufficiently large scale, an additional layer reduces the marginal costs of production, because workers become more efficient at solving problems with the help of the new manager. Hence, adding a layer of management is only profitable to firms producing at a sufficiently large scale, i.e. to those benefiting from favorable demand conditions.

We introduce to this analytical framework the feature that the minimum wage constrains the optimal decisions of firms: an increase in the minimum wage increases the wage floor and the costs of hiring low-skilled production workers, especially when training costs are large relative to managerial costs. This in turn distorts the way firms organize both in terms of hierarchical layers and knowledge across layers, which further affects both firms’ productivity and the structure of wages within firms. Our theoretical set-up thus generates a rich set of counterfactual predictions. In particular, one of the largest direct responses of firms to higher minimum wages is to raise the skill level of their production workers at the bottom of hierarchies, which means increasing their level of training. As a result, workers are more knowledgeable and require less help and direction, meaning that managers become less useful, so firms tend to select flatter organizations, saving on the “fixed costs” (associated with hierarchical levels) mentioned above and mitigating the cost associated with the minimum wage constraint. In addition, average firm size decreases and the industrial fabric is more fragmented.

We calibrate the model to key empirical moments and confirm that it replicates our regression results both qualitatively and quantitatively. We then simulate equilibrium outcomes under six scenarios: a baseline calibrated to French data; a no-minimum-wage case (equivalent to a 4% reduction); and four scenarios with minimum wages set 4%, 8%, 16%, and 24% above baseline. The 4% and 8% increases reflect actual changes observed for the second and first GMR groups (GMR2, GMR1) between 2003 and 2006, while the larger shocks are hypothetical but still realistic, with the 24% case mirroring the U.S. Fair Labor Standards Act

¹These estimates imply an elasticity of employment (measured by the number of jobs) with respect to the labor cost at the minimum wage of approximately -0.22 , and an elasticity of total hours worked of approximately -0.15 .

²The origins of this literature go back decades, to an analysis of span of control in a multi-layer firm in Rosen (1982), and even further to papers on team production in Marschak and Radner (1972) and a basic model of span of control in Lydall (1959). See Garicano and Rossi-Hansberg (2015) for a comprehensive review.

of 1966 (Bailey et al., 2021). We obtain that theory-based indicators of aggregate *quantity* productivity (Caliendo, Mion, et al., 2020) and profits deteriorate, but their decline is mitigated by firms’ organizational responses, which ultimately enhance aggregate *revenue* productivity. Market share reallocations between firms are significant, but the pure selection channel (Melitz, 2003) only plays a minor role. Workers benefit from large skill increases and pocket the associated productivity gains in the form of higher wages across all hierarchical layers. Our framework therefore implies that minimum wages, when set at moderate levels, are likely to induce firms to invest more in workers’ human capital and create higher-quality jobs, as in Acemoglu (2001) but via an entirely different channel. In addition, the model allows for an in-depth inspection of the detailed organizational channels through which the minimum wage affects the organization of production. These adjustments are made through extensive organizational redesign: most firms get flatter, with negative average impact on their size. Our simulations show that these organizational margins of adjustment make it possible to divide the impact of minimum wage constraints by a factor of four relative to a counterfactual economy where only skills, but not firms’ hierarchical organizations, could adjust.

Importantly, our framework also provides a rich description of firms’ technological environment through four parameters that capture distinct dimensions of technological change relevant to organizational choices: first, technologies augmenting homogeneously the productivity of all workers, second, communication technologies, third, information technologies, and fourth, technologies affecting the distribution of the difficulty level of production problems. These four dimensions allow us to document the likely impact of two on-going waves of technologies, namely, the implications of the huge spike of investment in communication technologies which occurred during the Covid-19 pandemic (Gibbs et al., 2023; Barrero et al., 2021; Bloom, Han, et al., 2024), and the implications of the AI-based wave of technologies, which are likely to resemble information or problem-solving technologies (Mullainathan and Obermeyer, 2021; Kleinberg et al., 2017; Noy and Zhang, 2023). We find that the relative cost of minimum wage constraints is amplified in economies with efficient communication technologies, which enhance the productivity of hierarchical firms (Garicano and Rossi-Hansberg, 2006): since flattening is the main organizational response to such constraints, firms are less able to exploit the potential gains generated by efficient communication technologies, which amplifies the aggregate output loss.³ On the other hand, our results are more reassuring regarding the likely impact of AI-based technologies. When interpreted as improvements in information processing or problem-solving capabilities, they both reduce the cost of strategies involving increases in workers’ skills and expand the set of problems that bottom, production workers are able to solve autonomously. These two mechanisms tend to mitigate the aggregate output cost associated with moderate minimum wage constraints, but they become ineffective when technological shocks or minimum wage constraints become large. Overall, our results imply that the current waves of technologies are likely to require a thorough re-examination of cost-benefit analyses of labor market institutions and regulations.

Our paper is related to several strands of the literature. As previously mentioned, it contributes directly to the research program documenting how corporate organizations affect economies (Garicano and Rossi-Hansberg, 2015). We do so in three important dimensions. First, we apply the toolbox to a new application, in labor economics, by investigating how firms’ organizational strategies mediate the impact of a major labor market institution: the minimum wage. The latter directly enters firms’ cost minimization programs which jointly determine firms’ labor demand and optimal organizational choices. We show that firms’ endogenous

³Importantly, our analysis thus contributes to clarifying why the empirical literature obtains mixed results about the productivity impact of remote work and the intense use of communication technologies which accompanies it: rather positive in Barrero et al. (2021), Bloom, Han, et al. (2024) or Angelici and Profeta (2024) and rather negative in Morikawa (2023) or Gibbs et al. (2023). Indeed, our framework shows that the impact might be heterogeneous across firms depending on their internal organization, as in Caliendo, Monte, et al. (2015), and that the overall impact is likely to be modulated by labor market institutions, such as minimum wages.

organizational adjustments are extremely powerful at dampening the impact of this constraint and that it directly translates into higher firm-level indices of revenue productivity. Secondly, we take this mainly theoretical toolbox and turn it into an operational quantitative framework, which we calibrate carefully to static moments of the French data. This is a critical step to provide a credible quantification of the organizational mechanisms at play as well as their firm-level and aggregate implications. We show that firms’ endogenous organizational responses to tighter minimum-wage constraints are quantitatively capable of explaining why minimum-wage increases might have only modest effects on aggregate output.⁴ Consistent with this mechanism, our empirical analysis of French firms shows that minimum-wage increases are associated with reductions in the number of hierarchical layers and increases in revenue productivity. These empirical patterns closely match the model’s predictions, both qualitatively and quantitatively, providing strong support for the economic relevance of the organizational channel. Finally, in a third contribution, we draw on our quantitative model to provide simulations and assess the potential impact of the current waves of technologies.

In addition, our paper relates to the vast literature documenting the different dimensions of the impact of the minimum wage. Besides the studies discussed above, those focusing on firm-level responses in terms of profit and productivity are particularly relevant to our research question. Draca et al. (2011) document that the introduction of the UK national minimum wage in 1999 led to wage increases as well as decreases in firms’ profitability. Rizov et al. (2016) and Riley and Rosazza Bondibene (2017) show that it also led on average to increases in productivity, while both Hau et al. (2020) and Mayneris et al. (2018) document that managerial practices might be an important driver of these productivity gains. Our theoretical and quantitative framework is able to fully reconcile all these patterns, since it predicts opposite impacts on average profits and revenue productivity, respectively, for the range of increases in the minimum wage observed in the data. Furthermore, it shows that selection and composition effects only become significant when minimum wage constraints are extremely stringent. More generally, our paper relates to the extremely large literature studying the effects of minimum wages on employment and wage distributions.⁵

The remainder of the paper is organized as follows. Section 2 describes the French minimum wage system and the institutional reforms that generated the large minimum wage increases between 2003 and 2006. Section 3 presents our empirical setting, identification strategies and regression results describing how firms adjusted to these increases. Section 4 develops the theoretical framework, while Section 5 details our calibration strategy as well as our simulation results.

2 Institutional Background

2.1 A Brief History of the French Minimum Wage

The minimum wage in France dates back to the end of World War II. It was first introduced into law in 1950 along with post-war reforms that restored workers’ freedom to negotiate their wages through collective

⁴The organizational channel also attenuates the effect on corporate employment relative to an economy without organizational adjustment. Moreover, an extension of our calibration that incorporates unemployment (presented in Online Appendix H) implies only modest effects on aggregate employment, consistent with the empirical evidence (Card and Krueger, 1994; Bailey et al., 2021; Cengiz et al., 2019; Dustmann et al., 2022). Other mechanisms proposed in the literature include labor-market monopsony, whereby minimum-wage increases primarily reduce firms’ rents rather than employment (Manning, 2011; Azar et al., 2023; Wiltshire et al., 2025), and price pass-through, which offsets part of the increase in labor costs (Ashenfelter and Juraïda, 2022; Lo Bello and Pesaresi, 2024; Sorkin, 2015; Harasztosi and Lindner, 2019; Renkin et al., 2022; Aaronson, French, et al., 2008). While these mechanisms may also contribute to the modest aggregate effects of minimum-wage increases, they do not explain the observed flattening of firms’ organizational hierarchies.

⁵Beyond those aforementioned, many extremely detailed studies could be cited, among which: Lee (1999) and DiNardo et al. (1996), Derenoncourt and Montialoux (2021) and Forsythe (2023) for the case of the US or Dickens, Machin, et al. (1999), Dickens and Manning (2004), Machin et al. (2003), Stewart (2004), Metcalf (2008), and Stewart (2012) for the case of the UK.

bargaining agreements (art. 31g/2 in loi n° 50-205 of February 11, 1950).⁶ Today, the minimum hourly wage in France is the Salaire Minimum Interprofessionnel de Croissance (SMIC). It applies uniformly across metropolitan France and covers workers in the French labor market, with the notable exception of civil servants.⁷ The latter represent around 20% of the workforce,⁸ but since their career paths are strictly regulated and take place in administrative environments that are largely disconnected from the for-profit sector, we exclude them from our analysis.

During our period of analysis, the SMIC was updated annually (every July 1st) according to the French Labor Code: (i) it was indexed to the national consumer price index (CPI), (ii) its annual gain in purchasing power could not be less than half that of blue-collar workers' average hourly wage (*Salaire Horaire de Base Ouvrier*, SHBO), and (iii) the government could grant additional discretionary increases. Formally, this implied the following dynamic relationship between the SMIC, the SHBO, and the CPI:

$$\frac{SMIC_{t+1}}{SMIC_t} = \frac{CPI_{t+1}}{CPI_t} + \frac{1}{2} \left(\frac{SHBO_{t+1}}{SHBO_t} - \frac{CPI_{t+1}}{CPI_t} \right) + \Delta_{t/t+1}GOV \quad (1)$$

where $\Delta_{t/t+1}GOV$ denotes discretionary increases to the minimum wage that the government can decide to implement between periods t and $t + 1$.

2.2 Workweek Reform: the Rise and Fall of Multiple Minimum Wages

Our empirical analysis exploits a unique transition period during which France moved from a system of multiple statutory minimum wages to a single nationwide minimum wage. The reform unfolded in two stages. The 1998 and 2000 “Aubry” laws reduced the statutory workweek from 39 to 35 hours (effective January 1, 2000 for firms with more than 20 employees, and January 1, 2002 for all others) while preserving the monthly earnings of minimum-wage workers through temporary minimum-wage schedules, known as the *Garanties Mensuelles de Rémunération* (GMR1–GMR5). Firms entered one of these schedules according to the date at which they adopted the shorter workweek, and the government subsequently reunified all schedules into a single statutory minimum wage between 2003 and 2005. These two stages play distinct roles in our analysis:

- Emergence of the GMRs between 1998 and 2002: this period determines how firms self-selected into the different minimum wage schedules – a phenomenon which will require a careful econometric treatment (see Section 3.1). It is however difficult to leverage econometrically, as it confounds both the organizational implications of the reduction of the legal workweek and those of the differentiated dynamics of the GMRs that were created in this process.
- Re-convergence of the multiple legal minimum wages to a single minimum wage between 2003 and 2005.

This episode generates the variation used in our regressions.

Online Appendix A provides a complete timeline of the reform and the corresponding legal references.⁹

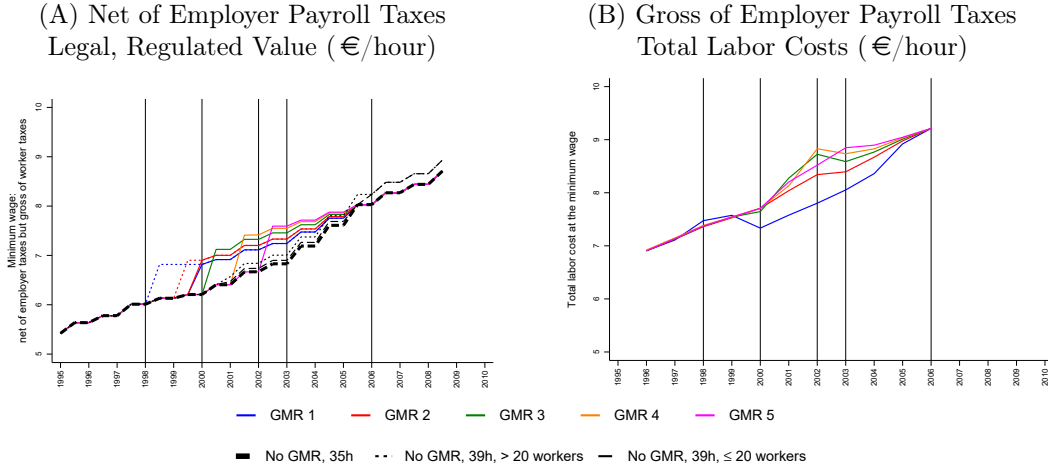
⁶Concerned that wages would be too low in sectors with weak unions, the government introduced the Salaire Minimum Interprofessionnel Garanti (SMIG), which determined the minimum hourly compensation in metropolitan France. Initially, the SMIG varied across metropolitan France: it was based on the standard budget of a single person living in Paris, and adjusted for workers living in the different regions of France. Moreover, following France's high inflation in 1951 and 1952, the SMIG was indexed to the consumer price index, so as to maintain minimum wage workers' standard of living (loi n° 52-834 of July 18, 1952). It was further reformed over the years, yet it continued to increase at a slower pace than average hourly wages, increasing inequality between minimum wage and other workers. As a result, it was replaced in 1970 by the Salaire Minimum Interprofessionnel de Croissance (SMIC), which remains in place until today (loi n° 50-205 of January 2, 1970).

⁷There are a few groups of workers, such as workers below the age of 18 and interns, that are able to earn wages below the SMIC. These workers are also outside of the scope of our analysis.

⁸See <https://www.insee.fr/fr/statistiques/5392034?sommaire=5392045>.

⁹The complete collection of relevant legal documents is available at <https://drive.google.com/file/d/1m4j3OawOvgc4rUchY0sM6ByTMln3hn45/view>. We thank Gilbert Cette, former advisor to Minister Martine Aubry during the reform, for his assistance in identifying the relevant legal sources. Further accounts in English include Askenazy (2008), Aeberhardt et al. (2016), and Estevão and Sá (2014) while Carry et al. (2025) studies the first stage of the reform

Figure 1: Minimum Wage and GMRs



(C) Average Changes in (ln) Total Labor Costs at GMR, across GMR, 2003 to 2006

	GMR1	GMR2	GMR3	GMR4	GMR5
In nominal terms	0.135	0.093	0.070	0.053	0.040
In real terms	0.079	0.037	0.014	-0.003	-0.016
(Standard deviation)	(0.027)	(0.018)	(0.014)	(0.013)	(0.000)

Notes: This figure shows the evolution of the statutory hourly minimum wage and the five temporary GMRs over the analysis period. The GMRs were introduced under the “Aubry 2” law of 2000, but for GMR1 and GMR2, their calculation required a retroactive reconstruction, which is represented by dashed lines between 1998 and 2000. The re-alignment of the GMRs with the official minimum wage occurred between July 2003 and July 2005. Panel (A) reports statutory hourly wages under the different minimum-wage schedules, expressed net of employer payroll contributions but gross of employee payroll contributions. These correspond to the legally regulated wage schedules (see https://baremes.ipp.eu/parameters/marche-du-travail/marche-travail.salaire_minimum.gmr/table/). For comparison, it also reports the schedules applying to firms not covered by the GMR system (generically referred to as the SMIC; see footnote 16 for details about the three different regimes). Panel (B) reports the corresponding average *total* hourly labor cost of a minimum-wage job in our estimation sample, obtained by combining the statutory GMR schedules with employer payroll taxes and payroll-tax reductions (see Section 3.2). Panels (A), (B), and the first row of Panel (C) are expressed in nominal terms. The second row of Panel (C) reports the same labor-cost changes after adjusting for the cumulative inflation rate of 5.6% between 2003 and 2006 (Insee online data, accessed August 2025). Labor costs are expressed in logarithms in Panel (C) to match the regression specifications in Section 3.4.

Assignment to GMR schedules (1998–2002). Between 1998 and 2002, two successive laws (loi n° 98-461 of June 13, 1998 and loi n° 2000-37 of January 19, 2000), known as the “Aubry” laws,¹⁰ mandated a reduction in the statutory workweek from 39 to 35 hours. In June 1998, the first law (Aubry 1) established the perspective of a future new legal standard that would be effective in 2000,¹¹ and provided incentives for firms to act before these deadlines. The question of the future implementation details — overtime, modulation, part-time work, minimum wage, etc. — was referred to a vote on the second law (Aubry 2), which was passed in early 2000, and based on the experience of agreements negotiated in the meantime.

These two laws also put into place financial incentives to encourage firms to sign collective bargaining agreements containing a reduction in working time (RWT) to 35 hours. The Aubry 1 subsidies were particularly generous (Decree of June 26, 1998),¹² but subject to ambitious hiring objectives, and eligibility thus

(1998–2002).

¹⁰The media named this law after the (left-wing) minister who led the reform.

¹¹More precisely, it would be effective January 1, 2000, for companies with more than 20 employees, and January 1, 2002, for others. Researchers interested in the precise definition of the 20 worker threshold can refer to article 1-II of loi n°2000-37, which provides all details about when (on January 1, 2000) and how the 20-worker employment threshold was to be legally computed (threshold not exceeded during the past 12 months). This threshold is however not critical to our estimation strategy, because our period of main interest is posterior to the 2000 and 2002 legal deadlines. Moreover, Garicano, Lelarge, et al. (2016) document that this threshold is not detectable in the data as of 2002, which limits the concern about firms’ strategic manipulation of their size, although we propose an IV strategy in Section 3.1 and in the Online Appendix (Section D.1) to address this potential issue.

¹²As shown in Panel (B) of Figure 1, these subsidies effectively stabilized labor costs at the minimum wage for firms which acted before the official date of working time reduction.

prohibited layoffs. This condition disappeared from the Aubry 2 law (see Article 19-III-2 and the discussion in Askenazy et al., 2004). Article 19-II of Law No. 2000-37 specifies the conditions under which agreements qualified for these subsidies. In particular, they had to be negotiated at the plant or firm level and validated by representative unions in firms employing more than 50 workers, while sectoral agreements were valid in smaller firms. By 2003, nearly 60% of private sector workers were in companies operating on a 35-hour workweek and 85% in companies with more than 200 workers (Askenazy, 2008) while overall, 90% of agreements were valid and benefited from subsidies (Bunel, 2004).¹³

Lastly, and most relevant to our analysis, the Aubry 2 law stipulated that all firms implementing the 35-hour workweek—regardless of whether they had signed a qualifying agreement and received the associated subsidies—had to maintain the monthly earnings of minimum wage workers. For firms that had adopted the 35-hour workweek before the 2000 legal deadline, this requirement was calculated retroactively.¹⁴ This implied that the hourly wage (net of employer payroll taxes) increased by $39/35 - 1 \approx 11.4\%$ upon the adoption of the 35 hour workweek, which further motivated the above subsidies to contain total labor costs.¹⁵ Ultimately, five distinct GMR schedules were created, each corresponding to the date at which firms adopted the 35-hour workweek: the first GMR (GMR1) for agreements signed between June 15, 1998 and June 30, 1999; GMR2 between July 1999 and June 2000; GMR3 between July 2000 and June 2001; GMR4 between July 2001 and June 2002; and GMR5 thereafter. In the first year of each schedule, a minimum wage worker employed under the corresponding GMR and working 35 hours per week earned exactly the same monthly wage as a worker employed for 39 hours at the statutory minimum wage (SMIC). In the subsequent years, each GMR was updated every July 1st, according to the following rule:

$$\frac{GMR_{t+1}^k}{GMR_t^k} = \frac{CPI_{t+1}}{CPI_t} + \frac{1}{2} \left(\frac{SMBO_{t+1}}{SMBO_t} - \frac{CPI_{t+1}}{CPI_t} \right) + \Delta_{t/t+1}GOV, \quad k = 1, \dots, 5 \quad (2)$$

where GMR_t^k denotes the minimum wage applicable to workers covered by GMR schedule k , $SMBO_t$ denotes the average nominal *monthly* wage of blue-collar workers (*Salaire Mensuel de Base Ouvrier*) in period t , and $\Delta_{t/t+1}GOV^k$ captures any discretionary increase in GMR k implemented by the government between periods t and $t + 1$. The key difference with Equation 1 is that GMRs were indexed to the growth of blue-collar workers' *monthly* wages (SMBO), whereas the SMIC was indexed to their *hourly* wages (SHBO). Because monthly wages grew more slowly than hourly wages during the transition to the 35-hour workweek, the SMIC and the GMR schedules followed significantly different paths between 1998 and 2002. Panel (A) of Figure 1 illustrates the evolution of the different minimum wages over time. It shows that, until 2002, all GMR schedules increased substantially more slowly than the SMIC ("No GMR" cases). Since they were updated according to the same indexation rule, the five GMR schedules evolved in parallel and therefore did not converge during this period.

Firms were, of course, not randomly assigned to the different GMR schedules. Rather, self-selection into the timing of the transition to the 35-hour workweek determined which minimum wage schedule subsequently applied to them. The literature provides limited insight on the mechanisms that may have driven these

¹³Article 5-IV of loi n° 2000-37 addressed the case of non-complying firms. It remained legal for such firms to exceed the statutory workweek by up to a fixed annual quota which in practice was set at 130 hours. In return, the first four overtime hours were subject to a 25% premium (10% only for firms with fewer than 20 employees), and any subsequent overtime to a 50% premium. In our main specifications, we exclude these non-complying firms and restrict the analysis to firms that explicitly negotiated a RWT agreement and benefited from the associated subsidies (see Table B1). This ensures that we know their actual workweek (35 hours) and can accurately compute their labor costs. In contrast, non-complying firms may operate under different working time regimes that are difficult to track in the (DADS) data and thus to control for in regressions. This also affects our ability to reconstitute their labor costs, since the latter depend on the exact amount of overtime worked annually by each worker.

¹⁴See Art. 32-I and Figure A1 in Appendix A.

¹⁵Notice that GMRs are defined in "net" terms, i.e. net of social payroll taxes - whereas the relevant concept for firms is the total cost of labor (incorporating such taxes, but also any subsidy such as those discussed above); see Section 3.2 for our strategy to simulate these costs at the job level.

patterns, beyond case studies pointing to the reluctance to be exposed to heavier administrative controls (Jorand and Tonneau, 2005). The Ministry of Labor also probably put a lot of pressure on large firms to shift early, in order to help launch the reform: Tables B2 and B4 of Appendix B actually show that “earlier” GMRs tend to be populated by larger firms, that they exhibit on average a lower share of workers paid at the minimum wage, and they consist of relatively more firms from manufacturing industries and fewer firms from trade, hotels or restaurants, which entered on average later GMRs. Section 3.1 will propose an empirical strategy (in triple differences) to solve the endogeneity problems generated by this critical point.

Reunification of minimum wages (2003–2005). As early as 2000, following the Council of State’s requirement that workers performing the same job should receive the same pay (“equal pay for equal work”), the Aubry 2 law stipulated that the SMIC and all GMR schedules should converge by July 1, 2005, while leaving the convergence mechanism unspecified. Designing this convergence plan was therefore left to the right-wing government that took office in 2002 (Figure A1 in Online Appendix A). The plan was announced in September 2002 and subsequently enacted through the Fillon Law (Law No. 2003-47 of January 17, 2003). Under the new rules, the most recent schedule, GMR5, was thereafter indexed only to inflation, preserving workers’ purchasing power but no longer tracking growth in blue-collar monthly wages as in Equation 2. At the same time, discretionary increases were granted each July 1 to the SMIC and the four earlier GMR schedules in order to bring them progressively into line with GMR5 by July 2005. These annual adjustments amounted to €40 for the SMIC, €18 for GMR1, €13.5 for GMR2, €7 for GMR3, and €2.25 for GMR4. In addition, the 2003 Fillon Law substantially reformed several other key provisions of the Aubry laws (Askenazy et al., 2004).¹⁶ Most importantly for our analysis, it initiated the harmonization of payroll subsidies and reductions in employers’ social security contributions, a process that was completed in 2005. As a result, not only the minimum wages received by workers but also the corresponding labor costs borne by firms converged between 2003 and 2005, as shown in Panel (B) of Figure 1; because our data are annual (see Section 3.2), we use 2006 as the final year of the analysis to ensure that the convergence process is fully reflected in the data.

These features make the 2003–2006 period particularly well suited to analyzing the mechanisms of interest. First, for firms that had signed a RWT agreement by 2002, the organizational adjustments associated with the transition to the 35-hour workweek were largely complete. Because these agreements typically followed lengthy negotiations and the Aubry reforms were phased in over several years, firms had sufficient time to stabilize their new organization before the reconvergence period (Askenazy et al., 2004). As a result, the organizational and productivity changes observed between 2003 and 2006 are more likely to reflect the subsequent increases in minimum wages than the RWT itself.¹⁷ Second, the reconvergence period generates substantial variation in firms’ exposure to minimum wage increases through their GMR schedule and their initial share of minimum wage workers.¹⁸ A critical empirical challenge is that firms self-selected into GMR schedules - an issue examined in detail in Sections 3.1 and 3.2.

¹⁶ While maintaining the statutory 35-hour workweek, the reform relaxed overtime regulations by increasing the annual overtime quota from 130 hours to 180 hours in 2003 and to 220 hours in 2004 (Decree of December 9, 2004). As a result, firms that had not adopted the 35-hour workweek could continue operating on a 39-hour schedule, provided they paid the statutory overtime premium. During our sample period, this premium depended on firm size: 10% for firms with fewer than 20 employees, 25% for larger firms (and 50% for overtime beyond eight hours per week). These schedules correspond to the two curves labeled “No GMR, 39 hours, >20 workers” and “No GMR, 39 hours, ≤20 workers” in Panel (A) of Figure 1.

¹⁷ Online Appendix D.2 excludes firms assigned to GMR5, which adopted the 35-hour workweek only in 2002, ensuring that all firms are observed at least one year after signing their RWT agreement. The results remain very similar to those reported in the main text despite the expected loss of precision from the smaller sample.

¹⁸ Aeberhardt et al. (2016) argues that the details of the reconvergence plan were at least partly unanticipated when firms signed their RWT agreements, and therefore when they became affiliated with a particular GMR schedule.

2.3 External Validity

While the institutional setting described above is specific to France, the mechanisms we study are likely to apply more broadly. Assessing the external validity of our analyses requires placing this episode in an international perspective. Two references are particularly informative in this regard. Du Caju et al. (2009) compare minimum-wage institutions across 25 countries as of 2006, while Bozio and Wasmer (2024) situates the French experience within a broader historical and international context.

Overall level of the minimum wage. Estimated minimum wage elasticities are local and may therefore depend on the initial level of the minimum wage, commonly measured using the Kaitz index. Because our analysis focuses on labor costs, we use the labor-cost version of the Kaitz index recently constructed by Bozio and Wasmer (2024) (see Figure A3 in Online Appendix A).¹⁹ This index declined from 48% in 2006 (the final year of our empirical analysis) to 44% in 2023 - placing France close to countries such as Spain, Germany, the Netherlands, Belgium, the UK, and the United States over the 2010–2018 period.

Magnitude of the shock. Panel (C) of Figure 1 shows that nominal minimum wages increased by 4.0% to 13.5% during the reconvergence period, corresponding to real changes ranging from -1.6% to $+7.9\%$. In addition, Table B2 indicates that, in 2002, 13.6% of workers earned less than the 2006 minimum wage. Compared with the existing literature, these shocks and exposure are moderate. They are substantially smaller than those studied by Bailey et al. (2021), whose analysis of the 1966 Fair Labor Standards Act relies on a 28% nominal (23% real) increase in the minimum wage,²⁰ and by Harasztosi and Lindner (2019), who examine a 60% real increase in Hungary’s minimum wage. By contrast, our setting is closer to those studied by Draca et al. (2011), who classify a UK industry-region cell as “low wage” when more than 10% of workers initially earned less than the new minimum wage, and by Dustmann et al. (2022), whose German reform affected about 15% of employees—figures that closely match the 13.6% observed in our data.

Role of unions. France combines very low unionization—less than 10% in 2006²¹—with broad collective bargaining coverage, unlike countries such as the United States or the United Kingdom, where coverage largely reflects union membership. Under normal circumstances, this disconnect arises because firm-level agreements apply to all employees, regardless of union membership, while industry agreements can be extended to all firms in a sector.²² In our setting, however, industry bargaining played only a limited role. Eligibility for RWT subsidies required large firms to negotiate firm-level agreements, precisely the level at which union representation tends to be weakest. This feature also reduced the incentive to bargain at the industry level (Askenazy et al., 2004; Askenazy, 2008). As a result, sector-level agreements covered only about 20% of the smallest firms, suggesting that unions had little influence on the implementation of the reform at either the firm or industry level (Askenazy, 2008).

¹⁹Du Caju et al. (2009) instead report the ratio of net minimum to net average wages. Since this measure excludes employers’ labor costs, it is less relevant for our purposes. In 2006, it stood at 55% in France and above 40% in Belgium, Cyprus, Finland, Iceland, the Netherlands, Portugal, the UK, and Slovenia. In our dataset—which excludes public-sector workers—it equals 50% (which is closer to the values observed in the countries mentioned), while the labor-cost-based measure is 42%.

²⁰According to the authors, 26% of workers earned less than the new minimum wage in highly exposed states such as Texas, compared with 11% in less exposed states such as New York.

²¹In 2006, Du Caju et al. (2009) classify France among the countries with “very low” unionization, alongside the United States, Germany, Spain, Japan, and five other European countries.

²²Industry agreements initially apply only to firms represented by the signatory employer organizations, but can subsequently be extended to the entire sector at the request of the government, employer organizations, or unions. Once extended, they become binding for all firms and workers. Such extensions are common in France and require no specific conditions (Avouyi-Dovi et al., 2013).

Implications of employment protection regulations. The Aubry 1 law initially made subsidy eligibility conditional on firms refraining from layoffs, although this requirement was removed by the Aubry 2 law in 2000. Overall however, French labor market regulations tend to render layoffs rather difficult in comparison with the US or UK: in 2006, the OECD employment protection index (on a 0–6 scale) reached 2.83 in France, compared with 0.67 in the United States and 1.76 in the United Kingdom, while remaining close to the levels observed in Germany (2.89), Spain (2.65), Italy (3.33), and Sweden (2.60). Stronger employment protection may lead firms to respond to minimum wage increases through internal rather than external adjustment margins. For example, firms may adjust labor “volume” through changes in hours worked per worker or internal labor market reallocation rather than through hiring and layoffs of workers. Likewise, skill adjustment may rely more on internal worker training than on external recruitment (and layoffs).

3 Empirical Evidence

3.1 Estimation Strategy

Discussion of causality. The institutional setting described in the previous section provides a natural source of variation to identify the effects of minimum wages on firms’ organization. During the 2003–2006 reconvergence period, firms faced different increases in the cost of employing minimum wage workers, depending both on their GMR schedule and on their initial share of minimum wage workers. Our objective is to exploit this variation to estimate the causal effects of minimum wages. A standard approach would be to estimate these effects using a difference-in-differences (DD) design comparing treated and untreated firms before and after a policy change (see, e.g., Card and Krueger, 1994; Draca et al., 2011; Drucker et al., 2021). This approach is not directly applicable here because all firms in our sample experienced increases in their applicable minimum wage (Figure 1). Instead, identification comes from differences in the magnitude of these increases across firms. We therefore estimate the following specification:

$$\Delta_{03/06} Y_{jt} = \alpha^{DD} \Delta_{03/06} \ln TC_{jt}^{MW} + \theta_{s(j)}^{DD} + \psi_{a(j)}^{DD} + \Delta_{03/06} \varepsilon_{jt}^{DD}, \quad (3)$$

where Y_{jt} denotes an outcome of firm j operating in industry $s(j)$ and area $a(j)$ in year t , $\Delta_{03/06}$ denotes the change between 2003 and 2006, TC_{jt}^{MW} is the total cost of employing a minimum wage worker in firm j at time t , and $\theta_{s(j)}^{DD}$ and $\psi_{a(j)}^{DD}$ are industry and commuting-zone fixed effects. Because the specification is estimated in first differences, it accounts for any time-invariant firm heterogeneity, while the fixed effects absorb industry- and area-specific trends. It is a generalization of the standard difference-in-differences setting to the extent that the main parameter of interest in equation (3), α^{DD} , is identified from firms in different GMRs that are *differentially* exposed to increases in the minimum wage rather than exposed *vs.* non exposed.

However, α^{DD} identifies the causal effect of interest only if, absent differential minimum wage increases, firms assigned to different GMR schedules would have followed parallel trends after controlling for industry- and area-specific shocks. This assumption may be violated because GMR assignment was not random. As discussed in Section 2 and documented in Tables B2 and B4 of Appendix B, firms self-selected into different GMR schedules and differ systematically in their characteristics. Earlier GMRs contain larger firms with, on average, a lower share of minimum wage workers—and therefore lower exposure to subsequent minimum wage increases—while being disproportionately concentrated in manufacturing rather than in trade, hotels, or restaurants. These systematic differences raise concerns that firms assigned to different GMR schedules may have followed different underlying trends.

To address this concern, we also estimate a triple-difference (DDD) specification that exploits within-GMR variation in firms’ exposure to minimum wage increases, as measured by their initial share of workers affected

by increases in the minimum wage (Draca et al., 2011; Riley and Rosazza Bondibene, 2017; Harasztosi and Lindner, 2019). Specifically, we estimate:

$$\begin{aligned}\Delta_{03/06}Y_{jt} = & \beta^{DDD}\Delta_{03/06}\ln TC_{jt}^{MW} \times SH_j^{MW} + \alpha^{DDD}\Delta_{03/06}\ln TC_{jt}^{MW} + \eta^{DDD}SH_j^{MW} \\ & + \theta_{s(j)}^{DDD} + \psi_{a(j)}^{DDD} + \Delta_{03/06}\varepsilon_{jt}^{DDD},\end{aligned}\quad (4)$$

where SH_j^{MW} measures the share of low-wage workers in firm j in 2002, i.e. those earning in 2002 less than the 2006 (unified) minimum wage. The identifying assumption is weaker than in the DD specification: while firms may have self-selected into different GMR schedules, it is plausible that, absent the policy, firms within the same GMR would have experienced similar trends between 2003 and 2006. Differences in wage structures nevertheless implied heterogeneous exposure to increases in the minimum wage.²³ In contrast to the DD specification, identification in the DDD strategy occurs exclusively within each GMR: the coefficient of interest, β^{DDD} , is estimated from firms facing the same GMR schedule but different initial exposure to minimum wage increases.²⁴

Implications of non-linear treatment effects. As shown in Panel (B) of Figure 1, all firms experienced the same cumulative increase in the minimum wage between 1998 and 2006, but its timing differed across GMR schedules. Firms in GMR1 faced relatively small increases before 2003 and larger increases thereafter, whereas the opposite pattern holds for GMR5, with GMR2–GMR4 lying in between. This heterogeneity in the timing of minimum wage increases across GMRs creates an endogeneity concern for the double-difference estimates, but not for the triple-difference estimates, which rely on within-GMR identifying variation.²⁵ However, the heterogeneity in the underlying shocks may still complicate the interpretation of the estimation results if the effects of minimum wage increases are non-linear. Indeed, coefficients estimated in the DD and DDD specifications correspond to weighted averages of local (linearized) treatment effects. Equations 3 and 4 formally assume linear responses to minimum wage increases and, in the DDD specification, to firms’ initial exposure. If this linearity assumption holds, the heterogeneity in the magnitude of shocks across GMRs should not affect either our estimates or their interpretation. If instead the true relationship is non-linear, the estimates from Equations 3 and 4 will reflect weighted averages of potentially heterogeneous effects, driven primarily by firms in GMR2 through GMR4, which together account for more than 90% of the sample (see Table B2).²⁶ More precisely, if the effect of minimum wage increases on the outcome of interest is increasing but concave, the 2003–2006 shock may have generated larger elasticities for firms in GMR1 than for firms in other GMRs. Conversely, if the effect is convex, the same shock may have been associated with lower elasticities for firms in GMR1 relative to other groups.

²³Equation 4 also controls for differential trends associated with firms’ initial share of minimum wage workers through the term $\eta^{DDD}SH_j^{MW}$.

²⁴One last concern lies in the fact that several of the subsidies provided by the Aubry laws varied slightly with the size of firms, e.g. below or above the 20 worker threshold. As a consequence, it may be the case that some firms manipulated their size during the pre-regression period (1998 to 2002), thus being able to endogenously lower their costs of employing a minimum wage worker in the subsequent years. To address this concern, we propose to instrument for $\Delta_{03/06}\ln TC_{jt}^{MW}$, firms’ total labor cost at the minimum wage, both in level and in interaction, using the change in the minimum wage actually paid out to workers, i.e. the minimum wage concept of panel (A) of Figure 1. This variable is net of the suspicious component, namely the potentially endogenous, firm-level varying, payroll taxes. The rationale of this strategy is that first, the instrumental variable is likely to have power, since workers’ “net” wages are the largest component determining their total cost to firms. Second, as discussed in Section 2, the sub-period surrounding the convergence of the different GMRs led to unanticipated and arguably more exogenous changes in the different “net” minimum wages in the French economy. The results of our IV specifications can be found in Online Appendix D.

²⁵This is also true of the possibility of anticipation effects: if firms adjusted their organization in advance of future GMR increases, the double-difference estimates would likely be lower bounds. But the triple-difference strategy should avoid such concerns due to its reliance on within-GMR variation.

²⁶Assessing such heterogeneity directly is difficult because estimating the model separately by GMR yields insufficient statistical power.

3.2 Data and Measurement

Data sources. Our dataset is based on comprehensive administrative records covering all French companies and employees. It extends the dataset used in Caliendo, Monte, et al. (2015) by incorporating newly available administrative sources.²⁷ The core source is the Annual Social Data Declarations (DADS), in their most detailed form at the job (“poste”) level (Insee, 1995–2008; Insee, 1993). These files span the full period of analysis and include each salaried worker’s occupation (Profession and Socio-Professional Category), hours worked, and wages (net of so-called “employer” payroll taxes).

A first important auxiliary dataset, compiled by the Ministry of Employment and Solidarity during the Aubry laws, is the Allègement des cotisations sociales (Reduction in Social Security Contributions) file (Insee, 2000). It is based on declaration forms (Cerfa No. 11499*01, Article 10 of Decree 2000-89) submitted by establishments that negotiated a working time reduction (RWT) agreement and applied for the corresponding payroll tax reductions under the Aubry 1 and Aubry 2 schemes.²⁸ This database records the dates of RWT agreements, which determine the GMR applicable to each firm, and the subsidies received under Aubry 1 and 2 schemes.

A second auxiliary source, the “2483 Declarations” (Ministère du Travail (DARES), 2002–2007), records the legally required vocational training reports submitted to the Ministry of Labor by companies with more than 10 employees (Lê, 2013; Aubert et al., 2009). These declarations are part of the “train or pay” system introduced by the Law of July 16, 1971, which mandates that firms devote at least 1.5% of their payroll to vocational training. Under this regulation, firms declare the amounts allocated to training so that compliance can be verified: when their training expenditures fall below the legal threshold, they must pay the difference either to public training bodies (Organismes Paritaires Agréés Collecteurs, OPAC) or as a tax. Unfortunately, the statistical files compiled by the Ministry of Labor during our period of analysis are not exhaustive, covering just over 30% of the eligible firms,²⁹ but they are representative, being random samples of companies with more than 10 employees.

Lastly, these files are matched with the exhaustive tax files (Fichier complet unifié SUSE-FICUS) containing the balance sheet data and profit and loss accounts of firms (Insee, 1996–2007). The latter allow us to compute various indicators of firms’ sizes beyond employment: sales and value-added. They also allow us to inspect factor shares and various proxies of revenue productivity.

Firms’ internal organization: hierarchical layers. To characterize firms’ internal organization, we follow Caliendo and Rossi-Hansberg (2012) and Caliendo, Monte, et al. (2015), whose concept of hierarchical layers also underpins the theoretical framework developed in Section 4. The French occupational classification (PCS-ESE) assigns each worker to one of five broad occupational categories (identified by their one-digit PCS code), which contain indications about the hierarchical layer of each worker:

2. Salaried company managers;
3. Senior staff: chief financial officers, chief technical officers, heads of human resources, and logistics or purchasing managers;

²⁷In particular, we integrate detailed job-level data on labor costs and worker training from administrative files recently made accessible to researchers.

²⁸Establishments belonging to firms that did not apply for the reduction are excluded, but the latter only represent about 10% of relevant firms according to complementary surveys (Bunel, 2004). Firms created directly at 35 hours in 2000 are also excluded from the “Allègement des cotisations sociales” dataset: this is one of the motivations to focus on firms that were already active in 2000 and continuously present until 2006 – see below.

²⁹Until 2004, only a random sample of 15,000 to 20,000 companies was entered each year. From 2004 onwards, the coverage rate of the source improved considerably with the entry of all valid forms returned to the tax authorities (Lê, 2013). The 2003 sample is thus relatively small in comparison with later years. To address this and preserve some statistical power, we estimate firm-level changes in training hours between 2003 and 2006 using the closest available data: 2003/06 when possible, and otherwise 2003/07, 2003/05, 2002/06, 2002/07, 2002/05, 2004/06, or 2004/07. Normalizations are made for differences in the number of years where required.

4. Supervisors: quality control, technical, accounting, and sales supervisors;
5. Clerical employees: qualified and non-qualified secretaries, human resources, sales, and accounting employees, and telephone operators;
6. Blue-collar workers: qualified and non-qualified welders, assemblers, machine operators, and maintenance workers.

We map the PCS classification into four hierarchical layers by assigning each occupational category to a separate layer, except PCS5 and PCS6, which are merged into a single production layer.³⁰ Firms are then classified according to the number of managerial layers they contain.³¹ Firms featuring only one of the above categories are not considered as hierarchical and are thus classified as 0- (managerial) layer firms, as in Caliendo and Rossi-Hansberg (2012). Firms featuring two layers, regardless of whether the occupied layers are contiguous, are classified as 1-(managerial) layer firms. Similarly, firms exhibiting three or four layers will be classified as 2-layer or 3-layer firms, respectively.

Measurement of labor costs. As discussed in Section 2, the DADS data do not report firms’ total labor costs. Instead, they record “salaires bruts”, which include “employees’ payroll contributions” but exclude “employers’ payroll taxes”.³² This distinction was largely irrelevant in Caliendo, Monte, et al. (2015), which uses wages to measure workers’ skills or productivity. In our setting, however, it is crucial because firms’ labor demand depends on total labor costs rather than on salaires bruts. In particular, the missing amount of “employer-level” payroll taxes depends on firm-level (industry, participation in a RWT agreement, etc) as well as worker-level (mainly wage level) characteristics. Fortunately, combining the DADS with the Reduction in Social Security Contributions files provides sufficient information to reconstruct labor costs by applying the relevant legislation, much as a firm’s human resources department would do. Following Cottet et al. (2012), we simulate the labor cost of every job in every GMR firm, incorporating the payroll-tax reductions associated with the Aubry 1, Aubry 2, and Fillon reforms, which account for the vast majority of changes in labor costs over our study period. We also implement the earlier ARTT-loi de Robien (1996) and the revised réduction bas salaires introduced by the Juppé government in 1996, which continued to affect labor costs until 2005 and 2003, respectively.³³

Measurement of training. A limitation of the DADS data is that they do not contain any direct information about education or skills. The complementary “2483 Declarations” dataset allows us to track firms’ internal training strategies, which in the French institutional context (marked by strict employment protection regulations, see Section 2.3) is expected to be the main channel of skill adjustment. While the financial data contained in this dataset do not always correspond to internal training — due to the “OPAC” balancing mechanism described above (Carriou and Jeger, 1997; Lê, 2013) — the declarations report the number of

³⁰Caliendo, Monte, et al. (2015) show that workers in these two categories have nearly identical wage distributions, suggesting similar knowledge levels and training requirements. We obtain the same result over our sample period.

³¹Our labeling convention is similar to Caliendo and Rossi-Hansberg (2012), and different from Caliendo, Monte, et al. (2015) (if we interpret their papers correctly!), but without impact on any aspect of the analysis: a L - (managerial) layer firm in our paper and in Caliendo and Rossi-Hansberg (2012) simply corresponds to a $L + 1$ - (production or managerial) layer firm in Caliendo, Monte, et al. (2015).

³²Of course, these legal definitions do not reflect the economic incidence of payroll taxes and contributions.

³³For an overview of the payroll-tax subsidy schemes that predated the Aubry reforms, see the report published by the French Senate: <https://www.senat.fr/rap/199-030/199-03093.html>. We are grateful to Sébastien Roux (INSEE) for sharing an early version of the programs developed by Cottet et al. (2012), which provided a first approximation of payroll-tax subsidies over our sample period. Our administrative data directly report subsidies received under the Aubry 1 and Aubry 2 programs. For the remaining schemes, subsidies must be imputed from the legislation because firm-level information on actual take-up is unavailable. We therefore assume that firms claim all subsidies for which they are legally eligible. Although this assumption may introduce some measurement error, we expect it to be limited given the strong incentives to claim eligible subsidies and the legal sanctions associated with fraudulent claims. Because simulated labor costs constitute our main explanatory variable, any remaining measurement error would tend to attenuate the estimated coefficients and could raise endogeneity concerns. We address this issue using the instrumental-variable strategy described in footnote 24. The corresponding estimates, reported in Online Appendix D, are quantitatively and qualitatively very similar to the baseline results.

employees who participated in training during the year, whether internal or external and financed by the employer. This indicator encompasses both participation in the Individual Right to Training (*Droit Individuel à la Formation*, DIF), a scheme granting employees a set number of training hours per year,³⁴ and training undertaken outside working hours but compensated by training allowances.³⁵ These declarations break down training participation by occupation and form our main empirical indicators of training activity.

Estimation sample. Our final estimation sample is fully documented in Online Appendix B. It is a balanced sample that contains all business organizations that:

- are profit-maximizing, i.e. active in the for-profit sector;
- report more than 5 jobs to the social administration: this condition ensures that non-trivial (i.e. sufficiently complex) organizations are inserted in the regressions;
- were created before 2000 (the year of change in legal working hours) and remained active until 2006. These “survival” conditions ensure that (i) all firms shared the same institutional framework and information set regarding the evolution of the GMR, as determined by the January 2000 Aubry 2 Law (Section 2, Appendix A), and that (ii) our sample consists of firms with a minimum degree of maturity, allowing us to abstract from organizational dynamics related to early growth or exit, which are not the focus of this paper;³⁶
- were subject to a GMR by 2002: this guarantees that all firms in the sample adopted the 35-hour workweek – thus harmonizing workweek duration across sample firms – and benefited from the Aubry 1 or 2 subsidies, ensuring that labor costs can be computed accurately at the job level (see Section 2 and Online Appendix B.1 for full details).

3.3 Descriptive Statistics

Appendix B contains comprehensive descriptive statistics about this population of firms in our estimation sample. We focus in this section on our main dimension of investigation: the interaction between labor costs and firms’ organizational strategies.

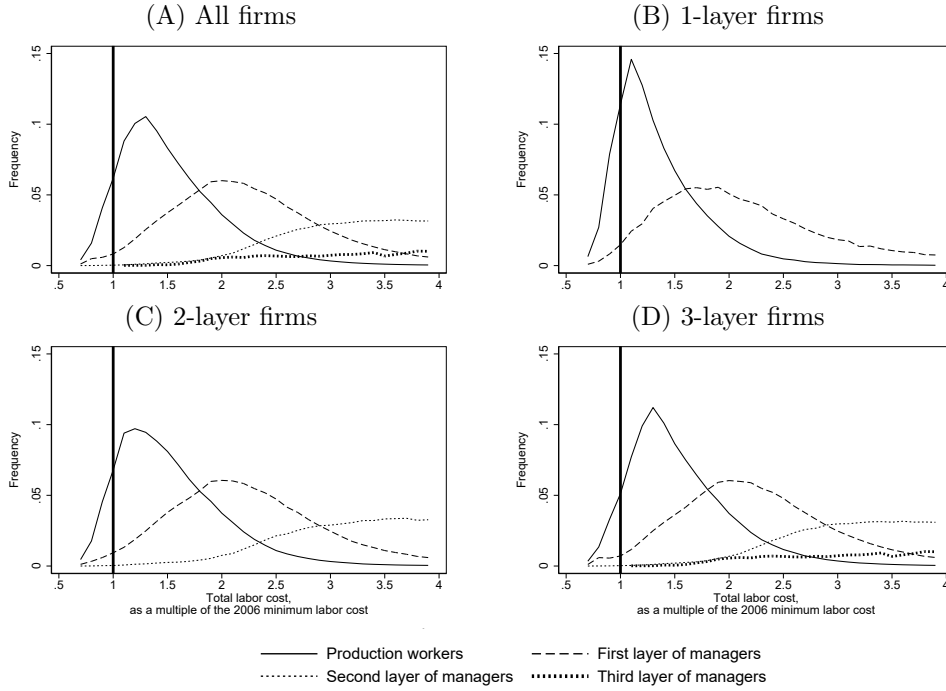
A key dimension to document is the extent to which the empirical methodology proposed by Caliendo, Monte, et al. (2015) to describe “production hierarchies” remains valid in our data. The answer to this question is positive, which comes as no surprise since we rely on the same data sources, over (almost) the same time period, although Caliendo, Monte, et al. (2015) focus on manufacturing industries only. Appendix C.1 replicate some of their main results in our sample: in our data as in theirs, firms form well-behaved hierarchies to the extent that the number of hours or jobs in the bottom, production layer (layer 0) is usually larger than in layer 1, which is itself larger than in layer 2, etc. Furthermore, the hierarchy also holds in terms

³⁴In September 2003, trade unions and company representatives signed an agreement creating an “individual right to training” (*Droit Individuel à la Formation*, DIF), which was later fully incorporated into the 2004 law on “professional training and social dialogue”. Under the terms of this agreement, employees on open-ended contracts or, under certain conditions, fixed-term contracts, were entitled to 20 (optional) hours of training per year, formally in a field of their choice - but subject to their employer’s agreement, implying that firms have in practice a margin to direct them in their best interest. It is important to acknowledge that this reform of vocational training affected all French firms during our estimation period. Our estimation strategy captures differences in training strategies across firms that are exposed to differential increases of their minimum wages (GMR) and thus naturally controls for such economy-wide shocks, which are captured by the $\theta_{s(j)}$ terms in Equations 3 and 4.

³⁵More anecdotally, they also report skills assessments and Validation of Acquired Experience (VAE) processes, which provide formal recognition of skills acquired through professional experience. Employees under apprenticeship or professionalization contracts are excluded from these counts.

³⁶We also exclude firms reporting negative values of value added, but this only discards very few observations (see Online Appendix B.1) and has mainly the benefit of harmonizing firms’ observation counts across tables. Results are unaltered when this cleaning is not implemented. More importantly, it should be noted that entries and exits, i.e. the firm selection margin, are fully accounted for in the calibration exercise of Sections 4 and 5, as the proportion of firms created over a seven-year period and the share of workers in these “new” firms are introduced as data moments to identify parameters δ , f_E and f as well as the distribution of α (and are thus critical to the endogenous determination of the selection threshold, $\bar{\alpha}$, see Sections 4 and 5.1).

Figure 2: Distributions of Hourly Labor Costs Across Layers in Organizations



Notes: The graphs correspond to histograms with bins of width 0.1 obtained for the distribution of hourly labor costs measured in 2002, but normalized by the total labor cost at the minimum wage in 2006. This norm is materialized by a vertical solid line in each panel.

of wages or labor costs, the latter concept being original in our paper. We document this feature in full detail in Figure 2, which reports histograms (by bins of width 0.1) of total labor costs as of 2002, normalized by the total labor cost at the minimum wage in 2006. Our analysis is performed for our entire sample in Panel (A), as well as separately for each category of firms featuring at least one managerial layer in Panels (B) to (D). In each case, we find that the support of the distributions of labor costs overlap across layers, but their mode is sequentially shifted to the right as one climbs hierarchical layers. In more technical terms, the distribution of wages at a given hierarchical layer has strong first order stochastic dominance over distributions of wages at lower layers: in the vast majority of firms, almost all managers earn more than production workers, and those at the top of the hierarchy earn more than those at intermediate layers. As a consequence, virtually all workers who are hit by the minimum wage constraint over our period of study, i.e. those earning in 2002 less than the minimum wage in 2006, belong to the bottom layer (Layer 0).

3.4 Organizational and Productivity Responses to Minimum Wages: Regression Results

We now turn to the analysis of how firms adjusted their internal organization to the minimum-wage increases implemented between 2003 and 2006. Throughout this section, we estimate the long-difference specifications given by Equations 3 (double differences, DD) and 4 (triple differences, DDD) using OLS.³⁷ For each specification, we report the estimated elasticities together with their implied economic effects. The DD coefficient, α^{DD} , measures the average elasticity of the outcome with respect to the labor cost of employing a minimum-

³⁷Online Appendix D replicates all specifications using the instrumental-variable strategy described in footnote 24 of Section 3.1. The IV estimates are both qualitatively and quantitatively similar to the OLS results reported here. The Online Appendix also reports a broad set of additional robustness checks.

wage worker. The DDD coefficient, β^{DDD} , captures how this elasticity varies with firms' initial exposure to minimum wage increases, measured by the share of workers earning less than the unified 2006 minimum wage in 2002. Since this interaction is scaled by exposure, the DDD coefficient corresponds to the differential response between a fully exposed firm (exposure equal to one) and a non-exposed firm (exposure equal to zero). Because the average exposure in our sample is only $\overline{SH}^{MW} = 0.136$ (Table B2), DDD coefficients are naturally larger in magnitude than the implied effects for the average firm. To facilitate interpretation, we further report the implied effect of the “medium” GMR2 shock, corresponding to a 3.7% real increase in the labor cost of employing a minimum-wage worker (Table 1). Under the DD specification, the implied effect is $\alpha^{DD} \Delta_{03/06} \overline{\ln TC}_{GMR2}^{MW}$. Under the DDD specification, the total implied effect for a firm with average exposure is $(\alpha^{DDD} + \beta^{DDD} \overline{SH}^{MW}) \Delta_{03/06} \overline{\ln TC}_{GMR2}^{MW}$. For transparency, we also report separately the contribution of the interaction term, $\beta^{DDD} \Delta_{03/06} \overline{\ln TC}_{GMR2}^{MW} \times \overline{SH}^{MW}$, which we refer to as the *exposure effect*. This component is identified exclusively from within-GMR variation in firms' initial exposure to the minimum wage and is therefore less sensitive to concerns that firms self-selected into different GMR schedules than the common component, $\alpha^{DDD} \Delta_{03/06} \overline{\ln TC}_{GMR2}^{MW}$, which is identified from variation across GMR groups.³⁸

Table 1 examines how increases in minimum wage labor costs propagated across firms' wage hierarchies. The estimates pool firms with different organizational structures. Appendix C.2 complements this analysis by estimating the same specifications separately for firms with one, two, and three managerial layers. Despite the smaller estimation samples, the qualitative patterns remain remarkably stable across organizational structures. The most robust finding concerns the bottom of the hierarchy. Both the DD and DDD specifications indicate that minimum-wage increases translated into substantially higher hourly labor costs in Layer 0. Under the “medium” GMR2 scenario, the DD estimates imply an increase of about 1.1% in labor costs, while the DDD exposure effect contributes to 0.2 percentage points for the average firm. Appendix C.2 shows that this bottom-layer response is present across all organizational structures, although it is estimated less precisely for firms with three managerial layers because of the smaller sample size. The evidence differs for higher hierarchical layers. While the DD estimates suggest positive wage spillovers throughout the hierarchy, the DDD interaction term becomes negative above Layer 0, indicating that firms with greater initial exposure to the minimum wage experienced relatively weaker labor-cost growth in managerial layers. This pattern is consistent with firms compressing wage growth higher up the hierarchy in response to the larger cost increases experienced at the bottom. Importantly, the same qualitative pattern emerges within each organizational category (Appendix C.2), indicating that this pattern is not driven by differences in firms' initial organizational complexity.

Table 2 next examines how firms adjusted their internal organization in response to minimum-wage increases. Panel (A) reports changes in the size of hierarchical layers, measured alternatively in jobs and in hours, while Panel (B) considers changes in the number of hierarchical layers. The baseline estimates pool firms with different organizational structures. Again, Appendix C.2 complements this analysis by estimating the same specifications separately for firms with one, two, and three managerial layers. Panel (A1) first considers employment measured in jobs. Higher minimum wages lead firms to reduce overall employment: according to the DD estimates, a 1% increase in the labor cost of employing a minimum-wage worker lowers total employment by about 0.22%, while the DDD interaction term is also negative and significant but implies a somewhat smaller impact at sample average. Under the “medium” GMR2 scenario, this corresponds to a decline of about 0.8% in the number of jobs according to the DD estimates, with a similar total DDD effect (and a smaller 0.3% decline implied by the DDD exposure term). The DD estimates suggest that this adjustment is concentrated in the production layer, where employment falls by about 1.7% whereas the estimated

³⁸At this stage, these calculations abstract from general-equilibrium effects, which are introduced in Section 4. We show there that, for the range of minimum-wage increases observed across GMRs 1 through 5, equilibrium price adjustments are limited, so that partial- and general-equilibrium impacts are quantitatively nearly identical.

Table 1: Impact of Minimum Wage Increases on Labor Costs Across Firms' Organizations

	(1)	(2)	(3)	(4)	(5)
$\Delta_{03/06}$ ln hourly labor cost at:	All layers	Layer 0	Layer 1	Layer 2	Layer 3
DD: Δ ln lab. cost at GMR	0.352*** (0.021)	0.286*** (0.024)	0.195*** (0.046)	0.405*** (0.060)	0.221 (0.150)
DDD: Δ ln lab. cost at GMR $\times$ Share MW workers in 2002	-0.085 (0.126)	0.334*** (0.123)	-0.625* (0.354)	-1.377** (0.561)	-0.127 (1.210)
Magnitude of the GMR2 shock at the sample average:					
DD implied effect	0.013	0.011	0.007	0.015	NS
DDD exposure effect	NS	0.002	-0.003	-0.007	NS
DDD total effect	0.014	0.012	0.006	0.012	NS
Mean of dependent variable:	0.095	0.110	0.097	0.073	0.106
Observations	55,345	55,345	49,480	33,203	8,484

Notes: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Standard errors, reported in parentheses, are clustered at the 1-digit industry $\times$ commuting-zone level. All regressions include 4-digit industry and commuting-zone fixed effects. The DD coefficient measures the average elasticity of the outcome with respect to the labor cost of employing a minimum-wage worker. The DDD coefficient measures this elasticity at full exposure to the minimum wage, where exposure is proxied by the share of workers earning less than the unified 2006 minimum wage in 2002. To facilitate interpretation, the table also reports the implied effect of the “medium” GMR2 reform at the sample mean. In the DD specification, the implied effect is $\alpha^{DD} \Delta_{03/06} \overline{\ln TC_{GMR2}^{MW}}$. In the DDD specification, the total implied effect is $(\alpha^{DDD} + \beta^{DDD} \overline{SH}^{MW}) \Delta_{03/06} \overline{\ln TC_{GMR2}^{MW}}$. The row labelled “DDD exposure effect” reports the contribution of the interaction term. “NS” indicates non-significance. The estimation sample varies across columns. Columns (1) and (2) include all firms. Columns (3)–(5) successively restrict the sample to firms exhibiting at least two (Layers 0 and 1), three, and four hierarchical layers, respectively, in both 2003 and 2006.

effects for managerial layers are small and statistically insignificant. The heterogeneity estimates reported in Appendix C.2 show that this pattern is remarkably stable across firms with different initial organizational structures. Taken together, these results indicate that firms primarily adjust by reducing the number of production workers rather than by downsizing their managerial workforce. Panel (A2) reports the corresponding estimates when layer size is measured in hours worked. The aggregate contraction is similar (slightly weaker), confirming that firms become smaller overall following minimum-wage increases. At the layer level, however, hours provide additional information on the intensive margin of adjustment: the DDD estimates point to a relative increase in hours worked in the second managerial layer across exposed firms, even though Panel (A1) shows no corresponding increase in managerial employment. This suggests that firms responded by relying more intensively on existing managers rather than by hiring additional supervisory workers. Appendix C.2 shows that this intensive-margin adjustment is driven primarily by firms with two managerial layers, while the aggregate contraction in the production layer remains a common feature across organizational structures.

Panel (B) turns to the extensive margin of organizational adjustment by examining changes in the number of hierarchical layers. Column (1) shows that firms exposed to larger minimum-wage increases adopted flatter organizational structures. Under the “medium” GMR2 scenario, the estimated reduction amounts to about 0.008 layers under the DD specification, and 0.007 layers for the exposure effect under the DDD specification. Columns (2)–(7) show that this flattening reflects both a lower probability of adding managerial layers and a higher probability of removing existing ones, resulting in a lower incidence of the most complex organizational structures.

Table 3 next examines the third type of organizational strategy captured in our data, namely training across hierarchical layers. Unfortunately, the training indicators are only available for a subset of firms (see Section 3.2). Although this restriction on the sample size reduces the statistical power of our regressions,³⁹ we find that workers in firms facing increases in their minimum wages tend to participate more frequently

³⁹The Online Appendix D.3 replicates Tables 1, 2 and 4 in order to check that the training data do not suffer from selection bias and are representative of the full population of firms.

Table 2: Impact of Minimum Wage Increases on Firms' Organizations

(A) Impact on the Relative Size of Hierarchical Layers

	(1)	(2)	(3)	(4)	(5)
Δ ln layer size, at:	All layers	Layer 0	Layer 1	Layer 2	Layer 3
Observations	55,345	55,345	49,480	33,203	8,484
(A1) Layer sizes in jobs					
DD: Δ ln lab. cost at GMR	-0.218*** (0.052)	-0.458*** (0.079)	-0.0904 (0.104)	-0.169 (0.113)	-0.00791 (0.159)
DDD: Δ ln lab. cost at GMR $\times$ Sh MW workers in 2002	-0.617* (0.371)	-0.093 (0.457)	-0.506 (0.668)	0.811 (0.798)	0.546 (1.270)
Magnitude of the GMR2 shock at the sample average:					
DD implied effect	-0.008	-0.017	NS	NS	NS
DDD exposure effect	-0.003	NS	NS	NS	NS
DDD total effect	-0.009	-0.018	NS	NS	NS
Mean of dependent variable:	0.009	0.006	-0.001	0.056	-0.004
(A2) Layer sizes in hours					
DD: Δ ln lab. cost at GMR	-0.154*** (0.050)	-0.397*** (0.085)	-0.045 (0.115)	-0.115 (0.137)	0.104 (0.277)
DDD: Δ ln lab. cost at GMR $\times$ Sh MW workers in 2002	-0.655* (0.366)	-0.105 (0.473)	-0.630 (0.843)	2.799** (1.180)	-1.617 (2.050)
Magnitude of the GMR2 shock at the sample average:					
DD implied effect	-0.006	-0.015	NS	NS	NS
DDD exposure effect	-0.003	NS	NS	0.014	NS
DDD total effect	-0.007	-0.015	NS	NS	NS
Mean of dependent variable:	0.014	0.016	-0.007	0.041	-0.043

(B) Impact on the Number of Hierarchical Layers

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
(55,345 obs.)	Δ nb layers	Prob. increase	Prob. decrease	Prob. stability	Δ Proba 3 layers	Δ Proba ≥ 2 layers	Δ Proba ≥ 1 layer
DD: Δ ln lab. cost at GMR	-0.205** (0.103)	-0.308*** (0.057)	-0.076 (0.064)	0.384*** (0.077)	-0.211*** (0.068)	0.015 (0.060)	-0.009 (0.035)
DDD: Δ ln lab. cost at GMR $\times$ Sh MW workers in 2002	-1.394** (0.647)	-0.125 (0.368)	0.993*** (0.374)	-0.868* (0.455)	0.218 (0.329)	-0.777* (0.402)	-0.836*** (0.316)
Magnitude of the GMR2 shock at the sample average:							
DD implied effect	-0.008	-0.011	NS	0.014	-0.008	NS	NS
DDD exposure effect	-0.007	NS	0.005	-0.004	NS	-0.004	-0.004
DDD total effect	-0.009	-0.011	NS	0.013	-0.007	NS	NS
Mean of dependent variable:	-0.036	0.165	0.193	0.642	-0.015	-0.013	-0.008

Notes: See Table 1. In Panel (A), the composition of the estimation sample evolves across columns: in the first two columns, all sample firms are included. In columns (3) to (5), only firms exhibiting at least 2 (Layers 0 and 1), 3 or 4 layers respectively, both in 2003 and 2006, are included.

in training programs (Arulampalam et al., 2004).⁴⁰ The effect is stronger for production workers in smaller (single-layer) firms, where the DDD interaction term is positive and statistically significant. It implies an increase of several percentage points in the probability of training, with an impact of roughly 2 percentage points in the medium GMR2 scenario. This represents around 6% of the baseline training incidence (approximately 30%, as shown in Table B2). Results across managerial layers are more mixed: the DD estimates remain consistently positive, but the DDD estimates are smaller in magnitude, and in two- and three-layer

⁴⁰As previously discussed, the “no redundancy” clause in the Aubry 1 law, along with the generally strong employment protection regulations in France, likely encouraged firms to favor internal training as the main channel for skill upgrading, rather than adjusting along the extensive margin—that is, replacing existing workers with new hires, as in e.g. Neumark and Wascher (2001).

Table 3: Impact of Minimum Wage Increases on Training

	(1)	(2)	(3)	(4)
Observations	All firms 17,407	1-layer firms 2,494	2-layer firms 9,689	3-layer firms 4,569
(A) Δ Proba. training at layer l_0				
DD: Δ ln lab. cost at GMR	0.575*** (0.147)	0.582 (0.473)	0.322 (0.205)	0.860*** (0.313)
DDD: Δ ln lab. cost at GMR $\times$ Sh MW workers in 2002	0.726 (0.946)	3.877* (2.148)	0.497 (1.379)	-0.498 (2.630)
Magnitude of the GMR2 shock at the sample average:				
DD implied effect	0.021	NS	NS	0.032
DDD exposure effect	NS	0.020	NS	NS
DDD total effect	0.022	NS	NS	0.031
Mean of dependent variable:	0.179	0.118	0.183	0.216
(B) Δ Proba. training at layers l_1 to l_3				
DD: Δ ln lab. cost at GMR	0.565*** (0.143)	0.372 (0.451)	0.399** (0.200)	0.489* (0.290)
DDD: Δ ln lab. cost at GMR $\times$ Sh MW workers in 2002	0.153 (0.916)	3.062 (2.133)	-0.297 (1.354)	-0.975 (2.565)
Magnitude of the GMR2 shock at the sample average:				
DD implied effect	0.021	NS	0.015	0.018
DDD exposure effect	NS	NS	NS	NS
DDD total effect	0.021	NS	0.014	NS
Mean of dependent variable:	0.173	0.087	0.182	0.215

Notes: See Table 1. Both in Panels (A) and (B), the composition of the estimation sample varies across columns: column (1) includes all firms, while columns (2) to (4) disaggregate the sample by organizational structure, focusing respectively on firms with one, two, and three managerial layers.

firms the sign is reversed (but insignificant). This suggests that the impact of the 2003–2006 shock may have been more heterogenous across organizations: in some firms, it may have been optimal to complement up-skilled production workers with better-trained managers, whereas in others, the enhanced skills of production workers may have reduced their dependence on skilled managerial supervision (Zamora, 2006).

Lastly, Table 4 examines the impact of increases in the minimum wage on firms’ scale in Panel (A), input mix in Panel (B), and productivity in Panel (C). Panel (A) compares the reductions in employment documented in Table 2 with the estimated impact on output. We obtain that value added and sales remain broadly unchanged. Thus, although firms reorganize their workforce, there is little evidence that they scale down production.

Panel (B) examines how firms adjusted production factors other than labor. The evidence points to only limited changes in the use of intermediate inputs. Although the DD estimates suggest a small increase in intermediate inputs per worker, this pattern is not confirmed by the DDD interaction term. By contrast, the DDD estimates indicate a statistically significant increase in both capital per worker and investment per worker (or per hour), suggesting that firms responded to higher labor costs by modestly increasing capital intensity.⁴¹ The implied magnitudes, however, remain economically small for the average firm.

Panel (C) shows that these changes translated into (revenue) productivity gains.⁴² Labor productivity rises significantly under both the DD and DDD specifications. Under the “medium” GMR2 scenario, the implied gain amounts to about 1% when productivity is measured per job and 0.9% per hour, of which

⁴¹ Around 15% of firms in the sample report no investment in either 2003 or 2006, making investment a particularly lumpy outcome.

⁴² Firm-level output prices are unavailable in the French administrative data, preventing the construction of physical TFP measures.

Table 4: Impact of Minimum Wage Increases on Firms' Size and Productivity

(A) Impact on Firms' Size

(55,345 obs.)	(1) $\Delta \ln$ jobs	(2) $\Delta \ln$ hours	(3) $\Delta \ln$ VA	(4) $\Delta \ln$ Sales
DD: $\Delta \ln$ lab. cost at GMR	-0.218*** (0.052)	-0.154*** (0.050)	0.052 (0.065)	0.006 (0.054)
DDD: $\Delta \ln$ lab. cost at GMR $\times$ Sh MW workers in 2002	-0.617* (0.371)	-0.655* (0.366)	0.135 (0.395)	-0.137 (0.337)
Magnitude of the GMR2 shock at the sample average:				
DD implied effect	-0.008	-0.006	NS	NS
DDD exposure effect	-0.003	-0.003	NS	NS
DDD total effect	-0.009	-0.007	NS	NS
Mean of dependent variable:	0.009	0.014	0.084	0.106

(B) Impact on Production Factors beyond Labor

	(1) $\Delta \ln$ inputs/jobs	(2) $\Delta \ln$ cap/jobs	(3) $\Delta \ln$ inv/jobs	(4) $\Delta \ln$ inputs/hrs	(5) $\Delta \ln$ cap/hrs	(6) $\Delta \ln$ inv/hrs
DD: $\Delta \ln$ lab. cost at GMR	0.166*** (0.062)	0.005 (0.078)	0.079 (0.263)	0.102* (0.059)	-0.060 (0.076)	0.031 (0.265)
DDD: $\Delta \ln$ lab. cost at GMR $\times$ Sh MW workers in 2002	0.207 (0.396)	1.504*** (0.492)	3.051* (1.791)	0.244 (0.379)	1.542*** (0.473)	3.161* (1.789)
Magnitude of the GMR2 shock at the sample average:						
DD estimate	0.006	NS	NS	0.004	NS	NS
DDD exposure effect	NS	0.008	0.015	NS	0.008	0.016
DDD total effect	0.007	NS	NS	0.004	NS	NS
Mean of dependent variable:	0.106	0.123	-0.026	0.118	-0.007	-0.034
Observations	55,329	55,345	47,181	55,329	55,345	47,181

(C) Impact on Productivity

$\Delta \ln$:	(1) Labor productivity	(2) Labor productivity	(3) Accounting TFP	(4) Accounting TFP	(5) Levinsohn–Petrin TFP	(6) Levinsohn–Petrin TFP
	Jobs	Hours	Jobs	Hours	Jobs	Hours
DD: $\Delta \ln$ lab. cost at GMR	0.271*** (0.062)	0.206*** (0.056)	0.078*** (0.023)	0.068*** (0.022)	0.074*** (0.024)	0.074*** (0.023)
DDD: $\Delta \ln$ lab. cost at GMR $\times$ Sh MW workers in 2002	0.752** (0.363)	0.790** (0.332)	0.268* (0.149)	0.261* (0.144)	0.056 (0.159)	0.086 (0.149)
Magnitude of the GMR2 shock at the sample average:						
DD implied effect	0.010	0.008	0.003	0.003	0.003	0.003
DDD exposure effect	0.004	0.004	0.001	0.001	NS	NS
DDD total effect	0.011	0.009	0.003	0.003	0.003	0.003
Mean of dependent variable:	0.075	0.070	0.025	0.023	0.022	0.023
Observations	55,345	55,345	53,437	53,437	53,437	53,437

Notes: See Table 1.

roughly 0.4 percentage points are attributable to the DDD exposure effect. Columns (3)–(6) show that these improvements persist after controlling for capital and intermediate inputs through alternative measures of revenue-based total factor productivity (TFP). The estimated TFP gains are remarkably consistent across specifications and estimation methods, although the Levinsohn–Petrin estimates are less precisely estimated. For the average firm, the “medium” GMR2 reform raises revenue TFP by roughly 0.3 percentage points, with larger gains predicted for firms more heavily exposed to the minimum wage.

4 Theoretical Framework

In this section, we propose a structural model of firms' organizations to assess whether the organizational adjustments that we observe in the data are likely to qualitatively and quantitatively rationalize the simultaneous firm-level productivity increases. It also allows us to investigate further general equilibrium implications of minimum wages, most importantly in terms of aggregate productivity.⁴³ This theoretical set-up is derived from the heterogeneous firm framework proposed by Caliendo and Rossi-Hansberg (2012), where we further assume that knowledge is cumulative in order to fit the French data parsimoniously.⁴⁴ The implications of this deviation and, above all, the introduction of a minimum wage constraint are better understood in a self-contained exposition of the model, which we provide in this section.

4.1 Optimal Firms' Organizations when Knowledge is Costly and Cumulative

Demand. We consider an economy that is populated by a measure N of homogeneous agents,⁴⁵ which we normalize to 1 without loss of generality. On the demand side, these agents correspond to consumers with CES utility function:

$$U(x) = \left(\int_{\Omega} \alpha^{\frac{1}{\sigma}} x(\alpha)^{\frac{\sigma-1}{\sigma}} M \mu(\alpha) d\alpha \right)^{\frac{\sigma}{\sigma-1}}, \quad (5)$$

where $\alpha \in \Omega$ denotes a variety which is valued more by consumers whenever α takes a high value, $\sigma > 1$ is the elasticity of substitution across varieties, x denotes consumption, M corresponds to the mass of products while μ is the probability density function over the set Ω of available varieties. In this expression, M and μ are endogenous objects which are derived later in the model. This CES utility function generates the following (inverse) demand schedule at the variety level:

$$p(\alpha) = q(\alpha)^{-\frac{1}{\sigma}} (\alpha I)^{\frac{1}{\sigma}} P^{\frac{\sigma-1}{\sigma}}, \quad (6)$$

where $q(\alpha)$ corresponds to aggregate demand, i.e. the summation of $x(\alpha)$ across all agents, and I corresponds to aggregate income, which is also endogenously determined in the model. In what follows, we normalize the CES price index $P \equiv \left(\int_{\Omega} \alpha p(\alpha)^{1-\sigma} M \mu(\alpha) d\alpha \right)^{\frac{1}{1-\sigma}}$ to 1 without loss of generality.

Supply: occupation choices. On the supply side, to keep the model as parsimonious as possible, only one production factor - labor - is taken into account. All costs are therefore delineated in terms of labor. Agents are each endowed with one unit of time that they supply inelastically. However, they can choose between three different types of occupations: entrepreneurs of firms (be they employers of other workers or simply self-employed entrepreneurs), paid workers directly contributing to production within firms (at the bottom production level, or as managers), or teachers.

Agents deciding to become entrepreneurs pay a fixed cost of entry $k f_E$, where k is the price of a unit of time, which is endogenously determined in equilibrium, and f_E is the number of workers required to perform the fixed-cost tasks. In the model, this enables agents to get a draw α from an exogeneously given cumulative

⁴³A standard Melitz (2003) model cannot account for the within-firm productivity gains documented in Section 3. By contrast, our framework accommodates this margin while preserving the selection mechanism of Melitz (2003), captured by the demand shifter α introduced in Equation 5. This allows us to compare the relative importance of within-firm productivity gains and between-firm selection in shaping the aggregate productivity effects of minimum wage increases, in Appendix I.

⁴⁴This variation ensures simultaneously that (1) wages are increasing across layers and, as a consequence, that (2) the legal minimum wage only bites at the bottom layer of firms' hierarchical organizations (Section 4.2). Both are features of the French data (Section 3.3) and the second characteristic eases numerical simulations considerably.

⁴⁵Our empirical and simulation analyses focus on firm-level outcomes, such as firm size, hierarchical structure, and productivity. This makes firm heterogeneity a central feature of the model and motivates our use of the framework developed by Caliendo and Rossi-Hansberg (2012). The main drawback of this approach is that incorporating *ex-ante* agent (worker) heterogeneity would render the assignment problem analytically intractable. We therefore assume that agents are *ex ante* homogeneous.

density function $G(\alpha)$ (with associated probability density function $g(\alpha)$). This draw determines the demand conditions faced by the candidate firms (according to Equation 5), and thus their level of profits. Depending on the value of the idiosyncratic α draw, the agent decides whether to actually become an entrepreneur and launch production, or not. If production actually starts, the entrepreneur has to incur an additional fixed cost kf .

Production consists in solving problems of varying “difficulty”,⁴⁶ which is indexed by $d \in \mathbb{R}^+$ and is drawn from a cumulative density function $F(d) = 1 - e^{-\lambda d}$, where $\lambda > 0$. By default, agents are not able to solve production problems autonomously ($z = 0$). However, it is possible for them to increase their skills to a higher level $z > 0$ subject to a training cost kcz . This level of ability enables them to solve all problems with difficulty $d \leq z$. The associated time cost is their entire time endowment (i.e., 1) when they have to solve the production problem autonomously, but only $h < 1$ whenever the problem has been pre-processed by another agent with skill $z' < d$, who was not able to solve it autonomously. Thus, agents featuring the minimal skill level $z' = 0$ are useful in production precisely because of their ability to pre-process production problems, which allows their managers to solve them at a reduced time cost ($h < 1$). In contrast to the formulation of the model in Caliendo and Rossi-Hansberg (2012), we assume that knowledge is cumulative: achieving knowledge level z requires mastering all intermediate levels between 0 and z .⁴⁷ The labor market is competitive, and in a mixed-strategy equilibrium workers must be compensated for their skill investment, such that workers earn the following gross wage:⁴⁸

$$w(z) = k(cz + 1) \quad (7)$$

As a result, after paying for training, all workers earn a net wage of k , where k is the equilibrium price of one unit of time.

Actual production within hierarchical organizations requires knowledge (aka the ability to solve problems) and labor (aka time). Bottom production workers (at level $l = 0$) spend their entire endowment of time studying the problem they pick, and produce A units of output when their level of knowledge allows them to solve it. When they do not know the solution, they ask a manager in the next layer above, at level $l = 1$. The latter spends h units of time studying the problem and solves it if d is less than the manager’s skill level (thus producing A units of output). In the opposite case, she asks a manager in the layer above (i.e., layer $l = 2$, if there is one) and the process repeats. Some workers are also assigned to perform the fixed-cost tasks embodied in f_E and f , which does not require acquiring knowledge ($z = 0$).⁴⁹

Lastly, teachers educate workers: learning one unit of knowledge requires c units of teachers’ time, who earn k per unit of time. The education sector is competitive such that education services are charged at

⁴⁶More precisely, production problems are heterogeneous in the sense that solving problems indexed by different d require different types of knowledge. The notion of “difficulty” is not completely appropriate, but is a convenient (yet approximate) label to describe the setting. In fact, the specification of $F()$ involves sorting problems according to their (decreasing) frequency (rather than according to their “difficulty”): $d = 0$ thus simply corresponds to the most frequent problems.

⁴⁷In the non-overlapping skill specification from Caliendo and Rossi-Hansberg (2012), managers can generically be allocated a smaller interval of skill than the workers in the layers below them. Such a situation would imply that they should receive a lower wage. This would also imply that the minimum wage could impact managers before production workers or lower-layer managers (and incidentally, complicate numerical simulations by a large amount). Figure 2 shows that in the data, there is a clear hierarchy of wages across layers and that minimum wages only bind at bottom layers. In practice, the cumulative knowledge assumption thus provides a better approximation to the French data.

⁴⁸In the specification of Equation 7, training costs are incorporated in workers’ compensation, which (broadly) corresponds to the French context, as discussed in Section 3. In fact, Equation 7 corresponds to the most parsimonious wage specification possible given our focus on the productivity implications of minimum wages (Caliendo and Rossi-Hansberg, 2012), rather than their impact on worker-level outcomes. Introducing richer mechanisms of wage formation, notably incorporating dimensions of *ex ante* worker-level heterogeneity as in Garicano and Rossi-Hansberg (2006) and capturing more sources of heterogeneity/inequality, would involve a (truly) significant increase in complexity and is beyond the scope of this paper. We leave this important task for future research.

⁴⁹This assumption is rather innocuous since it only affects the breakdown of workers between fixed costs and the educational sector, which is not the main focus of our article.

marginal cost, i.e. kcz in order to achieve level z .

Supply: firms' organizations. Each firm is set up by an entrepreneur who is located at the top of the hierarchy (as a CEO), and who decides:

- The total number of layers $L \in \{0, 1, 2, 3, \dots\}$ of her firm. The case $L = 0$ corresponds to self-employed managers.
- The number of workers n_L^l to be hired at each layer $l \in \{0, 1, \dots, L\}$
- The skill level z_L^l for each layer.

We derive the optimal firm structure in two steps: first, a cost-minimization problem conditional on the level of output q that has to be attained, and second, a profit-maximization step allowing us to determine the optimal level of output q conditional on the demand draw α .

Supply: cost minimization. First, the production conditions described above imply straightforwardly that the levels of skill must be increasing across layers – otherwise, there is no benefit to passing an unsolved problem to agents that are less competent than the considered worker. Second, it is necessary to hire enough workers at a given layer l in order to screen the total amount of problems that are transmitted at that hierarchical level. This implies the following inequalities:

$$\forall l \in \{0, 1, \dots, L-1\}, \quad \underbrace{h}_{\text{Time per problem}} \underbrace{n_L^0 e^{-\lambda z_L^l}}_{\substack{\# \text{ problems} \\ \text{transmitted} \\ \text{to layer } l+1}} \leq n_L^{l+1} \quad (8)$$

$$n_L^L = 1 \quad (9)$$

$$0 \leq z_L^0 \quad (10)$$

$$q \leq n_L^0 A(1 - e^{-\lambda z_L^L}) \quad (11)$$

The variable cost of producing a level q of output given wages k is:

$$C(q, k) = \min_{L \geq 0, (n_L^l)_l, (z_L^l)_l} C_L(q, k) = k \left(\sum_{l=0}^L n_L^l (cz_L^l + 1) \right) \quad (12)$$

where $C_L(q, k)$ corresponds to the cost of producing q with a firm organized into L layers. The resolution of each of these sub-programs is detailed in Appendix E. The proofs differ from those in Caliendo and Rossi-Hansberg (2012) because of two main deviations:

- First, as previously mentioned, we introduce an assumption of cumulative (or “overlapping”) knowledge imposing that any agent who is able to solve a problem of difficulty z is also able to solve any easier problem, which ensures that wages increase monotonically from the bottom to the top of the hierarchies.⁵⁰
- Second and more importantly, we do not impose their parametric restriction ensuring that bottom, production workers always acquire a strictly positive level of knowledge, $z_L^0 > 0$, because we later obtain from the model calibration that this restriction – which corresponds to $\frac{c}{\lambda} \leq h$ in our model – is rejected in the French data. This indicates that in France, given the prevalence of difficult problems captured by parameter λ , it is particularly expensive to train workers (c is high) as compared with transferring

⁵⁰This feature comes at the cost of a lower tractability, and in particular non-monotonic marginal costs (see Proposition 4 in Appendix E), depending on the number of layers and on whether firms find it optimal to train production workers or not. This in turn is determined by their respective demand draws, α .

problems upwards to skilled managers, given the relatively low managerial costs (h is low):⁵¹

$$\frac{c}{\lambda} > h \quad (13)$$

Technically, this feature complicates some of the proofs of Appendix E. Economically, it implies that in the firms' optimization problem described in Equation 12, it is relatively more cost-effective for them to rely on verticalized hierarchical organizations than to train workers within layers to achieve a given output target. Skills are so expensive that $L = 1$ -layer firms sometimes do not find it optimal to train production workers at the bottom ($z_1^0 = 0$) and even further reduce labor costs at the bottom ($l = 0$) layer by compressing its size below their manager's capacity. In such cases, the manager (at layer $l = 1$) does not exhaust his or her management time constraint.⁵²

However, several important characteristics of the baseline Caliendo and Rossi-Hansberg (2012) framework are preserved despite these two deviations. In Appendix E, we show in particular that the firms' cost function remains sufficiently well behaved: it remains increasing in q , and homogeneous of degree one in the baseline labor compensation k , such that the cost-minimizing firm organization for a given output target q is insensitive to k . Importantly, this implies that the Q -productivity index proposed in Caliendo and Rossi-Hansberg (2012) remains valid, which is key to our research question. It simply corresponds to the inverse of average variable costs (without considering the fixed costs of production), normalizing k to one so as to keep factor prices constant across the different general equilibrium scenarios that we will compare in our simulations:

$$a(q) = \frac{q}{C(q, 1)} = \frac{kq}{C(q, k)} \quad (14)$$

Figures 3 and 4 illustrate graphically these different characteristics of our set-up. Panel (A) of Figure 3 constructs the curves describing average costs of production in firms featuring different numbers of layers, from $L = 0$ (self-employed agents) to $L = 3$ (one bottom layer of production workers, and three managerial layers). It allows us to determine the optimal number of hierarchical layers, L , depending on the production target, q : it corresponds to the value associated to the lowest possible average cost. The overall cost function thus corresponds to the lower envelope of the curves of Panel (A) of Figure 3. Increasing the number of hierarchical layers deteriorates units costs for low scales of production, but becomes cost-effective as the production target becomes larger: adding a hierarchical layer is thus similar to incurring a higher fixed cost of production in order to achieve lower marginal costs of production for large scales of production, as shown in Panel (B) of Figure 3.

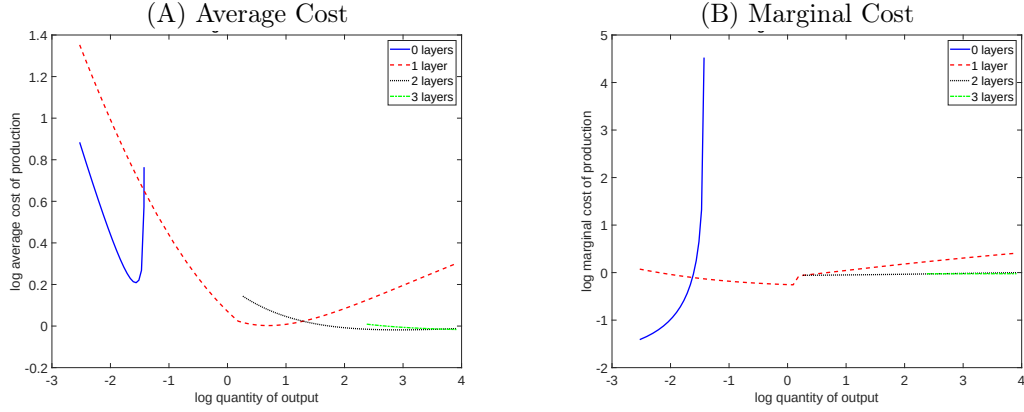
More technically, Panel (A) of Figure 3 also shows that average costs are functions that are continuous, but not continuously differentiable. In particular, 1-layer firms feature discontinuous marginal costs of production, and the average cost function features a point of non-differentiability, at the point where they start finding

⁵¹The condition that $z_L^0 > 0$ is harder to satisfy in our model with cumulative knowledge than it would have been in the original model of Caliendo and Rossi-Hansberg (2012). In addition, there are several reasons to expect that France may exhibit a high value of c (relative to h and λ). First, the quality and performance of the French vocational training system are widely viewed as relatively weak (see, for example, Brandt, 2015, and the OECD's "Getting Skills Right" reports), which contributes to skill mismatch. In the context of our model, this suggests a higher training cost c relative to the learning parameter λ . Second, measures of trust are relatively low in France (Algan and Cahuc, 2014). In such an environment, firms may fear that workers could leave with the returns to their human capital investment—a concern consistent with the findings of Blasco et al. (2012), who show that participation in workplace training increases both job mobility and the risk of poaching. This further raises the expected cost of training. In addition, Algan and Cahuc (2014) document that greater firm centralization is associated with lower trust, a pattern that is consistent with more intensive use of communication technologies in centralized hierarchies, and hence with a lower relative cost of communication, h , in France.

We also note that this parameter configuration ($c/\lambda > h$) makes the minimum wage constraint introduced in Section 4.2 more binding. Our quantitative estimates should thus be interpreted as providing an upper bound on the aggregate cost of minimum wages. Indeed, as we show below, mitigating minimum wage costs at the firm level requires substituting away from hierarchical structures toward greater reliance on training—that is, away from the relatively inexpensive technology and toward the relatively costly one.

⁵²In larger firms ($L > 1$), on the other hand, the time constraints of managers and entrepreneurs are always exhausted.

Figure 3: Firms' Cost Functions



Notes: This figure is constructed from the calibrated version of the model presented in Section 5 ($\lambda = 1$, $c = 0.220$, $h = 0.193$, $A = 0.240$), in a version which simulates the French economy in the absence of a minimum wage.

it optimal to train bottom workers, i.e. when z_1^0 switches from zero to a strictly positive value.⁵³ Meanwhile, employment at the bottom layer (n_1^0) increases from values which do not saturate the time constraint of the manager, to values where the latter works at full capacity (given z_1^0 and thus the total amount of problems that are transmitted to her, $hn_1^0 e^{-\lambda z_1^0} = 1$). In the range where the manager's time constraint is not saturated, the marginal cost of production decreases with q because the growing firm experiences increasing returns: as q increases, both z_1^1 and n_1^0 can increase, with n_1^0 increasing relatively faster until the managerial time constraint binds and slows the growth of n_1^0 .

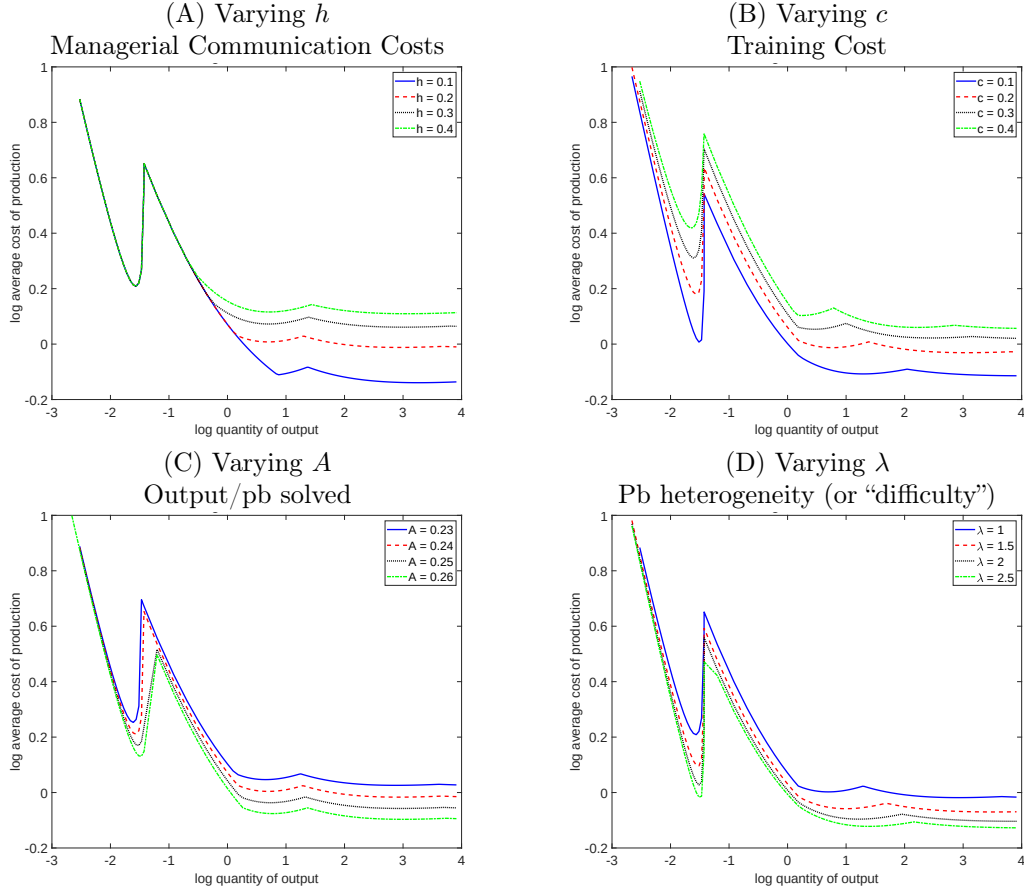
Figure 4 documents how average costs are shaped by the main technological parameters of the model. In Panel (A), we first consider the impact of an increase in the cost h of transmitting problems to managers. Unsurprisingly, production costs are unaffected when $L = 0$ (self-employment). They are also unaffected in firms with $L = 1$ hierarchical layer when the time (capacity) constraint of the manager is not saturated, i.e. $n_1^0 < \frac{\exp(\lambda z_1^0)}{h}$. In all other firms, costs curves are shifted upwards. As h becomes larger, Equation 13 no longer holds and the regime where the time constraint of managers in 1-layer firms is not saturated disappears: it is never optimal to waste managerial time when management costs are high relative to training costs. Panel (B) precisely considers the impact of the latter (parameter c). It affects all organizational forms and shifts all curves upwards. As c becomes large, Equation 13 generates a wider range of optimality for organizations which do not exhaust managers' time constraint. In contrast in Panel (D), higher values of λ , which corresponds to more favorable (more concentrated) distribution of production problems, affects all organizational forms but relaxes Equation 13, thus reducing the range of optimality for organizations which do not exhaust managers' time constraint. Lastly in Panel (C), the general efficiency parameter, A , affects all organizational forms but is more neutral in terms of the range of optimality for the different organizational forms.

Supply: profit maximization. Ultimately, all entrepreneurs solve the following profit maximization program:

$$\pi(\alpha, k) = \max_{q \geq 0} \underbrace{p(\alpha)q}_{q^{\frac{\sigma-1}{\sigma}} (\alpha k)^{\frac{1}{\sigma}}} - C(q, k) - kf, \quad (15)$$

⁵³As long as $L \geq 1$ and as shown in Panel (B) of Figure 3 (e.g. the red curve), marginal production costs are not always increasing in the production target, q . Note that this constitutes a difference with Caliendo and Rossi-Hansberg (2012) (see Appendix E).

Figure 4: Firms' Average Cost Function: Comparative Statics wrt Technologies



Notes: See Figure 3. Panel (A) displays the lower envelope of the layer-specific average cost curves, calculated for four different values of h , while $c = 0.220$, $\lambda = 1$, $A = 0.240$. In Panel (B), this exercise is replicated for 4 different values of c , while h is set to 0.193, $\lambda = 1$, $A = 0.240$. In Panel (C), this exercise is replicated for 4 different values of A , while $h = 0.193$, $\lambda = 1$ and $c = 0.220$. Lastly, the exercise is also replicated in Panel (D) for 4 different values of λ , while $h = 0.193$, $A = 0.240$, and $c = 0.220$.

which allows them to select the optimal organizational form $(L, (n_L^l)_l, (z_L^l)_l)$ as well as the optimal level of production q depending on their idiosyncratic demand conditions (captured by α). In such a CES framework with monopolistic competition, optimal pricing is a constant markup over marginal cost:

$$p(\alpha, k) = \frac{\sigma}{\sigma - 1} MC(q(\alpha); k) \quad (16)$$

such that the optimal quantity to be produced satisfies:

$$q(\alpha, k) = \alpha k \left(\frac{\sigma}{\sigma - 1} MC(q(\alpha); k) \right)^{-\sigma}. \quad (17)$$

General equilibrium. To close the model in general equilibrium, we assume an exogenous firm death rate of δ per period. The equilibrium is defined by the three following conditions:

1. Conditional on entry (and once the demand conditions α are revealed), entrepreneurs only produce if their profit in Equation 15 (which is net of entry fixed costs) is positive. Since profits are increasing in α , this implies that they only produce if their actual α draw is larger than a cutoff $\bar{\alpha}$ defined by the following equation:

$$\pi(\bar{\alpha}, k) = 0. \quad (18)$$

2. Entrepreneurs only enter (i.e. pay the fixed entry cost, f_E) if their expected profit (when α is not yet revealed) is positive:

$$\int_{\bar{\alpha}}^{+\infty} \frac{\pi(\alpha, k)}{\delta} g(\alpha) d\alpha = k f_E. \quad (19)$$

3. In a steady state, in each period, the $M\delta$ exiting firms are exactly replaced by the subset of entrants who remain in activity, $M^E (1 - G(\bar{\alpha}))$ (while all entrants, M^E , pay the fixed entry cost, f_E).

The labor market clearing equation is thus:⁵⁴

$$1 = \frac{M}{1 - G(\bar{\alpha})} \left[\delta f_E + \int_{\bar{\alpha}}^{+\infty} (C(q(\alpha); 1) + f) g(\alpha) d\alpha \right]. \quad (20)$$

The term on the left in Equation 20 corresponds to the supply of labor that is readily available to be hired as production workers or managers (which coincide with the mass of agents and is normalized to one), while the term on the right corresponds to total labor demand (with associated training time requirement directed to teachers, fixed costs of production, and fixed cost of entry in each period) by entrepreneurs who actually launch production.

This system of three equations determines the equilibrium values of the productivity entry threshold, $\bar{\alpha}$, the measure of active firms, M , and the equilibrium price of a unit of time, k .⁵⁵

4.2 Introducing the Legal Minimum Wage Constraint

We now introduce a minimum wage constraint in the previous setup. This constraint takes the form of a lower bound, denoted by $\bar{w}$, that is imposed on all (real) wages that are paid out by firms featuring $L \geq 1$. Specifically, we assume that the minimum wage constraint does not apply to self-employed workers (corresponding to $L = 0$), which is meant to reflect real-world characteristics, since self-employed agents can easily circumvent the minimum wage constraint by declaring their own compensation as dividends or other types of capital income rather than wages. Secondly, we also assume that the wage k of teachers is not subject to labor laws, and therefore unaffected by the introduction of the minimum wage constraint. This second hypothesis aims to account for the French institutional framework, where teachers are almost all civil servants while general labor laws (and the national minimum wage in particular) do not apply to civil servants.⁵⁶

Such a minimum wage constraint on real wages paid out by firms can equivalently be rewritten as a constraint on the minimum level of knowledge to be attained at the lower level of firms' hierarchies:

$$w(z_L^0) = k(cz_L^0 + 1) \geq \bar{w} \quad \Longleftrightarrow \quad z_L^0 \geq \underbrace{\frac{\bar{w} - k}{ck}}_{\bar{z}_L^0} \quad (21)$$

This condition reflects the fact that hiring a worker with a skill level below $\bar{z}_L^0$ is always dominated by hiring one at $\bar{z}_L^0$ for a given value $\bar{w}$ of the minimum wage. The minimum wage constraint thus simply alters the cost minimization program of firms with $L \geq 1$, by imposing a tighter constraint on z_L^0 in our previous Equation 10. The proof strategies are globally unaltered, as shown in Appendix E.3.1, but this addition to the set-up has non-trivial implications that we explain below.

⁵⁴This expression relies on the fact that the cost function C is homogeneous of degree 1 in k : see the proof in Appendix E.

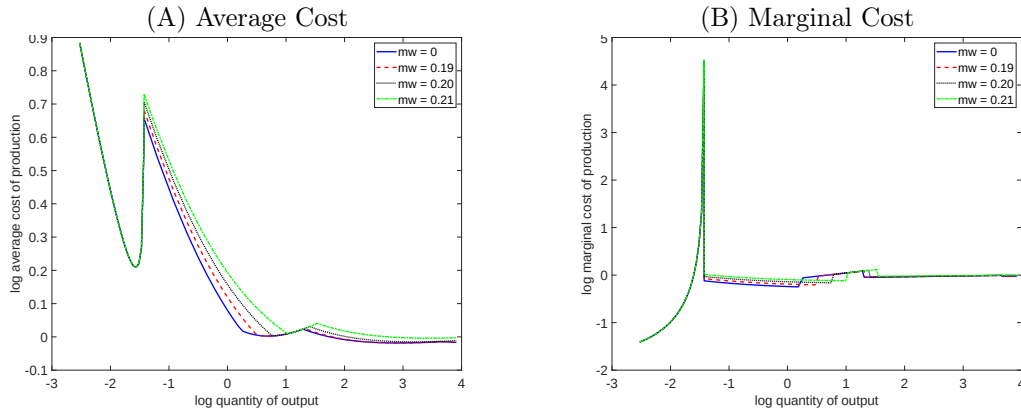
⁵⁵Notice that these equations also determine the endogenous objects that we introduced earlier. In particular, the probability density function of varieties that are available on the market relates to g and $\bar{\alpha}$ via the following formula: $\mu(\alpha) = \frac{g(\alpha)}{1 - G(\bar{\alpha})}$ for all $\alpha \geq \bar{\alpha}$. Secondly, good market clearing requires: $I = kN = k$, where I was introduced in Equation 6 and where the last equality follows from the fact that N was normalized to 1. This implies that Equation 6 can be rewritten as: $p(\alpha) = q(\alpha)^{-\frac{1}{\sigma}} (\alpha k)^{\frac{1}{\sigma}} P^{\frac{\sigma-1}{\sigma}}$.

⁵⁶We also assume that workers performing fixed-cost tasks are not subject to the minimum wage. None of these assumptions is critical, and our framework could easily be modified to describe a different institutional context.

Implications in partial equilibrium. Appendix E.3.2 first provides some important comparative statics results in partial equilibrium, i.e. holding k fixed, but endogenizing the optimal level of output, q . As illustrated in Figure 5, the minimum wage constraint leads to a general shift of all cost curves both upwards and to the right. This means that all else equal, in partial equilibrium, production costs increase as a consequence of the introduction of a binding minimum wage constraint, and the range of optimality of the different organizational forms is shifted to the right: for a given production target, q , optimal organizational forms tend to be (slightly) flatter. In addition, it increases the range of production targets for which it is optimal for 1-layer firms to remain “small” relative to the time capacity of their managers. This property is illustrated in Figure 5, where the range of production over which such organizational forms are optimal unambiguously increases as $\bar{w}$ increases. Proposition 1 formalizes these first results by considering firms who are “marginal” between organizational forms and investigating their responses to tighter minimum wage constraints.⁵⁷

Proposition 1. *If the minimum wage is binding on firms with M layers but not L layers, then any firm that is on the margin between choosing M or L will shift to production with L layers.*

Figure 5: Firms’ Cost Function: Comparative Statics wrt Minimum Wages, $\bar{w}$



Notes: The figure is constructed from the calibrated version model of Section 5 ($\lambda = 1$, $c = 0.220$, $h = 0.193$, $A = 0.240$), in versions which simulate the French economy with different levels of the minimum wage, $\bar{w}$ (measured in hundreds of €).

Panel (A) in Figure 5 shows that average costs always rise when firms are faced with stricter minimum wage constraints, while Panel (B) indicates that the same is not true for marginal costs, which sometimes decrease depending on the range of production that is considered. This is mainly a consequence of Proposition 1, because marginal costs are both non-monotonic and mainly determined by the number of layers (and, in 1-layer firms, by whether the time constraint of the manager is slack or tight): since minimum wages alter the range of optimality of these different regimes, they have non-monotonic implications in terms of marginal production costs - and hence, in terms of firms’ pricing strategies. Specifically, for a higher minimum wage, the value of q at which the manager’s time constraint binds with $L = 1$ is larger, because z_1^0 is larger and the manager can supervise more workers; therefore, the value of q at which the marginal cost jumps up is larger. Similarly, the marginal cost shifts down when the firm switches from $L = 1$ to $L = 2$, which also happens at a larger value of q .

Of course, higher minimum wages not only change the optimal number of hierarchical levels, but also

⁵⁷The proof of this proposition is reported in Appendix E.3.2. It is impossible to prove analytically that firms monotonically increase their number of layers with q , or with α in general equilibrium. However, such a monotonic pattern is true in all of our simulations. In such situations, and whenever $z_{L+1}^0 < z_L^0$ at the point of indifference, then Proposition 1 implies that the minimum wage causes marginal firms to remove a hierarchical layer.

the optimal level of knowledge across hierarchical layers since bottom, production workers become more knowledgeable (Equation 21). Appendix E.3.1 shows that whether it is optimal to increase or decrease skills across managerial layers depends in particular on the number of layers. For example, when the manager does not reach her time constraint in 1-layer firms, it is optimal to *decrease* her level of knowledge when production workers become more skilled because of a higher minimum wage, i.e. the skills of the manager and of the production workers are substitutes in this configuration, which limits the increase in marginal costs of production generated by minimum wages as much as possible. In contrast, when the manager exhausts her time constraint in 1-layer firms, it is optimal to *increase* her level of knowledge, which means that her skills are complements to those of production workers. This strategy allows maintaining low marginal costs for sufficiently large scales of production: over the corresponding range (which is narrow!), average and marginal costs are unaffected by minimum wages in Figure 5.⁵⁸ Proposition 5 in Appendix E formalizes these results by considering the impact of tighter minimum wage constraints for firms which are in a regime (i.e. a range of output targets, q) where it is not optimal to alter their hierarchical organizations in response.⁵⁹

Further implications in general equilibrium. In general equilibrium, an increase in the minimum wage lowers the profits of at least some firms such that the price of labor, k , must decrease to restore balance in the labor market. This in turn alters the partial equilibrium results: a reduction in k tends to lower marginal cost, but it also lowers q due to a reduction in income. The combined effect of these forces is ambiguous on firms' organizations and thus on the productivity implications of minimum wages. Quantifying them requires numerical simulation techniques, which are detailed in Section 5.

5 Calibration and Simulations

Using our model from Section 4, we now proceed to a careful calibration exercise, with three different objectives in mind. Firstly, we check that the proposed set-up is compatible with the cross-sectional structure of the French firm population, which is captured by our (static) calibration targets. Secondly, we check that our model is further "validated" by the French data, by verifying that the organizational mechanisms at the heart of our model are capable of rationalizing, both qualitatively and quantitatively, the responses of French firms to the tightening of minimum wage constraints that we documented in Section 3. These correspond to "dynamic" moments in the data that are not used as targets in the calibration procedure: for this reason, this verification is a powerful test of the framework's empirical relevance. Thirdly and finally, we use our calibrated framework to describe and quantify the impact of minimum wage constraints of different magnitudes on firms' strategies and on the resulting aggregate productivity. We extend the analysis to different counterfactual technological scenarios, with the aim to describe the possible implications of the current waves of technologies.

5.1 Baseline Calibration

Data calibration targets. We calibrate our model to a set of static moments measured from our dataset in 2006, which corresponds to the endpoint of our estimation period in Section 3 and to a stable period of the French economy:⁶⁰ at this date, all GMRs had converged, thus all firms faced the same level of

⁵⁸This non-monotonicity of the impact of minimum wages also implies that their impact on the prices charged by firms is not monotonic either, as a consequence of Equation 16.

⁵⁹Proposition 5 exhibits some well-defined cases where clear theoretical predictions arise in terms of the impact of minimum wages on marginal costs of production, ϕ , the size of the bottom production layer, n_L^0 , and the skills of their managers, z_L^l . However, as previously explained, they appear to be quite heterogeneous, which motivates the simulation techniques that are introduced in Section 5.

⁶⁰See Section 3.2 for the detailed description of our estimation sample. Appendix G complements the analysis shown in the main text by checking the robustness of our findings to seven alternative experiments, each relying on different (including virtually exhaustive) datasets or alternative data moments.

legal minimum wage, as in our theoretical set-up of Section 4. Panel (A) of Table 5 describes the precise moments of the data which we use as calibration targets. Since our mechanism of main interest involves firms’ optimal organizations, we introduce a rich set of indicators describing the firm size distribution and its articulation with production hierarchies: first, the breakdown of the firm population by classes of hierarchical organizations (except for the two bottom classes, which are more difficult to identify separately in the DADS data),⁶¹ and secondly, firms’ average size in each of these classes. All measures of firm size are calibrated in terms of jobs.⁶² These five moments contribute jointly to the identification of parameters $\frac{c}{\lambda}$, h , A as well as the distribution of α , which all determine firms’ sizes and organizations (see firms’ optimization programs in Section 4 and the characterization of optimal choices in Appendix E). The average hourly wage mainly contributes to identifying $\frac{c}{\lambda}$ and k , while the share of workers bound by the minimum wage mainly identifies $\bar{w}$.⁶³ All firms in our sample were continuously active over the 2000–2006 period. Parameter δ is therefore measured as the share of firms (0.266) that were created during the preceding seven-year period, 1993–1999. The share of workers in these “new” firms is introduced as an additional calibration target which contributes mainly to identifying the fixed costs f_E and f as well as the distribution of α (but also c , h and A).

Parameters that are calibrated to values from previous literature. We set the values of the other parameters of our model to values from previous literature. Specifically, the elasticity of substitution σ is set to 3.8 as in Caliendo and Rossi-Hansberg (2012), and as in the latter paper, we normalize λ to 1. In addition, the share of workers who are teachers reported in Table 5 could not be accurately estimated in the DADS, because education is largely in the public sector in France and is only poorly captured in the 2006 version of the DADS files. This is however a critical moment to identify the fixed costs (f_E , f) together with the distribution of α , such that we thus rely on external estimates made available by the French statistical institute.⁶⁴

Specification of the distribution of α . The distribution of α embodies the source of firm-level heterogeneity that is introduced in the model. It thus plays a critical role in terms of fit with the data. For this reason, we adopt a flexible specification: specifically, we set a wide support in terms of its distribution, by allowing values ranging between 1 and 1000. This implies that a firm earning a draw of 1000 can sell 1000 more quantity at any price than a firm earning a draw of 1: this wide range allows for possibly highly heterogeneous demand conditions. We also approximate non-parametrically the density $g(\alpha)$ by taking the exponential function of a cubic spline across four nodes, located at $\alpha = \{1, 50, 500, 1000\}$, where the values of the spline function at the four nodes $\{g_1, g_2, g_3, g_4\}$ are parameters set in our calibration – except for g_1 , which is fully determined conditional on the other three by the normalization condition requiring the density

⁶¹A difficulty with the DADS data is that it is not straightforward to distinguish firms with $L = 0$ from firms with $L = 1$. To be precise, we observe many firms in these files, which are registered as proper corporations, but which only feature one production layer, and no managerial layer. The latter are likely to either correspond to partnerships (which should probably be classified in the $L = 0$ class) or to small 1-layer firms with managers that are not paid in the form of wages. This is allowed in the smallest and simplest structures (smallest “SARL”, the equivalent of US LLCs), but not in standard corporations (large SARL, “SA” and “SAS”, the equivalent of US joint-stock companies), such that in practice this problem only affects the measurement of 1-layer firms. To mitigate this concern, the two classes ($L = 0$ and $L = 1$) are considered jointly in our main calibration. Appendix G also checks that our results are not sensitive to the fact the 2006 version of the matched DADS and fiscal files might underestimate the share of self-employed entrepreneurs ($L = 0$).

⁶²Appendix G performs an alternative calibration using hours, in the form of full-time equivalents, in column (v) of Table G1.

⁶³In our calibration, we follow the practice of the Ministry of Labor (Ananian and Calavrezo, 2011), which relies on a rather restrictive definition to assess this share: it amounts to focusing on workers in our sample earning less than 1.05 times the minimum wage. With such a definition, we obtain that around 5% of workers are bound by the minimum wage, as in Ananian and Calavrezo (2011). This share appears to almost coincide with the share of UK workers which were “hit” by the minimum wage introduced in 1999: Dickens and Manning (2004) estimate it to be around 6 to 7%.

⁶⁴INSEE’s Tableaux de l’Economie Française (2020 edition, data for 2018). The URL for these data is the following: <https://www.insee.fr/fr/statistiques/4277675?sommaire=4318291>. We excluded all sectors not covered in the DADS (administration, social action) and rescaled the proportion of teachers accordingly, thus attaining a value that is quite close to the US figure used in Caliendo and Rossi-Hansberg (2012). We were unfortunately not able to find data for 2006 and thus had to assume that the proportion of teachers is broadly stable over time.

Table 5: Baseline Calibration to French Data

(A) Empirical Targets

Moment	Data	Model
Share of agents who are teachers	0.098	0.098
Share of workers in new firms	0.203	0.222
Share of 2-layer firms	0.446	0.448
Share of 3-layer firms	0.228	0.229
Average firm size (jobs)	93.48	92.38
Average size (jobs) of 2-layer firms	87.23	87.11
Average size (jobs) of 3-layer firms	214.00	215.52
Average (hourly) wage (€)	20.21	20.29
Share of workers (in firms) bound by minimum wage	0.047	0.047

(B) Calibrated Values of Parameters

Parameter	Value
Cost of knowledge acquisition, c	0.220
Time cost of problem communication, h	0.193
Output per problem solved, A ($\sim 10^2$ €/hour)	0.240
Fixed cost of entry, f_E	5.693
Fixed cost of production, f	5.701
Minimum wage $\bar{w}$, relative to average wage	0.914
2nd-node of α distribution ($\alpha = 50$)	$g_2 = -12.14, e^{g_2} = 5.342E - 06$
3rd-node of α distribution ($\alpha = 500$)	$g_3 = -15.51, e^{g_3} = 1.837E - 07$
4th-node of α distribution ($\alpha = 1000$)	$g_4 = -24.38, e^{g_4} = 2.582E - 11$

Notes: In Panel (A), all moments are computed from the DADS, except the share of teachers, which is sourced from INSEE's Tableaux de l'Economie Française (2020 edition, data for 2018). e^{g_2} , e^{g_3} and e^{g_4} correspond to values of the density function $g(\alpha)$ for $\alpha = 50, 500$ and 1000 , respectively.

to integrate to one.⁶⁵

In the actual implementation of our optimization algorithm, we then solve for firm organization, output and profit across a grid of 250 values of $\alpha \in [1; 1000]$. We augment this grid with over 130000 values of $\alpha \in [1; 1000]$ but only approximate firms' outcomes (organization, output and profit) with a linear interpolation of the 250 previous values: bypassing actual numerical resolution increases computational speed. Overall, this strategy amounts to solving a discretized but highly accurate version of our initially continuous model.

Calibration criterion. The full list of parameters that we identify from the above list of nine data moments is the following: $\{c, h, A, f_E, f, \bar{w}, g_2, g_3, g_4\}$. There are as many moments as parameters, such that our procedure is just identified. We solve for the vector of parameter values that minimizes the following distance between the target moments and their values in the simulated model :

$$Q = \sqrt{\sum_{j=1}^9 \left(\frac{\hat{m}_j - m_j}{\min(\hat{m}_j, m_j)} \right)^2}$$

where m is the vector of target values for the moments, $\hat{m}$ their values in our numerical simulation, and j indexes the nine different moments used. For each vector of parameter values, we iterate over k (the endogenous price of labor) to find the value that satisfies general equilibrium.

⁶⁵In their quantitative exercise, Caliendo and Rossi-Hansberg (2012) only introduce one parameter, corresponding to the Pareto coefficient of the size distribution of firms (footnote 25, p. 1422). As we show below, this is too coarse to fit the French data in a detailed way, and it is also useful to relax the Pareto assumption in order to improve the fit with the calibration targets.

Calibration results. The results of the calibration are reported in Panel (B) of Table 5 while the values of the calibrated moments can be found in the final column of Panel (A). To facilitate comparisons with the parameter estimates reported by Caliendo and Rossi-Hansberg (2012), all monetary variables are expressed in hundreds of euros per hour. We first note that the model achieves an excellent fit for the data moments that are introduced in the calibration procedure: all of them are matched with a deviation that is inferior to 1.2% except for the share of workers in new firms. This indicates that the period of seven years that we use might be slightly too short (!) for the size distribution of new firms to converge to the firm size distribution of incumbents, but the fit remains very good since the distance between the model and the target is less than 10%.

In the French economy, training costs c are estimated at a relatively high value relative to h , such that $\frac{c}{\lambda} > h$. As explained in Section 4.1, this implies that some firms in the economy might find it optimal to adopt a “limit” behavior whereby they set the skill level of bottom, production workers at the lowest possible value. This behavior could emerge in any firm, and the smallest (1-layer) firms might in addition find it optimal to compress the size of their layer of production workers below the value that would saturate the manager’s time constraint.⁶⁶ The value (0.24) obtained for the quantity of output produced per problem solved, A , is slightly superior to the value of the average wage (0.20) that is used as a calibration target, which allows firms to earn a positive profit margin and cover their fixed costs of entry and production. The latter (f_E and f) are both estimated to be roughly equivalent to the wage bill of 5.7 agents. The results obtained for the distribution of α imply that the probability of a draw of 500 is only 3% of the probability of a draw of 50, while the probability of a draw of 1000 is only 0.0005% of the probability of a draw of 50. As a consequence, the distribution of α has thinner tails than a Pareto distribution, the standard benchmark in similar settings. Lastly, the binding minimum wage is estimated to be 8.6% lower than the average wage paid out by firms. This parameter is mostly identified from the share of workers bound by the minimum wage and is quite sensitive to the value of the corresponding data moment.

5.2 Simulation of Alternative Minimum Wage Scenarios

Considered scenarios. Having calibrated our model, we simulate its general equilibrium implications across six scenarios, ranging from the baseline (calibrated to French data) to hypothetical increases or decreases in the minimum wage. These include a no-minimum wage economy (numerically similar to a 4% cut, scenario (2)), two empirically grounded scenarios replicating the 4% and 8% real increases experienced by GMR2 and GMR1 between 2003 and 2006 (scenarios (3) and (4), respectively), and two more extreme variants involving 16% and 24% hikes (scenarios (5) and (6), respectively). While the latter exceed historical French changes, they remain within observed international bounds: for instance, the 1966 Fair Labor Standards Act in the U.S. led to a 28% nominal (23% real) increase, and Hungary experienced a 60% real hike in 2001, as documented by Bailey et al. (2021) and Harasztosi and Lindner (2019), respectively. Figure 6 and Table 6 describe the general equilibrium impact of minimum wages across each of these scenarios on aggregate outcomes, while Table F1 in Appendix F.1 describes firms’ organizational adjustments in full detail.

General equilibrium impact of Minimum Wages on aggregate output. The first row of Table 6 reports how the baseline price of labor, k , responds to increasingly stringent minimum wage constraints. Our simulations show that this price adjusts only marginally in scenarios (1) through (4), with a more pronounced decline appearing only in scenarios (5) and (6). This pattern is also illustrated graphically in Figure 6. Similarly, despite the fact that as many as 19% of firms and 5% of workers are directly bound

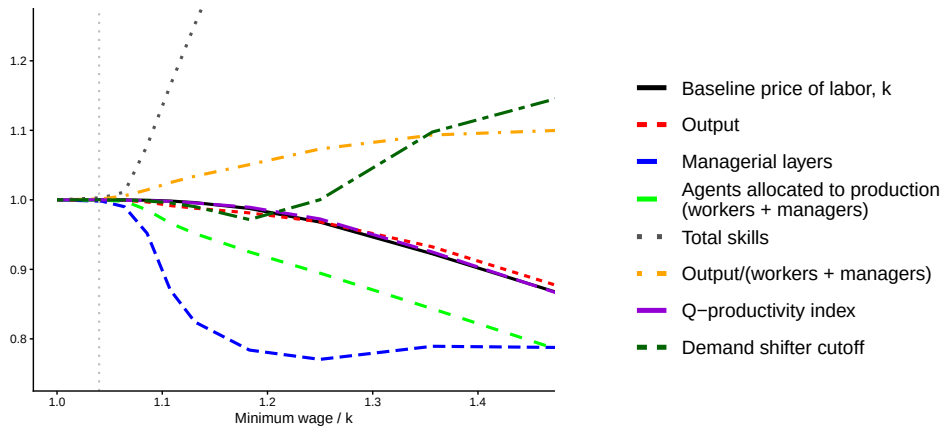
⁶⁶When minimum wage constraints become more stringent, as in our simulation exercises in Section 5.2 (see Table 6, Column (6)), such “limit” behaviors become increasingly likely because minimum wages implicitly set minimum constraints on the skills of production workers (Equation 21).

Table 6: Main General Equilibrium Aggregate Implications of Minimum Wages

	(1) Baseline	(2) No MW -4%	(3) + 4%	(4) + 8%	(5) + 16%	(6) + 24%
	<i>Wages</i>					
Net (real) wage, k (€/hour)	17.78	17.78	17.77	17.71	17.21	14.24
Minimum wage, relative to k	1.04	1	1.09	1.13	1.25	1.61
Minimum wage, relative to mean wage	0.91	0.88	0.94	0.95	0.98	1
Minimum wage, relative to median wage	0.98	0.94	1	1	1	1
Minimum wage, relative to maximum wage	0.56	0.54	0.58	0.61	0.68	0.89
<i>(as shares of total number of agents)</i>	<i>Occupational choices and skills</i>					
Workers/managers	0.63	0.63	0.62	0.60	0.57	0.45
– Share of workers bound by MW	0.05	0.00	0.46	0.85	0.93	0.99
Firms (entrepreneurs)	0.01	0.01	0.01	0.01	0.01	0.01
– Share of firms bound by MW	0.19	0.00	0.41	0.57	0.93	1
Teachers ($\propto$ total skills in economy)	0.10	0.10	0.11	0.12	0.17	0.29
Fixed-cost workers	0.26	0.26	0.26	0.27	0.26	0.25
<i>(as shares of total number of firms)</i>	<i>Structure of firm population</i>					
1-layer firms	0.32	0.32	0.36	0.43	0.53	0.50
1-layer firms with $n_1^0 < e^{\lambda \bar{z}_0}/h$	0.00	0.00	0.00	0.00	0.00	0.45
2-layer firms	0.45	0.45	0.46	0.57	0.47	0.50
3-layer firms	0.23	0.23	0.18	0.00	0.00	0.00
	<i>Production and productivity</i>					
Total output ($\sim$ €/hour)	13.30	13.31	13.27	13.15	12.89	10.79
Average firm size (in jobs)	92.38	92.43	90.39	84.98	82.99	86.11
Average wage (€/hour)	20.29	20.28	20.52	21.06	21.96	23.01
Average revenue per worker (€/hour)	28.42	28.37	28.90	29.68	29.96	30.48
Q -productivity index, $q/C_L(q, 1)$ (€/hour)	17.85	17.85	17.83	17.77	17.36	14.21
Production cutoff, $\bar{\alpha}$	52.03	52.03	51.95	51.52	52.04	62.60

Notes: This table presents simulations of alternative scenarios where the minimum wage is either removed (Column (2)), or increased by 4%, 8%, 16% or 24%, respectively, in Columns (3) to (6), relative to the baseline calibration presented in Column (1). The baseline scenario is described in detail in Section 5.1; parameter values are from Table 5. Among considered indicators, total output can be interpreted as per agent and per hour, in € : $P \int M\mu(\alpha)q(\alpha)d\alpha$, since P is normalized to 1. Average revenue per worker (and per hour) corresponds to $\frac{p(\alpha) \int M\mu(\alpha)q(\alpha)d\alpha}{\# \text{ Workers} + \# \text{ Managers}}$.

Figure 6: General Equilibrium Margins of Adjustment to Minimum Wages



Notes: This figure presents simulations results for alternative scenarios where the minimum wage is either removed or set at increasingly high levels (relative to k , the baseline price of labor in general equilibrium). Parameter values from Table 5.

by the minimum wage constraint in our baseline scenario (1), the aggregate output loss relative to the “no minimum wage” scenario (2) is negligible. As minimum wages further increase, we obtain that the output loss remains negligible in scenario (3), reaches 1% in scenario (4), 3% in scenario (5), and ultimately 19% in scenario (6). Figure 6 confirms that the output cost of minimum wage constraints remains extremely limited for values attaining up to 1.2 times the baseline price of labor, k .

Organizational adjustments. While the impact of a moderate tightening of the minimum wage constraint on aggregate output remains limited, its effect on firm organization is substantial. In fact, it is precisely these organizational adjustments that help contain the output decline, by boosting productivity. Figure 6 provides a detailed account of these changes. A first adjustment channel is a sharp increase in skill levels, which requires significant reallocation of workers from production and managerial tasks toward the education sector — capturing, in our model, the accumulation of human capital. As a result, wages rise across all hierarchical layers, as shown in Panel (A) of Appendix Table F1. A second channel is the rapid flattening of organizational structures: as minimum wage constraints tighten, the share of complex firms shrinks. In the most stringent scenario in Table 6, most one-layer firms find it optimal to reduce their production layer to the point that managerial time constraints are no longer saturated. Panel (B) of Appendix Table F1 documents that the overall decline in average firm size is mainly driven by layering and composition effects: firms reduce the number of hierarchical layers, while average firm size conditional on the number of layers varies non-monotonically for one- and two-layer firms, and increases for three-layer firms.

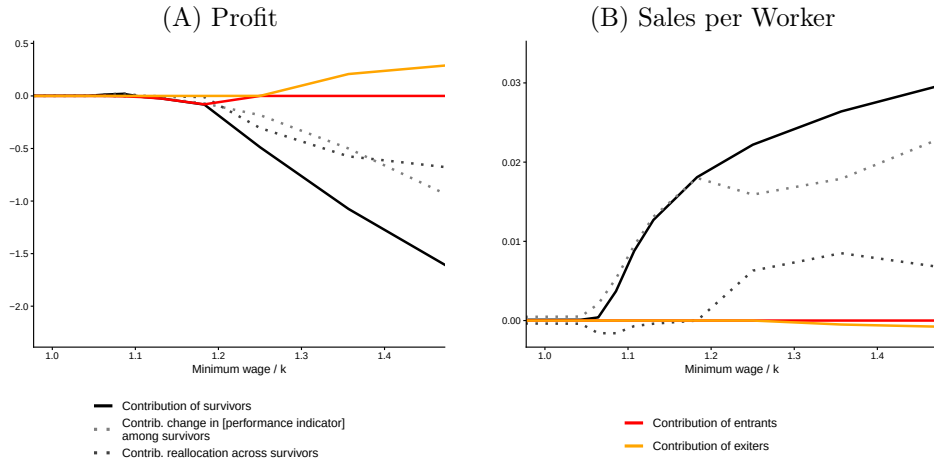
One striking feature of the simulations in Figure 6 is the speed and magnitude of these organizational responses, even in scenarios where the baseline labor cost, k , adjusts only slightly. For instance, in the “+4%” scenario, k increases by less than 0.1% compared to the baseline, yet the shares of firms with one, two, or three layers change by 12%, 2%, and 21%, respectively (Table 6). This is primarily because the minimum wage constraint (Equation 21) directly impacts skills and organization at the bottom of hierarchies. Additionally, internal training decisions (in terms of z) create multiplier effects across layers, thus amplifying the impact of small changes at the bottom, including in terms of which organizational form ultimately becomes optimal for a given demand draw, α .

Implications for aggregate productivity. Our simulations show that firms’ large organizational responses have significant R (evenue-) productivity implications. It is precisely this mechanism that mitigates the impact of (moderate) minimum wages on aggregate output. To be precise, in our set-up, aggregate R -productivity can be affected by three different channels. Firstly, the “pure” selection mechanism is driven by the fact that firms benefiting from more favorable idiosyncratic demand conditions (captured by α) are more likely to operate, as a consequence of Equations 18 and 19. This mechanism is identical to that of Melitz (2003).⁶⁷ Secondly, among active firms, firms benefiting from more favorable demand conditions tend to capture larger market shares: this “reallocation” channel affects aggregate productivity by reweighting the contribution of each individual firm to the aggregate. Thirdly, the endogenous response of firms’ organizations affects production costs and hence firm-level productivity. This third channel may counteract the first two. Our simulations allow us to assess the comparative importance of all of these mechanisms in response to minimum wage constraints. Table 6 contains some first insights. The magnitude of the “pure” selection channel is captured by the general equilibrium adjustment of $\bar{\alpha}$, the productivity cutoff. In scenarios (1) to (5), this parameter remains nearly constant, indicating that selection plays only a limited role. Moreover, the impact of the minimum wage appears non-monotonic: moderate increases in the minimum wage (scenarios (1) to (4))

⁶⁷We formulate exogenous heterogeneity at the firm level in terms of demand heterogeneity rather than marginal cost, in order to make it clear that everything referred to as “productivity” in our paper corresponds only to the endogenous component that is determined by firms’ organizational strategies (and not to ex ante exogenous differences), but this modeling choice is fully equivalent to the standard Melitz (2003) formulation.

are actually associated with excess firm entry due to the slight decline in k , while selection effects only become apparent under more stringent constraints (scenarios (5) and (6)). The organizational channel can be assessed through the different productivity indicators of Table 6. The Q -productivity index — a pure measure of productive efficiency (Equation 14) — systematically declines, since minimum wages constrain firms from implementing their optimal strategies (see Figure 5 and Panel (C) of Appendix Figure F1). In addition, at the macro level, more individuals are reallocated to teaching to meet the increased training demand, while fewer remain in direct production, which depresses output. However, the decline remains limited for moderate wage constraints and becomes steeper only in the more extreme scenarios. In contrast, revenue-based productivity indicators, such as average revenue per worker, increase steadily with tighter minimum wage constraints. Appendix F.2 further shows that this aggregate increase stems from a composition effect at the firm level. Specifically, the impact of minimum wage increases on revenue productivity is non-monotonic but generally positive for firms with moderate idiosyncratic productivity (α), due to price increases driven by higher marginal costs. For other firms, the effect is negative but limited in magnitude, resulting in a net positive aggregate outcome. This pattern is even more pronounced for sales per worker: as minimum wages rise, directly affected firms invest more in training, increasing both output and revenue per worker — at least under moderate minimum wage increases. Ultimately, we find that average revenue per worker increases by 1.7% in scenario (3), 4.4% in scenario (4), 5.4% in scenario (5), and ultimately 7.2% in scenario (6).

Figure 7: Decomposition of the Impact of Minimum Wages on Average Profits & Labor Productivity
(relative to the “No Minimum Wage” economy)



Notes: In each Panel, the decomposition (Equation 22) relies on output weights, but virtually identical results are obtained with employment weights. Panel (A) plots the contribution of the different terms to average profits in economies characterized by different tightness of the minimum wage constraints. Panel (B) replicates the same exercise for average sales per worker.

Figure 7 provides an even more comprehensive quantification of the three main channels that determine aggregate productivity: pure selection, market share reallocation, and the endogenous organizational channel. It relies on the decomposition framework originally proposed by Melitz and Polanec (2015),⁶⁸ which allows us to break down the changes in average profit or average revenue per worker in economies with minimum wages, relative to the “no minimum wage” scenario:

$$\Delta\Phi = \Delta\tilde{\phi}_S + \Delta COV_S + s_E(\Phi_E - \Phi_S) + s_X(\Phi_S^{no\ MW} - \Phi_X^{no\ MW}). \quad (22)$$

In this equation, Φ denotes the aggregate indicator of performance that is considered (average profits in Panel (A) and average revenue per worker in Panel (B)). $\tilde{\phi}_S$ is the unweighted average of this same indicator

⁶⁸See also Autor et al. (2020) for a recent and notable application of the same decomposition method.

across firms which are active under both the “no minimum wage” and the considered scenario, i.e. “survivors”, S . The difference between such averages computed in scenarios with *versus* without minimum wages thus captures the shift in the firm-level distribution of the considered indicator: in particular, when Φ measures productivity, this term thus quantifies the endogenous organizational channel. ΔCOV_S corresponds to the change in the covariance between market share and performance among “survivors” and captures market share reallocations. Lastly, the pure selection channel is captured by the last two terms in Equation 22. $s_E (\Phi_E - \Phi_S)$ captures the contribution of entrants (E), in scenarios where the $\bar{\alpha}$ threshold is lower than in the “no minimum wage” benchmark: this contribution is positive whenever the average performance of entrants is higher than that of survivors. Symmetrically, the term $s_X (\Phi_S^{no\ MW} - \Phi_X^{no\ MW})$ captures the contribution of exiters (X), in the opposite case where the $\bar{\alpha}$ threshold increases: the contribution of exits is positive whenever the average performance of exiters is lower than that of survivors.

Figure 7 displays the decomposition results for aggregate profits in Panel (A). Average output-weighted profits initially increase slightly due to reallocation effects (whereas unweighted average profits decline monotonically), but then decline as minimum wage constraints become more stringent. The main drivers of this decline are within-firm reductions and reallocation effects among “survivors”,⁶⁹ with the former becoming dominant under the most stringent constraints. Excess entry also contributes to depressing average profits at moderate levels of constraint, while firm exits tend to dampen aggregate profit losses under tighter constraints. In contrast, the patterns shown in Panel (B) for (revenue) productivity – measured as sales per worker – differ markedly. Aggregate sales per worker increase monotonically with the stringency of minimum wage constraints, and the main contributor to aggregate productivity gains is the strong within-firm increase, driven by firms’ organizational adjustments. Reallocation effects become more relevant under tighter constraints, but their magnitude never exceeds half that of the within-firm effect. Finally, the contribution of the pure selection channel (entries and exits) remains marginal.⁷⁰

Model performance on non-targeted data moments. Scenarios (1) to (3) in Table 6 are symmetric in the sense that they characterize economies with minimum wages that are either 4% lower (scenario (2), where the minimum wage does not bind) or 4% higher (scenario (3)) than the baseline calibration (scenario (1)). This configuration is useful, as it allows us to calculate the average (both upwards and downwards) impact of a 4% *variation* in the minimum wage.⁷¹ This directly compares to our regression results and thus provides an assessment of the model’s predictive performance on non-targeted data moments.⁷²

We first evaluate the model against the evolution of firm size, arguably the outcome that is measured most reliably in the data. Table F2 in Appendix F.3 summarizes the comparison. The DDD regressions imply that, for a firm with average exposure to the minimum wage, a 4% increase in the minimum wage reduces firm size by about 0.9% (total effect), of which one third (0.3 percentage points) is accounted for by the within-GMR exposure effect. The model predicts a remarkably similar response: average firm size declines from 92.38 workers in the baseline equilibrium to 90.39 after a 4% increase in the minimum wage, corresponding to an average contraction of 1.1% across the two counterfactual experiments. Given the parsimonious nature of the model, this degree of quantitative agreement is noteworthy.

The model performs equally well in matching productivity responses, which constitute the main object

⁶⁹These results are in particular consistent with Dustmann et al. (2022).

⁷⁰In addition, we document in Appendix F.2 that firms facing the least favorable demand conditions counteract them proactively via their organizations, which implies that they are not the least productive in terms of revenue per worker (Panel B of Figure 7). This implies in particular that “exits” do not always have a positive contribution to aggregate productivity in our set-up.

⁷¹As a reminder, in contrast, our calibration procedure only relies on static data moments (Table 5).

⁷²One point of caution is that in our setting, the regression results relate more to partial equilibrium than to general equilibrium comparative statics, although the different GMR groups are large enough to have the potential to affect general equilibrium prices (k). However, our simulations show that price effects are in fact negligible across scenarios (1) to (3), so that in this range, partial and general equilibrium quantifications almost coincide numerically.

of interest. The DDD estimates imply that a 4% increase in the minimum wage raises labor productivity by about 0.9% for the average firm, with roughly half of this effect (0.4 percentage points) attributable to the within-GMR exposure component. The model predicts an increase in revenue per worker of 0.9%; the quantitative correspondence is therefore almost exact.

Finally, the model also reproduces the main organizational adjustments observed in the data. A 4% increase in the minimum wage reduces the average number of hierarchical layers by about 0.05, decreases employment in the production layer by roughly 0.5%, and raises labor costs in this same layer by 0.8%. The corresponding regression estimates imply a reduction of about 0.01 layers, a 1.8% contraction of the bottom layer, and a 1.2% increase in bottom-layer labor costs for the average firm. Although the model understates the magnitude of the organizational responses, it correctly predicts their direction and captures their order of magnitude. Overall, the model successfully reproduces the qualitative organizational responses observed in the data while providing a remarkably close quantitative match for our indicator of main interest: productivity.

Comparison with a model specification without organizations (“unproductive managers”). The previous developments suggest that firms’ productivity responses to increases in the minimum wages are largely driven by their endogenous organizational responses. In Appendix I, we propose an even more direct strategy for assessing the importance of this channel – and more precisely within this channel, of the contribution of the responses of hierarchical organizations. There, we specify a version of the model in which workers’ skills remain useful to production, but (hierarchical) organizations do not. More precisely, we assume that managers are still useful to “bundle” teams of workers, but that this task exhausts their entire time constraint such that they are unable to contribute to actual problem solving. In such a setting, no firm finds it optimal to increase the number of managerial layers beyond $L = 1$, and managers do not need skills, but all firms still have to optimize their size in terms of number of production workers, as well as the level of skills of the latter. Appendix I proposes a calibration of such a framework to the French data: the comparison of its quantitative predictions to our main results provides a precise quantification of the mitigation impact of firms’ organizations to minimum wage constraints.

We obtain that firms find it optimal to train their bottom production workers (and pay them accordingly) even in absence of a minimum wage. As a consequence, the minimum wage only starts to be binding for values above 1.12 k , while it was binding for at least some workers for any value exceeding k in our main calibration. From this point, skills adjust even more quickly in the set-up without organizations than in the baseline set-up since this is the only possible mitigation channel. However, this only translates into mediocre increases in output per worker. As a consequence, the Q –productivity index and aggregate output both decrease four times faster than in the baseline set-up as the minimum wage constraint ($\bar{w}/k$) becomes more stringent. In other terms, the pure plasticity of firms’ organizations dampens the aggregate output cost of minimum wage constraints by dividing it by a factor of four, compared to a set-up where organizations are restricted to be fully rigid.

Further robustness checks. In the main specification of our model, the labor market is always in equilibrium. However, we investigate in Appendix H whether the employment cost of minimum wages could be amplified in a specification where we introduce job search costs, such that unemployment can emerge. This augmented version of the model thus corresponds to an economy with a more rigid labor market, a feature often attributed to the French case. In the calibration procedure, we target an unemployment rate as an additional moment while otherwise using the same baseline moments as in our central calibration. The main take-away of this experiment is that introducing a richer set of occupational “options” available to workers (i.e. an additional unemployment state) does not significantly alter the quantitative predictions of the model. In

particular, the unemployment margin absorbs very little of the minimum wage constraint, while productivity remains the main channel of adjustment. As a consequence, the overall employment impact is simulated to be very limited.⁷³

5.3 Simulation of Alternative Technological Shock Scenarios

Interaction between technologies and Minimum Wages. We now examine how the impact of the minimum wage, as described in the previous section, is modulated by different types of technologies. Our framework is particularly rich and well-suited to this analysis, as it includes four parameters that capture key aspects of the technological environment. First, parameter A reflects output per problem solved, and thus serves as a general measure of worker productivity. Second, while solving a problem occupies the full working time of a production worker, it only requires an additional h units of time from managers when they are consulted. As such, h captures a technological dimension related to managerial costs. Following Garicano and Rossi-Hansberg (2006), this can be interpreted as a proxy for a firm’s internal communication technology.⁷⁴ Third, parameter c reflects the cost of acquiring higher levels of problem-solving ability for both production workers and managers. In practice, this is likely shaped by access to and the efficiency of information technologies (Garicano and Rossi-Hansberg, 2006; Acemoglu, Aghion, et al., 2007). Finally, parameter λ governs the distribution of problem difficulty across production tasks. We hypothesize that emerging AI-based technologies, such as large language models (LLMs), may influence this dimension of production.⁷⁵

In the simulations that follow, we consider economy-wide technological shocks of the same nature as those studied in Garicano and Rossi-Hansberg (2006) and Caliendo and Rossi-Hansberg (2012), namely shocks that homogeneously and simultaneously affect all firms through changes in the model’s four key technological parameters (A , c , h , and λ).⁷⁶ Regarding the magnitude of the shocks, the literature offers limited guidance. An ideal experiment would calibrate variations in each of the four technological parameters so that they

⁷³This specification of the model delivers an aggregate employment elasticity which can be compared with previous results obtained in the empirical literature. It appears to be highly non-linear and in particular, close to 0 for limited increases in the minimum wage, since aggregate employment is barely affected over this range in our simulations. This result is consistent with many estimates of the empirical literature, from Card and Krueger (1994) to Bailey et al. (2021). For larger increases in the minimum wage, the employment elasticity is less precisely estimated (probably because of our numerical approximations). Contrasting the “+24%” scenario with the others, we find an employment elasticity of -0.09 if we compare to the baseline minimum wage, or -0.27 if we compare “+24%” with “+16%”. This compares closely to the synthesis of the literature’s estimates displayed in Figure A7 of Harasztosi and Lindner (2019) and Figure 6 of Bailey et al. (2021), which are most often negative and range roughly in the $[-1; +0.5]$ interval.

⁷⁴This dimension is especially relevant in the post-Covid context, where many firms were compelled to adopt or upgrade remote work tools to remain operational during lockdowns. Quantifying the long-term consequences of this unprecedented technological shift remains an open question (Barrero et al., 2021; Gibbs et al., 2023).

⁷⁵A number of recent contributions have explored the potential impact of “AI” technologies, but no definitive conclusions have been reached, as the effects may be uncertain, heterogeneous, and endogenous to the decisions of innovators themselves (Acemoglu and Johnson, February 6, 2023; Rotman, March 25, 2023). Initial case studies suggest that the on-going wave of AI-based technologies could enhance the capabilities of some workers (Mullainathan and Obermeyer, 2021), notably by reducing errors. In our framework, such improvements could be captured by a decrease in c —as was the case for standard information technologies in Garicano and Rossi-Hansberg (2006)—or by an increase in λ . For instance, the experiment conducted by Kleinberg et al. (2017), which shows that machine learning tools can reduce incarceration by up to 41.9% without increasing crime, could be interpreted as an increase in λ , since judicial decisions appear to become more accurate. Similarly, Noy and Zhang (2023) report that “ChatGPT substantially [raises] average productivity,” an effect that could, in our model, be attributed to an increase in either A or λ . Related findings appear in Brynjolfsson et al. (2025) and Peng et al. (2023). One common concern in this literature is the extent to which these technologies may replace labor when used primarily for automation (Eloundou et al., 2024; Rotman, March 25, 2023; Svanberg et al., 2024; Acemoglu, 2024). This automation margin remains outside the scope of our paper. In particular, we leave for future research the integration of additional production factors (such as capital or intermediate inputs, see Panel (B) of Table 4) and task-based approaches (Aaronson and Phelan, 2019) into the model. Here, we focus solely on the interaction between technological parameters and firms’ endogenous decisions regarding work organization (Garicano and Rossi-Hansberg, 2015).

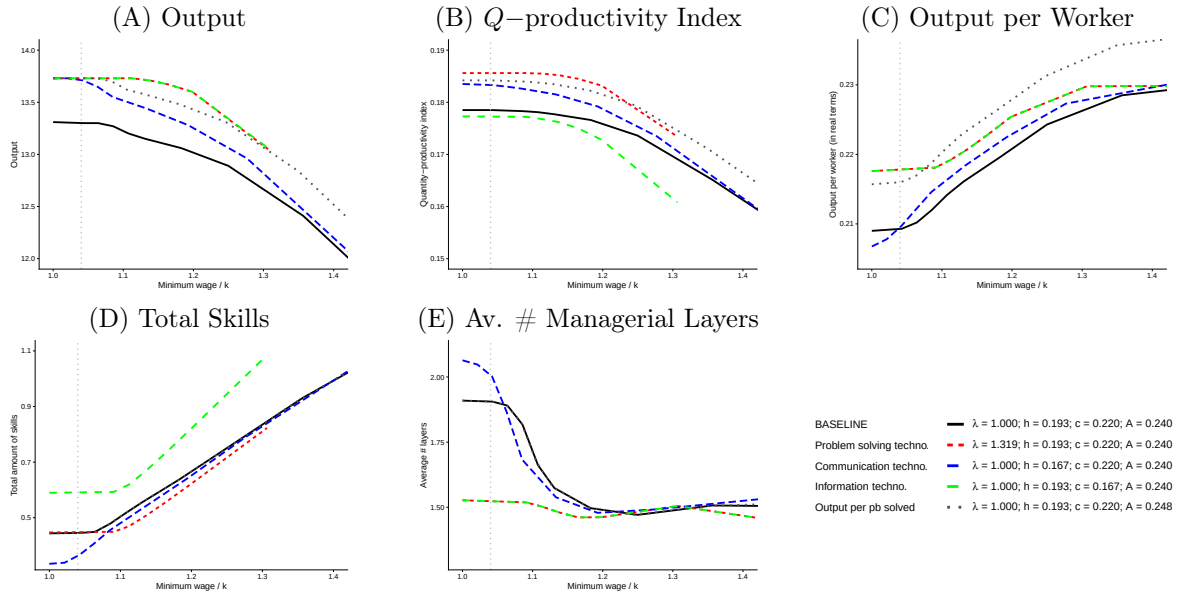
⁷⁶This type of comparative statics exercise is particularly relevant when considering the potential impact of recent AI technologies, such as ChatGPT (as a prototypical example of generative AI), which have been introduced at low cost—at least for now and at current service levels. A central question is whether their effects may be modulated by labor market institutions, such as minimum wage policies.

reflect equivalent social costs. However, to our knowledge, no such analysis is currently available. We therefore adopt a pragmatic approach, using benchmark values drawn from recent and empirically relevant episodes of technological change. In particular, Bloom, Sadun, et al. (2012) find that U.S. multinationals enjoy, on average, a productivity premium of 3.2 percent relative to non-U.S. multinationals, a gap the authors attribute to greater ICT intensity and accompanying organizational adaptations. Accordingly, we simulate four separate technology shocks — each involving one of the model’s parameters—calibrated so as to generate a 3.2 percent increase in aggregate output in the absence of a minimum wage policy. The resulting calibration scenarios are as follows:

- Baseline calibration: $\lambda = 1$, $h = 0.193$, $c = 0.220$, $A = 0.240$.
- Scenario 1 [varying λ , production problem difficulty]: $\lambda = 1.319$, $h = 0.193$, $c = 0.220$, $A = 0.240$.
- Scenario 2 [varying h , communication technologies]: $\lambda = 1$, $h = 0.167$, $c = 0.220$, $A = 0.240$.
- Scenario 3 [varying c , information technologies]: $\lambda = 1$, $h = 0.193$, $c = 0.167$, $A = 0.240$.
- Scenario 4 [varying A , output per problem solved]: $\lambda = 1$, $h = 0.193$, $c = 0.220$, $A = 0.248$.

In each of these scenarios, we replicate the same analyses as in Section 5.2 and consider different magnitudes of the minimum wage constraint. All results are reported in Figure 8, where the x -axis normalizes the value of the minimum wage relative to k , the baseline price of labor. Panel (A) first examines the effect of the minimum wage constraint on aggregate output. The results clearly show that the impact of minimum wages varies significantly depending on the technological environment: while output is systematically higher than in the baseline scenario in all cases, the degree of decline induced by increasingly stringent minimum wage constraints differs widely across technological settings.

Figure 8: Impact of Minimum Wage Across Different Technological Scenarios



Notes: This figure presents five different technological scenarios. The black curve corresponds to the baseline calibration described in Sections 5.1 and 5.2. All other scenarios reflect more efficient technologies, each normalized to produce an economy-wide output premium of 3.2% in the absence of a minimum wage (see main text and legend for parameter values). Panel (A) shows total aggregate output, while all other panels display firm-level averages. In Panel (B), the Q -productivity index corresponds to $q/C_L(q, 1)$, as in Equation 14 and Table 6. In Panel (D), “Total skills” correspond to the aggregate knowledge acquired by workers and managers, $(\sum n_L^t z_L^t)$, which also matches the total flow of knowledge generated by the equilibrium mass of active teachers, $\frac{\# \text{ Teachers}}{c}$. All measurement units are identical to those used in Table 6.

In the scenario where the aggregate productivity parameter A is increased, the output curve shifts upward, but its overall shape remains identical (i.e., homothetic) to that of the baseline. As a result, the relative

impact of the minimum wage remains unchanged. This “neutrality” result stems from the fact that A affects all agents uniformly, leaving the structure of the economy fundamentally unchanged, regardless of the level of the minimum wage constraint. In particular, the curves representing optimal firm organization and aggregate skill levels in Panels (D) and (E) are unaffected. As in the baseline case, firms respond to minimum wage constraints by hiring fewer but better-trained workers (Panel D) and adopting flatter organizational structures to economize on managerial “fixed” costs (Panel E). Consequently, output per worker rises and the curves for output per worker (Panel C) and for the Q -productivity index (Panel B) are simple upward translations of those in the baseline case, reflecting the uniform productivity premium applied to the entire economy.

Turning to the case of communication technologies (affecting parameter h), the situation appears significantly less favorable. In Panel (A) of Figure 8, the steeper slope of the curve implies that the adverse impact of minimum wages on aggregate output is amplified compared to the baseline scenario. Total skills adjust more markedly than in the reference case (Panel D), while firms’ organizational structures flatten more rapidly (Panel E). As previously shown, this organizational margin is particularly effective in mitigating the costs induced by minimum wage regulations. However, in this scenario, such a strategy proves socially more costly because it prevents the economy from fully realizing the gains associated with a lower value of h . Indeed, the benefits from improved communication technologies require sufficiently vertical firm structures to be fully effective. Minimum wage constraints, by encouraging rapid flattening, reduce firms’ capacity to leverage these technological improvements. As a result, the potential efficiency gains from enhanced communication technologies — reflected, for instance, in the Q -productivity index in Panel (B) — are diminished.

Finally, Figure 8 shows that the relative cost of minimum wage constraints tends to be mitigated in economies benefiting from cheaper knowledge (i.e., lower c) or less dispersed production problems (i.e., higher λ).⁷⁷ In terms of aggregate output (Panel (A)), the profiles of the curves corresponding to each of these scenarios actually coincide. These economies are also equivalent in terms of output per worker (Panel (C)) and firms’ organizations (Panel (E)),⁷⁸ but they differ in terms of the Q -productivity index (Panel (B)) and in terms of training decisions (Panel (D)).⁷⁹ Unsurprisingly, investment in knowledge is lower in economies where production problems are easier to solve (i.e., with higher λ), as fewer skills are needed. Conversely, such investments increase when acquiring knowledge is less costly. In these scenarios, the lower cost of the minimum wage constraint is simply driven by the fact that more efficient technologies — whether through lower c or higher λ — make it easier to enhance the productivity of production workers. This allows firms to meet the minimum wage constraint of their cost minimization programs more easily.

Varying the magnitude of technological shocks. In a final experiment presented in Online Appendix F.4, we examine how the previous results evolve under varying magnitudes of technological shocks. Specifically, we consider shocks that are two and three times larger than the 3.2% productivity premium analyzed previously. We find that the relative cost of the minimum wage constraint, in terms of aggregate output, remains unchanged in the scenarios in which A is increased, whereas the relative cost of the minimum wage tends to be higher in economies characterized by efficient communication technologies (i.e., low h), with a disadvantage that is amplified as the magnitude of the technology shock increases. Conversely, it remains true that the relative cost of moderate minimum wage constraints are smaller in economies characterized by low values of c or high values of λ , but the curves obtained for different magnitudes of technological shocks are not homothetic and imply that this advantage diminishes as the magnitude of the technological shocks increases.

⁷⁷This is partly driven by the fact that the economy switches from a technological environment where $\frac{c}{\lambda} > h$ to an environment where $\frac{c}{\lambda} < h$. As discussed in Section 4, the second regime prevents the emergence of inefficient regimes in which managerial time in 1-layer firms is (optimally) wasted. This aspect contributes to mitigating the minimum wage constraint.

⁷⁸This result was already present in Caliendo and Rossi-Hansberg (2012), who present their comparative static exercises of interest in terms of $\frac{c}{\lambda}$ directly rather than separately in terms of c and λ .

⁷⁹Note that the curves are simply translated, with the difference in skill responses exactly off-setting the difference in terms of Q -productivity.

These results show that minimum wages reduce the advantages for firms of organizing themselves into hierarchies, and the benefits lost are particularly significant when hierarchies are “technologically” efficient, i.e. when h is lowest. In addition, as the minimum wage constraint becomes particularly tight, firms tend to flatten their organizational structures regardless of the technological environment, such that the organizational adjustment margin ultimately disappears and no longer helps mitigate the regulatory constraint. In such a situation, the economy fails to benefit from the gains associated with technologies that are most effective, or “leveraged”, within hierarchical structures — such as those affecting c , λ , or h — whereas those generated by general labor-augmenting technologies (parameter A) are unaffected. More generally, the curves obtained for different magnitudes of technological shocks are not homothetic, implying that major technological transformations — such as the AI revolution now unfolding — most likely require a thorough reassessment of the costs and benefits associated with labor market institutions and regulations.

6 Conclusion

In this paper, we exploit a unique institutional setting and a rich administrative information system to study how minimum wages affect firms’ strategies and performance. Our empirical analysis reveals that the firm-level (revenue) productivity channel plays a central role in mitigating the costs associated with minimum wage constraints. This productivity response primarily takes the form of endogenous organizational adjustments, i.e. a flattening of firms’ organizations, and increases in the frequency of worker-level training programs. To rationalize these findings, we develop a quantitative version of a structural model of optimal hierarchies, which captures both qualitatively and quantitatively how firms’ internal organizations respond to increases in the minimum wage. We calibrate the model to key moments in the French data and show that its predictions are consistent with our (non-targeted) empirical estimates. We then use the model to run simulations demonstrating that while aggregate *quantity* productivity and profits deteriorate, their decline is mitigated by firms’ training and organizational strategies, which ultimately enhance aggregate *revenue* productivity. Reallocations across firms are significant but the contribution of the “pure” selection channel (Melitz, 2003) is quantitatively weak. Further simulations show that the aggregate cost of minimum wage constraints is shaped by the technological environment. These costs are amplified in economies with more efficient communication technologies, while they are dampened in those with more advanced information or problem-solving technologies. However, this pattern is non-homothetic and tends to reverse when both the technological shocks and the minimum wage constraints are large. Overall, these findings suggest that the current wave of AI-driven innovations, as well as the massive investments in communication technologies prompted by the recent COVID-19 pandemic, are likely to require a thorough re-evaluation of the cost-benefit analyses surrounding labor market institutions and regulations. In particular, our framework highlights the importance of accurately quantifying the magnitude and the specific technological nature of such shocks, as these are crucial for drawing robust policy implications for the labor market.

We leave several important aspects for future research. These include, notably, analyzing how our quantitative results are affected when departing from the assumptions of monopolistic competition in output markets, and perfect competition in input (labor) markets. These assumptions, drawn from benchmark models in the productivity literature (e.g., Melitz (2003)), were adopted to facilitate interpretation. However, a recent literature suggests that firms in at least some sectors may exercise greater market power — both in output markets (see e.g., De Loecker et al., 2020) and in labor markets.⁸⁰ We nonetheless expect that, when faced with minimum wage constraints, firms will first fully exploit the organizational and productivity margins —

⁸⁰See for instance Azar et al. (2023) and Berger et al. (2025) for recent analyses of monopsony power in the presence of minimum wages. Separately quantifying the contribution of the productivity vs. market power channels would in particular require detailed information about firm-level output prices, which were not available in 2003/2006.

as evidenced by the observed organizational — before adjusting markups or wage markdowns, in order to preserve as much profit as possible. A second challenge⁸¹ lies in a richer modeling of wages, incorporating a broader range of dimensions of heterogeneity — especially *ex ante* worker heterogeneity as in Garicano and Rossi-Hansberg (2006) — that may interact with minimum wage policies and help to further assess their welfare implications. Finally, detailed firm-level data on technology use, if it became available, would make it possible to endogenize (AI) technology adoption behavior and determine whether firms’ organizational optimization problems are an important driver of technological change.

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⁸¹The technical complexity of such extensions will be significant: to our knowledge, the full theoretical framework of Garicano and Rossi-Hansberg (2006) has never been calibrated to real-world data.

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Online Appendix for

The Minimum Wage in Firms' Organizations: Productivity Implications

Nicholas Lawson, Claire Lelarge and Grigorios Spanos

- Appendix [A](#): Details of the Institutional Setting
- Appendix [B](#): Estimation Sample
- Appendix [C](#): Empirical Results: Robustness Checks and Extensions
- Appendix [D](#): Regression Results: Alternative Estimation Strategies and Specifications
- Appendix [E](#): Details and Proofs of the Theoretical Model
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- Appendix [G](#): Sensitivity Analyses of Simulation Results
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- Appendix [I](#): Framework without Organizations: Unproductive Managers

A Details of the Institutional Setting

This appendix provides the institutional details supporting the synthetic exposition in the main text. The reform unfolded in two stages: firms first entered different GMR schedules when they adopted the 35-hour workweek between 1998 and 2002; the schedules were then reunified between 2003 and 2005. Figure A1 records the complete sequence of legal and political events and the information available to firms. A separate legal appendix reproduces the relevant statutory texts.¹

Political Context The implementation of the 35-hour workweek and the associated wage schemes took place against a shifting political backdrop. The *Jospin government* (left-wing), in office from June 1997 to May 2002, introduced and enacted both *Aubry 1* (1998) and *Aubry 2* (2000) laws. These reforms established the statutory reduction of the legal workweek and created the framework for wage guarantees (GMR).

Notably, the *Aubry 2* law anticipated the creation of all GMRs up to 2002 and set the re-convergence process to be completed by July 1st, 2005. As a result, uncertainty was limited since only the precise conditions of the convergence process remained unknown until January 2003. Figure A2 actually indicates that Economic Policy Uncertainty (Economic Policy Uncertainty, 2025; Baker et al., 2016; Ghirelli et al., 2019) was very limited at the time, particularly when compared with current levels. It is therefore unlikely that such a degree of uncertainty constrained or “froze” firm-level decisions—such as hiring—given that government measures had been announced several years in advance.

Following the 2002 presidential and legislative elections, a succession of *right-wing governments*—*Raffarin I–III* (2002–2005) and later *De Villepin* (from May 2005)—continued to implement the reform. While these administrations did not repeal the 35-hour workweek, they introduced significant adjustments. Most notably, the *Fillon Law* (2003) initiated the convergence of the various GMR wage floors toward a unified minimum wage and allowed late adopters to remain *de facto* at 39 hours by expanding the use of overtime, albeit with wage premiums.

Adjustments to SMIC and GMR Rates Between 1997 and 2006, both the *SMIC* (national minimum wage) and the various *GMR* levels were updated annually, typically on July 1st.

GMR Groups and Workweek Reduction Agreements The timeline also distinguishes five *GMR groups* (GMR1 to GMR5), classifying firms according to the date of their workweek reduction agreement:

- **GMR1:** before June 30, 1999
- **GMR2:** between July 1, 1999 and June 30, 2000
- **GMR3:** between July 1, 2000 and June 30, 2001
- **GMR4:** between July 1, 2001 and June 30, 2002
- **GMR5:** after July 1, 2002

This classification is central to our empirical analysis. It determines how firms self-selected into different GMR categories (see Section 3.1 for a detailed discussion of the associated endogeneity issues and our identification strategy) and shaped their subsequent exposure to minimum wage increases between 2003 and 2006, which are the focus of our regression analysis in Section 3.4.

Running example: an early and a late adopter. Table A1 tracks two otherwise similar firms through the reform. Firm A adopted the 35-hour workweek in January 1999 and entered GMR1; Firm B adopted it only in July 2002 and entered GMR5. The example separates the assignment stage, during which adoption

¹The legal appendix is available at <https://drive.google.com/file/d/1m4j3OawOvgc4rUchY0sM6ByTMln3hn45/view>.

timing determined GMR affiliation, from the convergence stage used in the empirical analysis. By the beginning of our estimation period, both firms operated under the same statutory workweek, but they faced different minimum-wage trajectories. Between 2003 and 2005, GMR1 had to rise substantially to catch up with GMR5, whereas GMR5 was essentially held to inflation. Firm A therefore experienced a much larger increase in the statutory wage floor and in the associated labor cost. Our DD strategy uses this cross-GMR difference; our DDD strategy additionally compares firms within the same GMR according to the share of workers initially (as of 2002) exposed to the unified 2006 minimum wage.

Table A1: Illustrative GMR Trajectories for an Early and a Late Adopter

Date	Firm A: early adopter	Firm B: late adopter
January 1999	Adopts the 35-hour workweek	Has not yet adopted the 35-hour workweek.
January 2000	GMRs are created; enters GMR1	Has not yet adopted the 35-hour workweek.
July 2002	Operates at 35 hours under GMR1.	Adopts the 35-hour workweek and enters GMR5.
2003–2005	Faces comparatively large increases as GMR1 catches up with GMR5.	Faces much smaller increases because GMR5 is the convergence target.
July 2005 onward	Both firms face the unified statutory minimum wage.	

Notes: This stylized example isolates the institutional mechanism. Actual firms differed in sector, size, wage structure, and other characteristics, which motivates the controls and the within-GMR exposure design described in Section 3.1.

Kaitz indices across countries and over time. Figure A3 replicates the Kaitz indices presented in Bozio and Wasmer (2024). Panel (A) reports the ratio of total labor costs at the minimum wage to total labor costs at the median wage, which corresponds to the relevant concept in our framework. Notice however that these estimates are based on the OECD Tax-Ben model, which does not account for firm-level heterogeneity in labor costs (such as payroll taxes and subsidies) – unlike our proxies, which rely on quasi-exhaustive firm-level data (see Section 3.2). These figures show that the French index steadily declined from 48% in 2006 (our calibration year) to 44% in 2023.² Cross-country comparisons suggest that France is not an outlier and aligns well with other developed economies such as Spain, Germany, the Netherlands, Belgium, the UK, and even with the United States over the 2010–2018 period.

Panel (B) of Figure A3 shows the ratio of total income at the minimum wage (including relevant social benefits) to total income at the median wage. We report this indicator because it corresponds to the most commonly published one, but it is not directly relevant to firms, as it includes not only wage income but also social transfers, and does not take employer payroll taxes into account. These are particularly significant in France at the minimum wage level: in our estimation sample, the ratio of the minimum to median wage (both net of employer payroll taxes) reaches 65% in 2006, while the OECD indicator, which includes social benefits, is 68% for the same year.³ According to this “income-based” measure, likely more aligned with public perception, France appears at the upper end of the distribution among European countries, with an index around 20 percentage points higher than that of the United States. Again, this is however not the relevant index in our set-up.

²Another study by Bozio, Guillot, et al. (2024) focuses only on France, but during a longer time period of 1967 to 2019, and their Figure 1(b) shows that the “pre-tax wage” measure (which corresponds to labor costs) declined in the early 1990s, but then stayed relatively constant during the late 1990s and throughout the 2000s.

³In recent years, this gap has widened, especially between 2019 and 2022. In January 2019, as part of the emergency economic and social measures following the Yellow Vest movement, the “bonus” component of the working tax credit (*prime d’activité*) was significantly increased—by €90 per month for a single person earning the minimum wage (SMIC). In 2021 and 2022, exceptional “inflation” allowances also contributed to the rise in the Kaitz index.

Figure A1: Timeline of Relevant Events, 1997– 2006

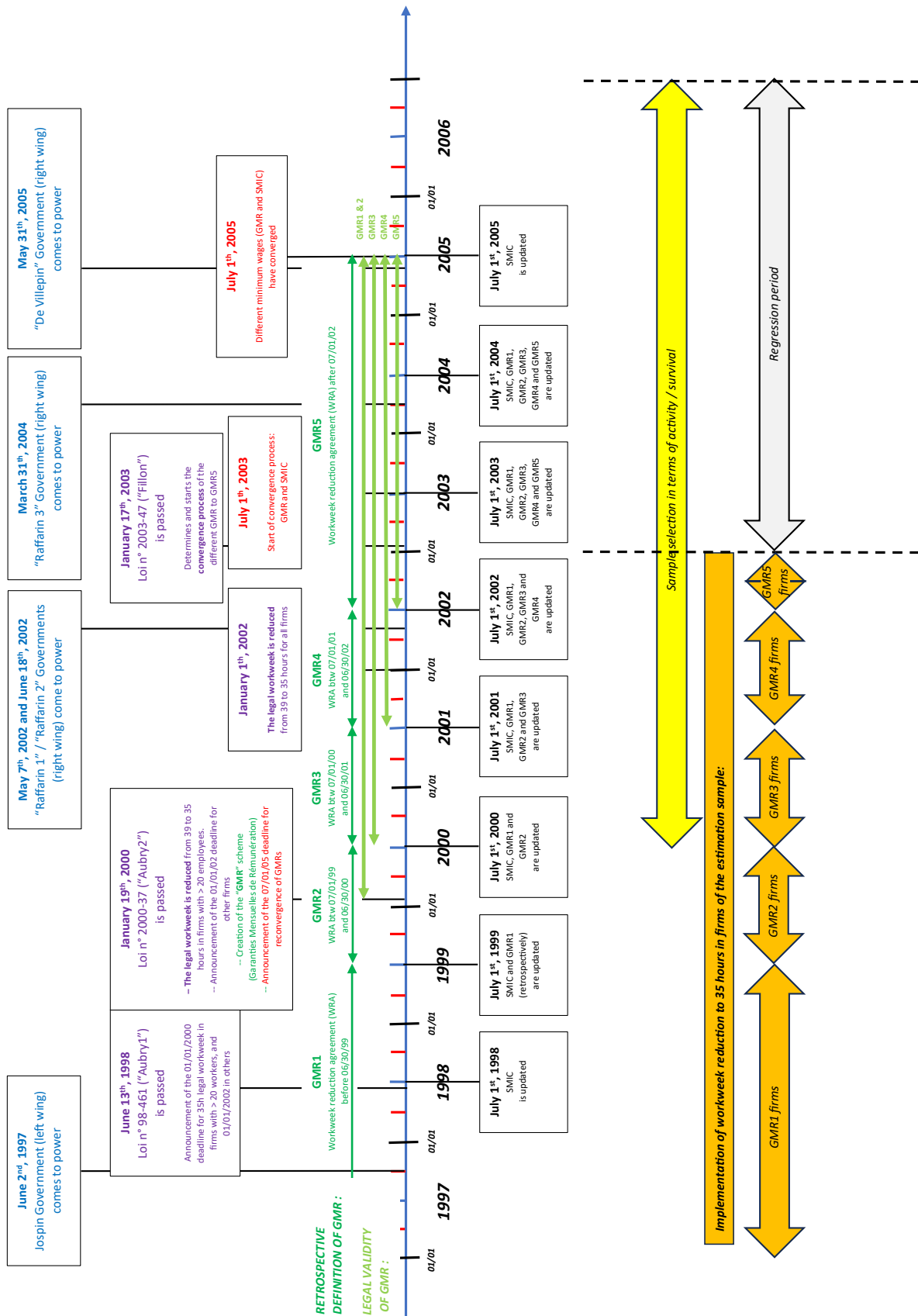
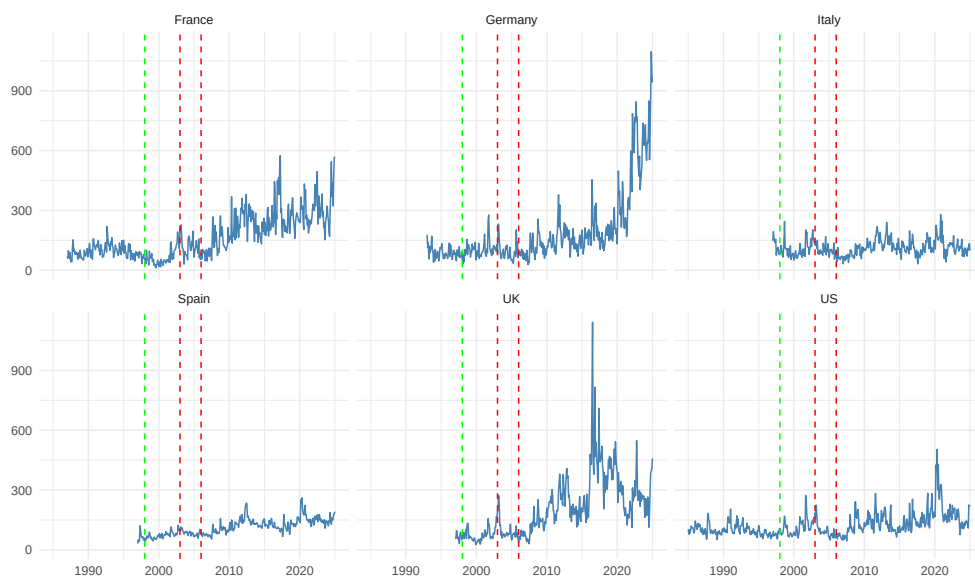
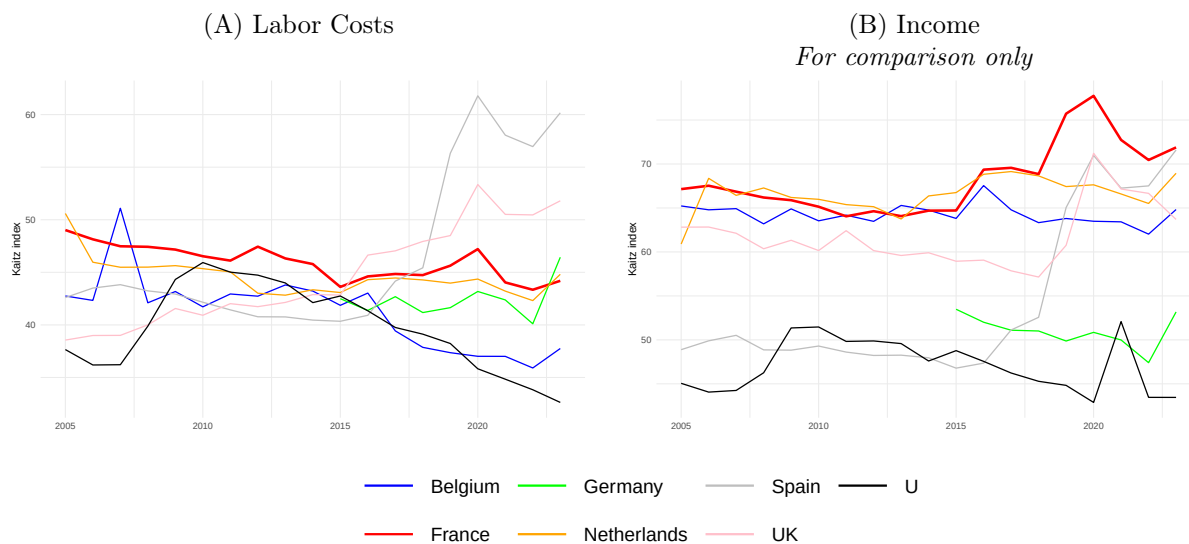


Figure A2: Economic Policy Uncertainty Index (Baker et al., 2016)



Notes: This figure reports the evolution of the Economic Policy Index for France, Germany, Italy, Spain, UK and the US; the data can be found at Economic Policy Uncertainty (2025), and are sourced from Baker et al. (2016) for all countries except Spain, and from Ghirelli et al. (2019) for Spain. The green dashed vertical line materializes 1998, the year in which the Aubry 1 law was passed. The two red dashed vertical lines materialize our regression period (2003 to 2006).

Figure A3: Kaitz Indices (Bozio and Wasmer, 2024)



Sources: Bozio and Wasmer (2024) using the Tax-Ben model developed by the OECD. The simulations rest on the assumption of a household composed of a single individual without children, and rely on modeled representations of the policies in force across OECD member countries. Panel (A) reports the ratio of total labor costs at the minimum wage to total labor costs at the median wage. Panel (B) reports the ratio of total income at the minimum wage (including complementary social benefits when relevant) to total income at the median wage.

B Estimation Sample

This Appendix complements Section 3.3 by providing a comprehensive description of the regression sample.

B.1 Sample Construction

Table B1: Sample Construction

Year	(A) Gross FICUS files			(B) Business corporations (CJ == 5)		
	Nb firms (thousands)	Value Added (FICUS) (billions)	Employment (FICUS, ~ FTE) (millions)	Nb firms (thousands)	Value Added (FICUS) (billions)	Employment (FICUS, ~ FTE) (millions)
2000	2,291	946	14.46	908	796	11.97
2001	2,226	906	14.92	891	756	12.41
2002	2,344	884	14.83	957	723	12.39
2003	2,401	879	14.32	999	752	12.46
2004	2,509	960	15.30	1,078	813	12.96
2005	2,447	1,218	14.98	1,082	1,064	12.74
2006	2,653	1,067	15.43	1,214	912	13.22
Average	2,410	980	14.89	1,018	831	12.59

Year	(C) Employment >= 5 workers (EFFSALM)			(D) Consistently active 2000/06		
	Nb firms (thousands)	Value Added (FICUS) (billions)	Employment (FICUS, ~ FTE) (millions)	Nb firms (thousands)	Value Added (FICUS) (billions)	Employment (FICUS, ~ FTE) (millions)
2000	330	744	11.15	191	437	7.75
2001	333	700	11.62	191	461	8.10
2002	341	660	11.55	191	476	8.27
2003	347	683	11.58	191	492	8.32
2004	357	742	12.02	191	513	8.46
2005	360	981	11.80	191	533	8.47
2006	378	813	12.16	191	563	8.60
Average	349	760	11.70	191	496	8.28

Year	(E) Missing or negative VA, etc			(F) Gross DADS x FICUS files			
	Nb firms (thousands)	Value Added (FICUS) (billions)	Employment (FICUS, ~ FTE) (millions)	Nb firms (thousands)	Value Added (FICUS) (billions)	Jobs (~ heads) (DADS) (millions)	Hours (DADS) (billions)
2000	187	439	7.49	167	412	8.66	12.00
2001	187	462	7.81	167	433	9.07	12.50
2002	187	477	8.00	167	447	9.33	12.81
2003	187	494	8.07	167	463	9.23	12.85
2004	187	518	8.20	167	482	9.29	12.94
2005	187	535	8.23	167	499	9.34	13.13
2006	187	569	8.37	167	528	9.48	13.29
Average	187	499	8.03	167	466	9.20	12.79

Year	(G) Sample with non-GMR firms				(H) Sample with GMR firms only			
	Nb firms (thousands)	Value Added (FICUS) (billions)	Jobs (~ heads) (DADS) (millions)	Hours (DADS) (billions)	Nb firms (thousands)	Value Added (FICUS) (billions)	Jobs (~ heads) (DADS) (millions)	Hours (DADS) (billions)
2000	154	307	6.89	9.40	55	199	4.73	6.28
2001	154	326	7.21	9.81	55	209	4.95	6.54
2002	154	337	7.43	10.06	55	218	5.10	6.71
2003	154	352	7.36	10.11	55	226	5.03	6.74
2004	154	369	7.41	10.20	55	238	5.04	6.77
2005	154	380	7.48	10.43	55	246	5.06	6.90
2006	154	402	7.64	10.61	55	260	5.17	7.00
Average	154	353	7.35	10.09	55	228	5.01	6.71

Sources: FICUS, DADS, BRN, Allègement des cotisations sociales (Reduction in Social Security Contributions). The different panels describe the implications of each of the selection criteria, starting from the full FICUS (fiscal) files.

Table B1 first describes the construction of the estimation (and calibration) sample from the main data

sources presented in Section 3.2.

- Panel (A) begins with the FICUS files, a widely used dataset compiling fiscal declarations (IS/BRN, IS/RSI, IR/BRN, IR/RSI, BNC/DECC) for virtually all taxable organizations, with the notable exception of the MICRO tax scheme.⁴ It contains on average 2.4 million observations per year.
- Panel (B) excludes sole proprietors (code 1), who are not organized as firms, and restricts the sample to for-profit entities. This restriction is required by our focus on profit/cost-minimizing firms and implies excluding non-profit entities (codes 0, 2, 4, and 6–9) as well as foreign legal entities (code 3) operating under a different legal framework. The resulting sample consists solely of firms classified under legal code 5.

Code	French	English
0	Organisme de placement collectif en valeurs mobilières sans personnalité morale	Undertaking for Collective Investment in Transferable Securities (UCITS) without legal personality
1	Entrepreneur individuel	Sole trader / Individual entrepreneur
2	Groupeement de droit privé non doté de la personnalité morale	Private-law association without legal personality
3	Personne morale de droit étranger	Foreign legal entity
4	Personne morale de droit public soumise au droit commercial	Public-law legal entity subject to commercial law
5	Société commerciale	Commercial company (e.g. Public Limited Company, Private Limited Company)
6	Autre personne morale immatriculée au RCS	Other legal entity registered in the Trade and Companies Register (RCS)
7	Personne morale et organisme soumis au droit administratif	Legal entity or body governed by administrative law
8	Organisme privé spécialisé	Specialized private organization
9	Groupeement de droit privé	Private-law association or consortium

The sample of relevant, profit-maximizing business organizations contains on average 1 million observations per year.

- Panel (C) restricts the sample to firms with at least five employees, ensuring that we analyze non-degenerate organizations. Because the size distribution of firms is close to Pareto, this step removes many small firms and reduces the number of observations by roughly two-thirds, while having little impact on coverage in terms of value added. The sample size reduces to 349 thousand observations per year, but it covers more than 90% of value added and employment.
- Panel (D) further restricts the sample to firms active in 2000 and continuously present until 2006. This serves two purposes. First, it ensures that all firms were subject to the same institutional framework and shared the same information set regarding the evolution of the GMR, as determined by the January 2000 Aubry 2 Law (Section 2, Appendix A). Second, it ensures that our sample consists of firms with a minimum degree of maturity, allowing us to abstract from organizational dynamics related to early growth or exit, which are not the focus of this paper. This restriction retains 55% of the firms in Panel (C), representing 65% of value added and 70% of employment.

Note that entries and exits, i.e. the firm selection margin, are nonetheless fully accounted for in the calibration exercise, as the proportion of firms created over a seven-year period and the share of workers in these “new” firms are introduced as data moments to identify parameters δ , f_E and f as well as the distribution of α (and are thus critical to the endogenous determination of the selection threshold, $\bar{\alpha}$, see Sections 4 and 5.1).

- Panels (E), (F) and (G) involve minor cleaning steps that have negligible effects on estimation results (see replication programs). These include ensuring non-missing and positive values for sales, value added, and capital stock; restricting the sample to firms covered by both FICUS and DADS (which are theoretically exhaustive but in practice sometimes inconsistent); and at the job level, excluding records with missing hours or wages, as well as home-based workers.

⁴As a consequence, the FICUS files (unlike the later, improved FARE datasets) provide only partial coverage of self-employed individuals; see Appendix C.

- Panel (H) defines our main estimation sample. It includes only firms covered by the “Allègement des cotisations sociales” (Reduction in Social Security Contributions) dataset. This guarantees that all firms in the sample adopted the 35-hour workweek by 2002 – thus harmonizing workweek duration across sample firms – and benefited from the Aubry 1 or 2 subsidies, ensuring that labor costs can be computed accurately at the job level. In contrast, firms outside GMR1 to 5 either transitioned to 35 hours during the regression period – likely implying organizational changes not of direct interest – or effectively remained at 39 hours under the 2003 Fillon Law.⁵

B.2 Detailed Descriptive Statistics

Tables B2 and B3 describe the main characteristics of firms in 2002, the year preceding the minimum-wage reconvergence analyzed in our regressions.⁶ Beyond documenting the sample, these statistics illustrate the two sources of variation underlying our empirical strategy: differences in statutory minimum wages across GMRs, exploited by the DD specification, and differences in firms’ initial exposure to minimum-wage increases, exploited by the DDD specification.

Table B2 first reports descriptive statistics by GMR. As discussed in Section 2, firms were not randomly assigned to the different GMRs. Consistent with this institutional background, firms in GMR1 and GMR2 are, on average, larger, more productive, and more hierarchical than firms in the remaining GMRs. Although they face lower statutory minimum wages, they also pay higher wages throughout the wage distribution and provide substantially more training, both to production workers and managers. These systematic differences strongly suggest that firms self-selected into GMRs, making the parallel-trends assumption underlying a simple DD specification less plausible and motivating the within-GMR identification strategy developed in Section 3.1.

Table B3 instead groups firms according to their initial exposure to minimum-wage increases, measured by the share of workers in 2002 earning less than the unified 2006 minimum wage. Perhaps surprisingly, this dimension of treatment intensity is only weakly related to GMR assignment. While GMR1 firms are 8 percentage points more likely than the average firm to belong to the lowest exposure quartile and GMR5 firms are 9.5 percentage points less likely to do so, GMR2 to GMR4 display much more balanced exposure distributions. Accordingly, every GMR is represented in each exposure quartile, providing substantial within-GMR variation to identify the DDD specification. Table B3 also shows that firm size, measured either by employment or by the number of hierarchical layers, is more strongly associated with minimum-wage exposure than with GMR affiliation, whereas productivity and labor costs exhibit the opposite pattern.

Lastly, Tables B4 to B6 provide information about the industry composition of our estimation sample, together with its correlation with the two previous variables: GMRs and exposure to minimum wage constraints. All industries are represented in all GMRs, although their respective industry structure exhibits significant heterogeneity. Most notably, extractive and manufacturing industries tend to be over-represented in GMR1 and 2, while hotels and restaurants concentrate in GMR3 to 5. Conversely, manufacturing industries are over-represented in the first quartile of firms least exposed to minimum wages, while hotels and restaurants are over-represented in the two intermediate quartiles and the extractive industries as well as business services are over-represented in the upper quartile of minimum wage exposure.

⁵As explained above, this law allowed late adopters to remain at 39 hours by paying overtime at higher rates, which varied by firm size (10% for firms with fewer than 20 employees; 25% for firms with more than 20 employees; 50% above eight overtime hours). These differences in working times, that are impossible to track in our data, and the potential for endogenous cost manipulation justify their exclusion from our regression analyses (see the robustness checks proposed in the replication programs).

⁶The regression tables report the average changes in these variables over the estimation period, 2003–2006.

Table B2: Descriptive Statistics by GMR, 2002

GMR # Observations	GMR1 3,501	GMR2 14,810	GMR3 10,945	GMR4 25,220	GMR5 869	All firms 55,345
Jobs	245 (5,866)	178 (1,065)	88 (281)	24 (102)	41 (205)	92 (1,583)
Hours/1607	166 (2356)	148 (779)	79 (256)	21 (97)	34 (155)	75 (731)
Value Added (10^6€)	7.535 (74.326)	8.271 (47.283)	3.957 (19.137)	0.960 (10.365)	1.380 (5.494)	3.932 (32.853)
Value Added (10^3€) / hour	0.0313 (0.0688)	0.0331 (0.189)	0.0298 (0.0219)	0.0275 (0.0669)	0.0257 (0.0127)	0.0297 (0.109)
Total factor productivity, Accounting method, hours	0.843 (0.281)	0.836 (0.505)	0.886 (0.598)	0.842 (0.308)	0.837 (0.325)	0.849 (0.434)
Total factor productivity, LevPet method, hours	5.636 (33.54)	4.636 (17.97)	4.009 (25.94)	3.113 (11.62)	3.044 (2.847)	3.849 (18.75)
Number of layers per firm	2.196 (0.773)	2.176 (0.778)	2.027 (0.790)	1.562 (0.865)	1.636 (0.888)	1.860 (0.870)
Share of 0 layer firms	0.030	0.030	0.036	0.113	0.113	0.070
Share of 1 layer firms	0.131	0.139	0.191	0.350	0.304	0.248
Share of 2 layer firms	0.453	0.455	0.483	0.399	0.418	0.434
Share of 3 layer firms	0.386	0.376	0.290	0.138	0.166	0.248
Layer 0:	0.643	0.687	0.671	0.707	0.708	0.690
Share in jobs	(0.254)	(0.240)	(0.246)	(0.245)	(0.244)	(0.245)
Layer 1	0.233	0.208	0.223	0.211	0.204	0.214
Share in jobs	(0.175)	(0.165)	(0.179)	(0.182)	(0.175)	(0.177)
Layer 2:	0.113	0.096	0.098	0.075	0.080	0.088
Share in jobs	(0.142)	(0.135)	(0.130)	(0.117)	(0.123)	(0.127)
Layer 3:	0.010	0.009	0.008	0.008	0.008	0.008
Share in jobs	(0.022)	(0.019)	(0.020)	(0.025)	(0.022)	(0.022)
Layer 0:	0.627	0.670	0.652	0.684	0.689	0.671
Share in hours	(0.258)	(0.241)	(0.249)	(0.251)	(0.248)	(0.249)
Layer 1:	0.241	0.218	0.236	0.225	0.217	0.226
Share in hours	(0.176)	(0.166)	(0.184)	(0.188)	(0.179)	(0.181)
Layer 2:	0.120	0.102	0.102	0.082	0.085	0.094
Share in hours	(0.149)	(0.139)	(0.132)	(0.125)	(0.128)	(0.132)
Layer 3:	0.012	0.010	0.010	0.009	0.009	0.010
Share in hours	(0.026)	(0.021)	(0.024)	(0.029)	(0.027)	(0.026)
Hourly labor cost at GMR level	7.803 (0.171)	8.342 (0.147)	8.725 (0.131)	8.830 (0.126)	8.520 (0.132)	8.609 (0.322)
Gross hourly wage at GMR level (legal, regulated concept)	7.177 (0)	7.266 (0)	7.391 (0)	7.423 (0)	6.917 (0)	7.352 (0.111)
Share of firms with MW workers representing:						
$[0; \frac{1}{4}]$ of total workforce	0.885	0.817	0.806	0.786	0.710	0.804
$[\frac{1}{4}; \frac{1}{2}]$ of total workforce	0.073	0.100	0.111	0.135	0.155	0.117
$[\frac{1}{2}; \frac{3}{4}]$ of total workforce	0.031	0.068	0.074	0.064	0.104	0.065
$[\frac{3}{4}; 1]$ of total workforce	0.012	0.016	0.008	0.015	0.031	0.014
Share of 2002 workers below the 2006 min. wage	0.094 (0.157)	0.132 (0.195)	0.135 (0.190)	0.144 (0.197)	0.194 (0.229)	0.136 (0.194)
Hourly labor cost:	18.750	18.490	17.850	17.480	16.810	17.890
Average	(6.165)	(6.653)	(5.949)	(6.228)	(5.622)	(6.297)
Hourly labor cost:	14.020	14.030	13.660	13.550	13.100	13.720
Layer 0	(3.331)	(3.509)	(3.069)	(3.088)	(2.790)	(3.222)
Hourly labor cost:	21.160	21.980	21.070	21.970	20.500	21.710
Layer 1	(6.081)	(9.301)	(8.032)	(9.401)	(7.564)	(8.904)
Hourly labor cost:	37.760	39.740	37.910	37.820	37.100	38.450
Layer 2	(12.610)	(15.560)	(14.060)	(15.940)	(14.410)	(15.170)
Hourly labor cost:	70.250	72.290	67.390	58.590	60.940	67.350
Layer 3	(35.670)	(38.740)	(35.220)	(30.370)	(40.320)	(36.090)
Training programs (Y/N)	0.455	0.443	0.417	0.273	0.392	0.412
Training at layer L_0	0.340	0.317	0.297	0.178	0.236	0.294
Training at layers L_1 to L_3	0.340	0.334	0.309	0.189	0.296	0.308

Source: Estimation sample, constructed from matched DADS, FICUS, BRN, Allègement des cotisations sociales (Reduction in Social Security Contributions) and “CERFA 2483” (French Ministry of Labor) files. In each cell of the table, we report averages together with standard deviations (in parentheses). In this table, the “Gross hourly wage at GMR level” has no within-GMR variation, since it corresponds exactly to the legally binding GMR value.

Table B3: Descriptive Statistics by Sub-samples of Varying Shares of Low-Wage Workers, 2002

Share of LW workers: # Observations	$[0; \frac{1}{4}[$ 44,475	$[\frac{1}{4}; \frac{1}{2}[$ 6,481	$[\frac{1}{2}; \frac{3}{4}[$ 3,617	$[\frac{3}{4}; 1]$ 772	All firms 55,345
Jobs	100 (1,758)	58 (274)	71 (441)	37 (66)	92 (1,583)
Hours/1607	84 (810)	41 (164)	44 (238)	25 (39)	75 (731)
Value Added (10^6€)	4.593 (36.531)	1.316 (4.843)	1.221 (6.070)	561 (752)	3.932 (32.854)
Value Added (10^3€) / hour	0.032 (0.122)	0.022 (0.016)	0.019 (0.007)	0.017 (0.008)	0.030 (0.109)
Total factor productivity, Accounting method, hours	0.862 (0.461)	0.812 (0.300)	0.772 (0.274)	0.776 (0.409)	0.849 (0.434)
Total factor productivity, LevPet method, hours	3.827 (19.51)	4.110 (17.49)	3.798 (12.32)	3.177 (5.249)	3.849 (18.75)
Number of layers	1.960 (0.836)	1.538 (0.857)	1.411 (0.892)	0.861 (0.845)	1.860 (0.870)
Share of 0 layer firms	0.052	0.109	0.158	0.389	0.070
Share of 1 layer firms	0.214	0.376	0.393	0.408	0.248
Share of 2 layer firms	0.455	0.381	0.330	0.157	0.434
Share of 3 layer firms	0.279	0.133	0.119	0.047	0.248
Layer 0:	0.650	0.823	0.891	0.937	0.690
Share in jobs	(0.249)	(0.150)	(0.102)	(0.088)	(0.245)
Layer 1:	0.238	0.140	0.089	0.055	0.214
Share in jobs	(0.181)	(0.129)	(0.092)	(0.081)	(0.177)
Layer 2:	0.102	0.033	0.018	0.007	0.088
Share in jobs	(0.135)	(0.054)	(0.030)	(0.022)	(0.127)
Layer 3:	0.010	0.004	0.003	0.001	0.008
Share in jobs	(0.024)	(0.014)	(0.009)	(0.004)	(0.022)
Layer 0:	0.632	0.795	0.863	0.928	0.671
Share in hours	(0.253)	(0.160)	(0.116)	(0.104)	(0.249)
Layer 1:	0.248	0.162	0.111	0.063	0.226
Share in hours	(0.185)	(0.139)	(0.106)	(0.098)	(0.181)
Layer 2:	0.109	0.039	0.023	0.008	0.094
Share in hours	(0.141)	(0.060)	(0.037)	(0.024)	(0.132)
Layer 3:	0.011	0.004	0.004	0.001	0.010
Share in hours	(0.028)	(0.016)	(0.013)	(0.006)	(0.026)
Hourly labor cost at GMR level	8.599 (0.329)	8.650 (0.294)	8.653 (0.272)	8.614 (0.297)	8.609 (0.322)
Gross hourly wage at GMR level (legal, regulated concept)	7.351 (0.111)	7.358 (0.113)	7.347 (0.114)	7.335 (0.125)	7.352 (0.111)
Share of firms:					
in GMR1	0.070	0.040	0.030	0.048	0.063
in GMR2	0.272	0.228	0.276	0.302	0.268
in GMR3	0.198	0.187	0.225	0.118	0.198
in GMR4	0.446	0.525	0.444	0.497	0.456
in GMR5	0.014	0.021	0.025	0.035	0.016
Share of 2002 workers below the 2006 min. wage	0.053 (0.067)	0.362 (0.072)	0.608 (0.069)	0.836 (0.071)	0.136 (0.194)
Hourly labor cost:	19.250	13.140	11.430	9.911	17.890
Average	(6.228)	(2.256)	(1.392)	(1.052)	(6.297)
Hourly labor cost:	14.510	10.940	9.994	9.233	13.720
Layer 0	(3.081)	(1.014)	(0.696)	(0.615)	(3.222)
Hourly labor cost:	22.290	19.280	18.690	19.070	21.710
Layer 1	(8.888)	(8.330)	(8.578)	(9.696)	(8.904)
Hourly labor cost:	38.960	35.490	34.730	35.190	38.450
Layer 2	(14.990)	(16.170)	(15.450)	(16.300)	(15.170)
Hourly labor cost:	68.620	57.080	52.880	51.470	67.350
Layer 3	(36.510)	(28.670)	(31.220)	(22.600)	(36.090)
Training programs (Y/N)	0.424	0.318	0.267	0.214	0.412
Training at layer L_0	0.311	0.161	0.106	0.029	0.294
Training at layers L_1 to L_3	0.324	0.166	0.104	0.053	0.308

Source: Estimation sample, constructed from matched DADS, FICUS, BRN, Allègement des cotisations sociales (Reduction in Social Security Contributions) and “CERFA 2483” (French Ministry of Labor) files. The variable defining the table columns corresponds to the “share of low-wage workers”, as measured in 2002 by the total number of hours worked by employees earning in 2002 less than the 2006 minimum wage (in nominal terms). In each cell of the table, we report averages together with standard deviations (in parentheses).

Table B4: Industry Structure of the Estimation Sample, by GMR Group

	NACE 1-digit	GMR1 3,501	GMR2 14,810	GMR3 10,945	GMR4 25,220	GMR5 869	All GMR 55,345
Extractive industries, food, textiles, wood	1	0.092	0.086	0.067	0.053	0.062	0.067
Manufacturing industries: paper, chemicals, minerals, metals	2	0.188	0.179	0.184	0.126	0.115	0.155
Manufacturing industries: equipment, machinery, transportation	3	0.085	0.060	0.060	0.044	0.037	0.054
Electricity, gas, water, construction	4	0.113	0.097	0.088	0.146	0.110	0.119
Trade, hotels, restaurants	5	0.207	0.268	0.385	0.400	0.435	0.350
Transport, communications, finance	6	0.037	0.056	0.037	0.022	0.039	0.035
Real estate, business services, administration	7	0.222	0.177	0.110	0.130	0.140	0.145
Education, health, social action	8	0.029	0.056	0.040	0.036	0.046	0.042
Other services (personal, domestic)	9	0.027	0.021	0.028	0.042	0.015	0.032
All industries		1	1	1	1	1	1

Source: DADS files.

Table B5: Industry Structure of the Estimation Sample, by Share of Low-Wage Workers

	NACE 1-digit	$[0; \frac{1}{4}[$ 44,475	$[\frac{1}{4}; \frac{1}{2}[$ 6,481	$[\frac{1}{2}; \frac{3}{4}[$ 3,617	$[\frac{3}{4}; 1]$ 772	All firms 55,345
Extractive industries, food, textiles, wood	1	0.063	0.093	0.062	0.132	0.067
Manufacturing industries: paper, chemicals, minerals, metals	2	0.180	0.074	0.030	0.023	0.155
Manufacturing industries: equipment, machinery, transportation	3	0.060	0.040	0.014	0.021	0.054
Electricity, gas, water, construction	4	0.143	0.029	0.006	0.006	0.119
Trade, hotels, restaurants	5	0.290	0.570	0.671	0.499	0.350
Transport, communications, finance	6	0.041	0.016	0.008	0.006	0.035
Real estate, business services, administration	7	0.152	0.083	0.147	0.216	0.145
Education, health, social action	8	0.042	0.045	0.032	0.038	0.042
Other services (personal, domestic)	9	0.029	0.050	0.032	0.058	0.032
All industries		1	1	1	1	1

Source: DADS files. In this table, low-wage workers are defined as workers earning in 2002 less than the 2006 minimum wage.

Table B6: Share of Low-Wage Workers, by Industry, in the Estimation Sample

	NACE 1-digit	$[0; \frac{1}{4}[$ 44,866	$[\frac{1}{4}; \frac{1}{2}[$ 6,293	$[\frac{1}{2}; \frac{3}{4}[$ 3,473	$[\frac{3}{4}; 1]$ 712	All firms 55,344
Extractive industries, food, textiles, wood	1	0.751	0.161	0.060	0.027	1
Manufacturing industries: paper, chemicals, minerals, metals	2	0.930	0.056	0.012	0.002	1
Manufacturing industries: equipment, machinery, transportation	3	0.890	0.088	0.017	0.005	1
Electricity, gas, water, construction	4	0.968	0.028	0.003	0.001	1
Trade, hotels, restaurants	5	0.664	0.191	0.125	0.020	1
Transport, communications, finance	6	0.931	0.052	0.015	0.003	1
Real estate, business services, administration	7	0.845	0.068	0.067	0.021	1
Education, health, social action	8	0.812	0.127	0.049	0.012	1
Other services (personal, domestic)	9	0.728	0.183	0.064	0.025	1
All industries		0.804	0.117	0.065	0.014	1

Source: DADS files. In this table, low-wage workers are defined as workers earning in 2002 less than the 2006 minimum wage.

C Empirical Results: Robustness Checks and Extensions

C.1 Production Hierarchies in the Estimation Sample

The analysis in Section 3.4 relies on the mapping between the French occupational classification and hierarchical layers proposed by Caliendo, Monte, et al. (2015). This appendix documents the organizational structures obtained using this procedure and assesses whether they exhibit the properties of production hierarchies assumed in our theoretical framework. Table C1 shows that the resulting organizations overwhelmingly display the pyramidal structure predicted by hierarchical production models. Panels (A) and (B) report the share of firms in which each hierarchical layer is at least as large as the layer immediately above it, using jobs and hours, respectively, as measures of layer size. Measured in jobs, more than 90% of firms with one managerial layer have more production workers than managers. Among firms with two managerial layers, about 72% satisfy all pyramid conditions simultaneously, while the corresponding figure is 66% for firms with three managerial layers. The evidence is qualitatively similar when layer size is measured using hours rather than jobs, although the hierarchy conditions are satisfied somewhat less frequently, reflecting heterogeneity in working hours across layers. Panel (C) reports the share of firms in which hourly labor costs increase monotonically across hierarchical layers. This property is even more prevalent than the size hierarchy: it holds for 94% of firms with one managerial layer, 90% of firms with two managerial layers, and 82% of firms with three managerial layers. These descriptive patterns provide empirical support for the modeling assumptions adopted in Section 4. In particular, the assumption of cumulative knowledge implies that higher hierarchical layers command (weakly) higher wages, a property that is observed for the overwhelming majority of firms in the French data.

Table C1: Check of Organizational Hierarchy Conditions

(A) Using Jobs (2003)				
Condition tested:	$N_L^\ell \geq N_L^{\ell+1}$ for all ℓ	$N_L^0 \geq N_L^1$	$N_L^1 \geq N_L^2$	$N_L^2 \geq N_L^3$
1-layer firms	0.921	0.921		
2-layer firms	0.722	0.825	0.874	
3-layer firms	0.658	0.809	0.831	0.968

(B) Using Hours (2003)				
Condition tested:	$h_L^\ell \geq h_L^{\ell+1}$ for all ℓ	$h_L^0 \geq h_L^1$	$h_L^1 \geq h_L^2$	$h_L^2 \geq h_L^3$
1-layer firms	0.901	0.901		
2-layer firms	0.621	0.796	0.793	
3-layer firms	0.537	0.787	0.794	0.877

(C) Using Hourly Labor Costs (2003)				
Condition tested:	$w_L^{\ell+1} \geq w_L^\ell$ for all ℓ	$w_L^1 \geq w_L^0$	$w_L^2 \geq w_L^1$	$w_L^3 \geq w_L^2$
1-layer firms	0.940	0.940		
2-layer firms	0.899	0.947	0.951	
3-layer firms	0.822	0.972	0.962	0.881

Notes: The table reports the share of firms satisfying the hierarchy conditions of Caliendo, Monte, et al. (2015). Panels (A) and (B) consider the condition that each hierarchical layer is at least as large as the layer immediately above it, measured using the number of jobs and total hours, respectively. Panel (C) reports the share of firms for which average hourly labor costs (measured by total labor costs per hour) increase monotonically across hierarchical layers.

C.2 Heterogeneity by Initial Organizational Structure

The baseline specifications reported in Tables 1 and 2 pool firms with different initial organizational structures. While the regressions are estimated separately for each hierarchical layer, each estimation sample includes firms with one, two, and three managerial layers. This appendix assesses the robustness of the baseline results by estimating the specifications separately for firms with one, two, and three managerial layers that are observed in both 2003 and 2006, thereby eliminating any composition effects arising from pooling firms with different organizational structures.

Table C2 reports the corresponding estimates for hourly labor costs. The qualitative patterns documented in the main text remain remarkably stable across organizational structures. Minimum-wage increases primarily affect labor costs in the lowest hierarchical layer, while the estimated spillovers to upper layers remain limited. Although statistical precision decreases as the sample is partitioned, the magnitude and sign of the estimated effects are broadly consistent with the baseline results. Tables C3 and C4 report analogous estimates for the size of hierarchical layers, measured respectively in jobs and in hours. Across organizational structures, the adjustment continues to be concentrated in the production layer, with comparatively small changes in managerial employment. Overall, these results indicate that the organizational responses documented in the main text are not driven by changes in the composition of firms across hierarchical structures, but characterize firms with different initial numbers of managerial layers.

Table C2: Heterogeneity by Organizational Structure: Hourly Labor Costs

$\Delta_{03/06}$ ln hourly labor cost at:	(1) All layers	(2) Layer 0 (A) 1-layer firms (16,195 obs.)	(3) Layer 1	(4) Layer 2	(5) Layer 3
DD: Δ ln lab. cost at GMR	0.299*** (0.047)	0.221*** (0.056)	0.001 (0.130)		
DDD: Δ ln lab. cost at GMR $\times$ Share MW workers in 2002	0.020 (0.237)	0.501** (0.234)	-0.360 (0.696)		
(B) 2-layer firms (24,613 obs.)					
DD: Δ ln lab. cost at GMR	0.374*** (0.030)	0.313*** (0.034)	0.306*** (0.055)	0.402*** (0.076)	
DDD: Δ ln lab. cost at GMR $\times$ Share MW workers in 2002	0.193 (0.204)	0.342* (0.197)	-0.838** (0.426)	-1.455** (0.672)	
(C) 3-layer firms (8,484 obs.)					
DD: Δ ln lab. cost at GMR	0.440*** (0.048)	0.362*** (0.053)	0.335*** (0.069)	0.350*** (0.094)	0.221 (0.150)
DDD: Δ ln lab. cost at GMR $\times$ Share MW workers in 2002	-0.323 (0.392)	0.277 (0.392)	-0.750 (0.803)	-0.259 (0.902)	-0.127 (1.210)

Notes: See Table 1.

Table C3: Heterogeneity by Organizational Structure: Relative Layer Sizes in Jobs

$\Delta \ln \text{ jobs, at:}$	(1) All layers	(2) Layer 0	(3) Layer 1	(4) Layer 2	(5) Layer 3
(A) 1-layer firms (16,195 obs.)					
DD: $\Delta \ln \text{ lab. cost at GMR}$	-0.075 (0.107)	0.021 (0.165)	-0.158 (0.224)		
DDD: $\Delta \ln \text{ lab. cost at GMR}$ $\times \text{ Sh MW workers in 2002}$	-0.387 (0.595)	-0.456 (0.790)	0.007 (1.044)		
(B) 2-layer firms (24,613 obs.)					
DD: $\Delta \ln \text{ lab. cost at GMR}$	-0.319*** (0.076)	-0.612*** (0.117)	-0.044 (0.143)	-0.248* (0.138)	
DDD: $\Delta \ln \text{ lab. cost at GMR}$ $\times \text{ Sh MW workers in 2002}$	-0.433 (0.547)	0.152 (0.752)	-0.310 (1.062)	0.825 (0.886)	
(C) 3-layer firms (8,484 obs.)					
DD: $\Delta \ln \text{ lab. cost at GMR}$	-0.190 (0.118)	-0.498** (0.195)	-0.063 (0.213)	0.019 (0.210)	-0.008 (0.159)
DDD: $\Delta \ln \text{ lab. cost at GMR}$ $\times \text{ Sh MW workers in 2002}$	-1.890 (1.254)	-0.868 (1.596)	-1.695 (2.218)	0.206 (2.039)	0.546 (1.270)

Notes: See Table 1.

Table C4: Heterogeneity by Organizational Structure: Relative Layer Sizes in Hours

Layers measured in hours

$\Delta \ln \text{ jobs, at:}$	(1) All layers	(2) Layer 0	(3) Layer 1	(4) Layer 2	(5) Layer 3
(A) 1-layer firms (16,195 obs.)					
DD: $\Delta \ln \text{ lab. cost at GMR}$	-0.113 (0.102)	-0.031 (0.186)	-0.107 (0.264)		
DDD: $\Delta \ln \text{ lab. cost at GMR}$ $\times \text{ Sh MW workers in 2002}$	-0.375 (0.561)	0.030 (0.843)	-0.251 (1.337)		
(B) 2-layer firms (24,613 obs.)					
DD: $\Delta \ln \text{ lab. cost at GMR}$	-0.242*** (0.074)	-0.502*** (0.127)	-0.037 (0.156)	-0.244 (0.171)	
DDD: $\Delta \ln \text{ lab. cost at GMR}$ $\times \text{ Sh MW workers in 2002}$	-0.889* (0.529)	-0.878 (0.777)	-0.004 (1.234)	1.946 (1.287)	
(C) 3-layer firms (8,484 obs.)					
DD: $\Delta \ln \text{ lab. cost at GMR}$	-0.212* (0.113)	-0.593*** (0.205)	-0.149 (0.214)	0.074 (0.236)	0.104 (0.277)
DDD: $\Delta \ln \text{ lab. cost at GMR}$ $\times \text{ Sh MW workers in 2002}$	-1.980* (1.146)	-1.028 (1.593)	-2.856 (2.292)	5.083* (3.002)	-1.617 (2.050)

Notes: See Table 1.

D Regression Results:

Alternative Estimation Strategies and Specifications

D.1 Instrumental Variable Specifications

Tables D1–D4 re-estimate all baseline specifications using the IV strategy described in Footnote 24, which instruments changes in firms’ labor costs with the statutory evolution of the corresponding GMR schedules. The first stage is extremely strong in both the DD and DDD specifications. More importantly, the IV estimates closely resemble the baseline OLS results throughout the paper. The effects of minimum-wage increases on labor costs, firms’ organizational structures, training, employment, and productivity remain qualitatively unchanged, while the magnitudes of the estimated coefficients are generally similar and, in several cases, somewhat larger in absolute value. This pattern is consistent with the presence of classical measurement error in our reconstructed labor-cost measure, which would attenuate the baseline OLS estimates. Overall, these results indicate that the paper’s conclusions are robust to instrumenting labor-cost changes and are unlikely to be driven by measurement error in the reconstructed labor-cost variable.

Table D1: Impact of Minimum Wage Increases on Labor Costs Across Firms’ Organizations

	(1)	(2)	(3)	(4)	(5)
$\Delta_{03/06}$ hourly labor cost at:	All layers	Layer 0	Layer 1	Layer 2	Layer 3
Observations	55,345	55,345	49,480	33,203	8,484
IV-DD: $\Delta \ln$ lab. cost at GMR	0.234*** (0.028)	0.0987*** (0.031)	-0.0217 (0.061)	0.265*** (0.077)	0.049 (0.184)
IV-DDD: $\Delta \ln$ lab. cost at GMR × Share MW workers in 2002	0.006 (0.162)	0.413*** (0.155)	-0.292 (0.453)	-1.244* (0.691)	-1.125 (1.502)

Notes: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Standard errors clustered at the 1-digit sector-commuting zone level in parentheses. All specifications contain 4-digit industry fixed effects and commuting zone fixed effects. Coefficients of main interest from Equations 3 (line DD) and 4 (line DDD), estimated by 2SLS, using the increase in legal GMRs as an instrumental variable for the evolution of the corresponding total labor costs. Value of the Stock-Yogo Cragg-Donald Wald F -statistic for the DD specification: 1.0e+05. Value of the Stock-Yogo Cragg-Donald Wald F -statistic for the DDD specification: 4.6e+04. The composition of the estimation sample evolves across columns: in the first two columns, all sample firms are included. In columns (3) to (5), only firms exhibiting at least 2 (Layers 0 and 1), 3 or 4 layers respectively, both in 2003 and 2006, are included.

Table D2: Impact of Minimum Wage Increases on Firms' Organizations

(A) Impact on the Relative Size of Hierarchical Layers

(A1) $\Delta \ln$ jobs, at: Observations	(1) All layers 55,345	(2) Layer 0 55,345	(3) Layer 1 49,480	(4) Layer 2 33,203	(5) Layer 3 8,484
IV-DD: $\Delta \ln$ lab. cost at GMR	-0.403*** (0.069)	-0.839*** (0.104)	-0.115 (0.134)	-0.172 (0.142)	-0.176 (0.192)
IV-DDD: $\Delta \ln$ lab. cost at GMR $\times$ Sh MW workers in 2002	-1.320*** (0.480)	-0.566 (0.581)	-0.726 (0.845)	-0.349 (0.987)	0.098 (1.489)
(A2) $\Delta \ln$ hours, at: Observations	(1) All layers 55,345	(2) Layer 0 55,345	(3) Layer 1 49,480	(4) Layer 2 33,203	(5) Layer 3 8,484
IV-DD: $\Delta \ln$ lab. cost at GMR	-0.322*** (0.068)	-0.760*** (0.113)	-0.007 (0.148)	-0.098 (0.173)	-0.155 (0.321)
IV-DDD: $\Delta \ln$ lab. cost at GMR $\times$ Sh MW workers in 2002	-1.073** (0.474)	-0.353 (0.597)	-0.919 (1.041)	2.214 (1.382)	-1.635 (2.545)

(B) Impact on the Number of Hierarchical Layers

(55,345 obs.)	(1) Δ nb layers	(2) Prob. increase	(3) Prob. decrease	(4) Prob. stability	(5) Δ Proba 3 layers	(6) Δ Proba ≥ 2 layers	(7) Δ Proba ≥ 1 layer
IV-DD: $\Delta \ln$ lab. cost at GMR	-0.193 (0.133)	-0.516*** (0.074)	-0.297*** (0.081)	0.814*** (0.096)	-0.286*** (0.087)	0.063 (0.078)	0.029 (0.043)
IV-DDD: $\Delta \ln$ lab. cost at GMR $\times$ Sh MW workers in 2002	-1.911** (0.818)	-0.004 (0.469)	1.511*** (0.458)	-1.507*** (0.560)	0.020 (0.416)	-1.150* (0.509)	-0.781** (0.387)

Notes: See Table D1. In Panel (A), the composition of the estimation sample evolves across columns: in the first two columns, all sample firms are included. In columns (3) to (5), only firms exhibiting at least 2 (Layers 0 and 1), 3 or 4 layers respectively, both in 2003 and 2006, are included.

Table D3: Impact of Minimum Wage Increases on Training

Observations	(1) All firms 17,407	(2) 1-layer firms 2,494	(3) 2-layer firms 9,689	(4) 3-layer firms 4,569
(A) Δ Proba. training at layer l_0				
IV-DD: $\Delta \ln$ lab. cost at GMR	1.101*** (0.183)	0.471 (0.565)	0.808*** (0.255)	1.417*** (0.380)
IV-DDD: $\Delta \ln$ lab. cost at GMR $\times$ Sh MW workers in 2002	1.471 (1.144)	3.223 (2.556)	2.682 (1.706)	-0.989 (3.340)
(B) Δ Proba. training at layers l_1 to l_3				
IV-DD: $\Delta \ln$ lab. cost at GMR	1.118*** (0.178)	0.687 (0.512)	0.990*** (0.253)	1.017*** (0.367)
IV-DDD: $\Delta \ln$ lab. cost at GMR $\times$ Sh MW workers in 2002	-0.080 (1.112)	0.146 (2.334)	0.763 (1.664)	-1.684 (3.324)

Notes: See Table D1. Both in Panels (A) and (B), the composition of the estimation sample evolves across columns: in the first column, all sample firms are included. In columns (2) to (4), only firms exhibiting at least 2 (Layers 0 and 1), 3 or 4 layers respectively, both in 2003 and 2006, are included.

Table D4: Impact of Minimum Wage Increases on Firms' Size and Productivity

(A) Impact on Firms' Size

	(1) $\Delta \ln$ # jobs	(2) $\Delta \ln$ hours	(3) $\Delta \ln$ VA	(4) $\Delta \ln$ Sales
(55,345 obs.)				
IV-DD: $\Delta \ln$ lab. cost at GMR	-0.403*** (0.069)	-0.322*** (0.068)	-0.079 (0.084)	-0.031 (0.069)
IV-DDD: $\Delta \ln$ lab. cost at GMR × Sh MW workers in 2002	-1.320*** (0.480)	-1.073** (0.474)	0.166 (0.520)	-0.103 (0.437)

(B) Impact on Production Factors beyond Labor

	(1) $\Delta \ln$ inputs/jobs 55,329	(2) $\Delta \ln$ cap/jobs 55,345	(3) $\Delta \ln$ inv/jobs 47,181	(4) $\Delta \ln$ inputs/hrs 55,329	(5) $\Delta \ln$ cap/hrs 55,345	(6) $\Delta \ln$ inv/hrs 47,181
Observations						
IV-DD: $\Delta \ln$ lab. cost at GMR	0.387*** (0.081)	-0.022 (0.102)	0.287 (0.336)	0.306*** (0.077)	-0.103 (0.100)	0.236 (0.338)
IV-DDD: $\Delta \ln$ lab. cost at GMR × Sh MW workers in 2002	0.672 (0.520)	2.223*** (0.616)	3.358 (2.288)	0.427 (0.495)	1.977*** (0.595)	3.187 (2.304)

(C) Impact on Productivity

	(1) $\Delta \ln$ Labor Productivity	(2) $\Delta \ln$ Labor Productivity	(3) $\Delta \ln$ Labor Productivity	(4) $\Delta \ln$ TFP Accounting	(5) $\Delta \ln$ TFP Lev. Pet.	(6) $\Delta \ln$ TFP Lev. Pet.
Observations	Jobs 55,345	Hours 55,345	Jobs 53,437	Hours 53,437	Jobs 53,437	Hours 53,437
IV-DD: $\Delta \ln$ lab. cost at GMR	0.324*** (0.079)	0.244*** (0.072)	0.063** (0.030)	0.051* (0.029)	0.061* (0.031)	0.068** (0.030)
IV-DDD: $\Delta \ln$ lab. cost at GMR × Sh MW workers in 2002	1.486*** (0.467)	1.239*** (0.419)	0.603*** (0.184)	0.526*** (0.176)	0.185 (0.216)	0.216 (0.196)

Notes: See Table D1.

D.2 Excluding GMR5 Firms

An important aspect of our empirical strategy lies in its ability to exclude the impact of the legal reduction in working hours which occurred in the years preceding our regression period, in order to estimate only the impact of minimum wage increases on our variables of interest. To this end, we restrict the estimation sample to firms with a uniform official working time (35 hours) by focusing on those that concluded reduction agreements by 2002. The fact that these agreements were negotiated (sometimes at length) and the two-stage phasing of the Aubry laws, allowing companies to anticipate the switch to 35 hours, enabled companies to quickly implement the working time reduction measures (Askenazy et al., 2004). In this appendix, however, we assess the robustness of our main results by focusing on firms that had at least one year to “digest” their RWT agreements, thereby excluding companies covered by GMR5, which switched to the 35-hour workweek only in 2002.

Tables D5–D8 confirm that the baseline results are robust to excluding GMR5 firms, despite the loss in statistical power resulting from the smaller estimation sample. The estimated effects on labor costs, the concentration of employment and hours adjustments at the production layer, and the flattening of firms’ organizational hierarchies are all closely aligned with the estimates reported in the main text. The positive responses of training, labor productivity, and accounting-based TFP are likewise preserved, while value added and sales remain largely unaffected. Some DDD estimates—most notably those for aggregate employment and productivity—are smaller and less precisely estimated in the restricted sample, but their signs and economic magnitudes remain consistent with the baseline evidence. These findings indicate that the main results are not driven by firms that moved to the 35-hour workweek in 2002 and are therefore unlikely to capture delayed organizational adjustments to the reduction in working time.

Table D5: Impact of Minimum Wage Increases on Labor Costs Across Firms’ Organizations

	(1)	(2)	(3)	(4)	(5)
$\Delta_{03/06}$ hourly labor cost at:	All layers	Layer 0	Layer 1	Layer 2	Layer 3
Observations	54,474	54,474	48,732	32,782	8,411
DD: $\Delta \ln$ lab. cost at GMR	0.353*** (0.022)	0.291*** (0.024)	0.177*** (0.047)	0.385*** (0.061)	0.234 (0.152)
DDD: $\Delta \ln$ lab. cost at GMR $\times$ Sh MW workers in 2002	-0.106 (0.129)	0.332*** (0.126)	-0.753** (0.359)	-1.365** (0.573)	0.089 (1.217)

Notes: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Standard errors clustered at the 1-digit sector-commuting zone level in parentheses. All specifications contain 4-digit industry fixed effects and commuting zone fixed effects. Coefficients of main interest from Equations 3 (line DD) and 4 (line DDD), estimated by OLS. The composition of the estimation sample evolves across columns: in the first two columns, all sample firms are included. In columns (3) to (5), only firms exhibiting at least 2 (Layers 0 and 1), 3 or 4 layers respectively, both in 2003 and 2006, are included.

Table D6: Impact of Minimum Wage Increases on Firms' Organizations

(A) Impact on the Relative Size of Hierarchical Layers

(A1)	(1)	(2)	(3)	(4)	(5)
$\Delta \ln$ jobs, at:	All layers	Layer 0	Layer 1	Layer 2	Layer 3
Observations	54,474	54,474	48,732	32,782	8,411
DD: $\Delta \ln$ lab. cost at GMR	-0.213*** (0.052)	-0.469*** (0.080)	-0.090 (0.105)	-0.155 (0.114)	-0.018 (0.161)
DDD: $\Delta \ln$ lab. cost at GMR $\times$ Sh MW workers in 2002	-0.510 (0.376)	-0.033 (0.464)	-0.442 (0.681)	0.936 (0.803)	0.697 (1.294)

(A2)	(1)	(2)	(3)	(4)	(5)
$\Delta \ln$ hours, at:	All layers	Layer 0	Layer 1	Layer 2	Layer 3
Observations	54,474	54,474	48,732	32,782	8,411
DD: $\Delta \ln$ lab. cost at GMR	-0.147*** (0.051)	-0.412*** (0.086)	-0.044 (0.116)	-0.095 (0.138)	0.064 (0.280)
DDD: $\Delta \ln$ lab. cost at GMR $\times$ Sh MW workers in 2002	-0.569 (0.372)	-0.047 (0.482)	-0.592 (0.857)	3.113*** (1.192)	-1.395 (2.077)

(B) Impact on the Number of Hierarchical Layers

(54,474 obs.)	(1) Δ nb layers	(2) Prob. increase	(3) Prob. decrease	(4) Prob. stability	(5) Δ Proba 3 layers	(6) Δ Proba ≥ 2 layers	(7) Δ Proba ≥ 1 layer
DD: $\Delta \ln$ lab. cost at GMR	-0.196* (0.104)	-0.308*** (0.058)	-0.082 (0.065)	0.390*** (0.077)	-0.206*** (0.069)	0.016 (0.061)	-0.006 (0.035)
DDD: $\Delta \ln$ lab. cost at GMR $\times$ Sh MW workers in 2002	-1.372** (0.658)	-0.123 (0.373)	0.985*** (0.381)	-0.862* (0.462)	0.229 (0.331)	-0.803** (0.409)	-0.797** (0.325)

Notes: See Table D5. In Panel (A), the composition of the estimation sample evolves across columns: in the first two columns, all sample firms are included. In columns (3) to (5), only firms exhibiting at least 2 (Layers 0 and 1), 3 or 4 layers respectively, both in 2003 and 2006, are included.

Table D7: Impact of Minimum Wage Increases on Training

	(1)	(2)	(3)	(4)
Observations	All firms 17,214	1-layer firms 2,446	2-layer firms 9,580	3-layer firms 4,539
(A) Δ Proba. training at layer l_0				
DD: $\Delta \ln$ lab. cost at GMR	0.599*** (0.149)	0.573 (0.480)	0.364* (0.207)	0.845*** (0.316)
DDD: $\Delta \ln$ lab. cost at GMR $\times$ Sh MW workers in 2002	0.652 (0.969)	3.620 (2.210)	0.396 (1.416)	-0.281 (2.644)
(B) Δ Proba. training at layers l_1 to l_3				
DD: $\Delta \ln$ lab. cost at GMR	0.571*** (0.145)	0.334 (0.462)	0.418** (0.203)	0.444 (0.294)
DDD: $\Delta \ln$ lab. cost at GMR $\times$ Sh MW workers in 2002	0.055 (0.935)	3.011 (2.208)	-0.569 (1.384)	-0.763 (2.578)

Notes: See Table D5.

Table D8: Impact of Minimum Wage Increases on Firms' Size and Productivity

(A) Impact on Firms' Size

	(1) $\Delta \ln$ # jobs	(2) $\Delta \ln$ hours	(3) $\Delta \ln$ VA	(4) $\Delta \ln$ Sales
(54,474 obs.)				
DD: $\Delta \ln$ lab. cost at GMR	-0.213*** (0.052)	-0.147*** (0.051)	0.060 (0.066)	0.016 (0.054)
DDD: $\Delta \ln$ lab. cost at GMR $\times$ Sh MW workers in 2002	-0.510 (0.376)	-0.569 (0.372)	0.167 (0.400)	-0.081 (0.344)

(B) Impact on Production Factors beyond Labor

	(1) $\Delta \ln$ inputs/jobs	(2) $\Delta \ln$ cap/jobs	(3) $\Delta \ln$ inv/jobs	(4) $\Delta \ln$ inputs/hrs	(5) $\Delta \ln$ cap/hrs	(6) $\Delta \ln$ inv/hrs
Observations	54,458	54,474	46,476	54,458	54,474	46,476
DD: $\Delta \ln$ lab. cost at GMR	0.170*** (0.063)	-0.002 (0.079)	0.035 (0.266)	0.104* (0.059)	-0.069 (0.077)	-0.011 (0.268)
DDD: $\Delta \ln$ lab. cost at GMR $\times$ Sh MW workers in 2002	0.141 (0.404)	1.414*** (0.500)	3.111* (1.814)	0.199 (0.387)	1.472*** (0.481)	3.256* (1.813)

(C) Impact on Productivity

	(1) $\Delta \ln$ Labor Productivity	(2) $\Delta \ln$ Labor Productivity	(3) $\Delta \ln$ Labor Productivity	(4) $\Delta \ln$ TFP Accounting	(5) $\Delta \ln$ TFP Lev. Pet.	(6) $\Delta \ln$ TFP Lev. Pet.
Observations	Jobs 54,474	Hours 54,474	Jobs 52,579	Hours 52,579	Jobs 52,579	Hours 52,579
DD: $\Delta \ln$ lab. cost at GMR	0.274*** (0.063)	0.207*** (0.057)	0.079*** (0.023)	0.0686*** (0.022)	0.076*** (0.024)	0.076*** (0.023)
DDD: $\Delta \ln$ lab. cost at GMR $\times$ Sh MW workers in 2002	0.677* (0.368)	0.735** (0.336)	0.278* (0.153)	0.269* (0.147)	0.068 (0.163)	0.101 (0.153)

Notes: See Table D5.

D.3 Robustness Checks on the Subsample of Firms with Training Data

This Appendix replicates Tables 1, 2 and 4 in order to check that the training data do not suffer from selection bias and are representative of the full population of firms. All results are broadly preserved, both qualitatively and quantitatively, despite the loss in statistical power resulting from the smaller sample size (affecting in particular the regressions with estimated TFP indices, which are by construction noisier than variables that are directly reported and measured in the accounting data).

Tables D9–D11 show that firms for which training data are available display broadly similar responses to those observed in the full sample. The estimated pass-through of statutory minimum-wage increases to labor costs is quantitatively close to the baseline estimates and remains significant across the lower hierarchical layers. The main organizational responses are also preserved: employment and hours decline, particularly at the bottom of the hierarchy, and firms become less likely to maintain or expand complex organizational structures. The estimated contractions in aggregate employment and hours are, if anything, somewhat larger in the DDD specifications for the training subsample. The DD estimates likewise continue to indicate increases in labor productivity and TFP of comparable magnitude to those reported in the main text. By contrast, the corresponding DDD productivity coefficients are smaller and no longer statistically significant, which is due to the substantial loss of precision associated with the reduced sample and to the greater noise inherent in estimated TFP measures. Overall, the close similarity of the labor-cost, employment, and organizational estimates suggests that the availability of training data does not select firms with fundamentally different responses to minimum-wage increases, although the smaller sample provides less precise evidence on heterogeneous productivity effects.

Table D9: Impact of Minimum Wage Increases on Labor Costs Across Firms' Organizations

	(1)	(2)	(3)	(4)	(5)
$\Delta_{03/06}$ hourly labor cost at: Observations	All layers 17,407	Layer 0 17,407	Layer 1 17,059	Layer 2 14,410	Layer 3 4,569
DD: $\Delta \ln$ lab. cost at GMR	0.388*** (0.033)	0.389*** (0.034)	0.311*** (0.064)	0.420*** (0.085)	0.323 (0.219)
DDD: $\Delta \ln$ lab. cost at GMR $\times$ Share MW workers in 2002	-0.010 (0.205)	0.255 (0.191)	-0.103 (0.544)	-0.901 (0.829)	0.082 (1.768)

Notes: See Table 1.

Table D10: Impact of Minimum Wage Increases on Firms' Organizations

(A) Impact on the Relative Size of Hierarchical Layers

(A1) $\Delta \ln$ jobs, at: Observations	(1) All layers 17,407	(2) Layer 0 17,407	(3) Layer 1 17,059	(4) Layer 2 14,410	(5) Layer 3 4,569
DD: $\Delta \ln$ lab. cost at GMR	-0.262*** (0.082)	-0.256** (0.120)	-0.101 (0.155)	-0.146 (0.167)	0.282 (0.246)
DDD: $\Delta \ln$ lab. cost at GMR $\times$ Sh MW workers in 2002	-0.872* (0.504)	-0.965 (0.630)	-1.861 (1.138)	0.890 (1.202)	1.932 (1.969)

(A2) $\Delta \ln$ hours, at: Observations	(1) All layers 17,407	(2) Layer 0 17,407	(3) Layer 1 17,059	(4) Layer 2 14,410	(5) Layer 3 4,569
DD: $\Delta \ln$ lab. cost at GMR	-0.224*** (0.077)	-0.228* (0.121)	-0.108 (0.163)	-0.180 (0.190)	0.494 (0.409)
DDD: $\Delta \ln$ lab. cost at GMR $\times$ Sh MW workers in 2002	-1.121** (0.470)	-1.198* (0.618)	-1.887 (1.309)	2.327 (1.677)	-0.441 (3.046)

(B) Impact on the Number of Hierarchical Layers

	(1) Δ nb layers (17,407 obs.)	(2) Prob. increase	(3) Prob. decrease	(4) Prob. stability	(5) Δ Proba 3 layers	(6) Δ Proba ≥ 2 layers	(7) Δ Proba ≥ 1 layer
DD: $\Delta \ln$ lab. cost at GMR	-0.409** (0.164)	-0.277*** (0.097)	0.083 (0.106)	0.194 (0.130)	-0.217 (0.133)	-0.140* (0.080)	-0.051 (0.032)
DDD: $\Delta \ln$ lab. cost at GMR $\times$ Sh MW workers in 2002	-1.714 (1.197)	-0.806 (0.694)	0.695 (0.705)	0.111 (0.857)	0.0899 (0.793)	-1.508** (0.741)	-0.296 (0.398)

Notes: See Table 2. In Panel (A), the composition of the estimation sample evolves across columns: in the first two columns, all sample firms are included. In columns (3) to (5), only firms exhibiting at least 2 (Layers 0 and 1), 3 or 4 layers respectively, both in 2003 and 2006, are included.

Table D11: Impact of Minimum Wage Increases on Firms' Size and Productivity

(A) Impact on Firms' Size

	(1) $\Delta \ln$ # jobs	(2) $\Delta \ln$ hours	(3) $\Delta \ln$ VA	(4) $\Delta \ln$ Sales
(17,407 obs.)				
DD: $\Delta \ln$ lab. cost at GMR	-0.262*** (0.082)	-0.224*** (0.077)	-0.059 (0.103)	-0.088 (0.088)
DDD: $\Delta \ln$ lab. cost at GMR $\times$ Sh MW workers in 2002	-0.872* (0.504)	-1.121** (0.470)	-0.793 (0.627)	-1.610*** (0.517)

(B) Impact on Production Factors beyond Labor

	(1) $\Delta \ln$ inputs/jobs 17,403	(2) $\Delta \ln$ cap/jobs 17,407	(3) $\Delta \ln$ inv/jobs 16,495	(4) $\Delta \ln$ inputs/hrs 17,403	(5) $\Delta \ln$ cap/hrs 17,407	(6) $\Delta \ln$ inv/hrs 16,495
Observations						
DD: $\Delta \ln$ lab. cost at GMR	0.098 (0.105)	-0.085 (0.119)	-0.659 (0.419)	0.061 (0.102)	-0.122 (0.117)	-0.690 (0.420)
DDD: $\Delta \ln$ lab. cost at GMR $\times$ Sh MW workers in 2002	-1.019 (0.693)	1.362 (0.839)	3.027 (3.109)	-0.773 (0.680)	1.612** (0.817)	3.273 (3.100)

(C) Impact on Productivity

	(1) $\Delta \ln$ Labor Productivity	(2) $\Delta \ln$ Labor Productivity	(3) $\Delta \ln$ Labor Productivity	(4) $\Delta \ln$ TFP Accounting	(5) $\Delta \ln$ TFP Lev. Pet.	(6) $\Delta \ln$ TFP Lev. Pet.
Observations	Jobs 17,407	Hours 17,407	Jobs 16,618	Hours 16,618	Jobs 16,618	Hours 16,618
DD: $\Delta \ln$ lab. cost at GMR	0.202** (0.101)	0.165* (0.096)	0.112*** (0.037)	0.105*** (0.036)	0.089** (0.038)	0.091** (0.037)
DDD: $\Delta \ln$ lab. cost at GMR $\times$ Sh MW workers in 2002	0.079 (0.602)	0.328 (0.564)	0.018 (0.281)	0.069 (0.270)	-0.344 (0.278)	-0.329 (0.260)

Notes: See Table 4.

E Details and Proofs of the Theoretical Model

This Appendix contains the details of the equilibrium determination for Section 4.

E.1 Supply: Cost Minimization Programs

We consider in sequence the cases of firms with 0, 1, and 2 or more layers, respectively and solve the problem embodied in Equation 12.

Firms with 0 layers (self-employed entrepreneurs). The expected output of a self-employed entrepreneur with knowledge level z_0^0 is:

$$q \leq A(1 - e^{-\lambda z_0^0}) \iff \frac{1}{\lambda} \ln \frac{A}{A - q} \leq z_0^0 \quad (\text{E.1})$$

It is always optimal to saturate this output constraint, such that the cost function is:

$$\forall q \in [0; A[, \quad C_0(z_0^0; q, k) = k(cz_0^0 + 1) = k \left(\frac{c}{\lambda} \ln \frac{A}{A - q} + 1 \right) \quad (\text{E.2})$$

Firms with 1 layer. If $n_1^0 \leq \frac{e^{\lambda z_1^0}}{h}$, the manager has enough time to deal with with all the problems that are transferred to her and the expected output of a firm with one layer is:

$$q \leq n_1^0 A(1 - e^{-\lambda z_1^1}) \iff \frac{1}{\lambda} \ln \frac{n_1^0 A}{n_1^0 A - q} \leq z_1^1 \quad (\text{E.3})$$

Notice that n_1^0 and z_1^1 have to be strictly positive to achieve strictly positive levels of output (while $n_1^1 = 1$ by definition of a 1-layer firm). The cost function to be minimized is:

$$\forall q \in \mathbb{R}^+, \quad C_1(n_1^0, z_1^0, z_1^1; q, k) = k \left(\underbrace{cz_1^1 + 1}_{\text{Entrepreneur}} + \underbrace{n_1^0(cz_1^0 + 1)}_{\text{Production workers}} \right), \quad (\text{E.4})$$

subject to the following three additional constraints:

$$q \leq n_1^0 A(1 - e^{-\lambda z_1^1}) \quad (\text{E.5})$$

$$n_1^0 \leq \frac{e^{\lambda z_1^0}}{h} \quad (\text{E.6})$$

$$0 \leq z_1^0 \quad (\text{E.7})$$

The full Lagrangian associated with this program is thus:

$$\mathcal{L}_1(n_1^0, z_1^0; q, k) = k [cz_1^1 + 1 + n_1^0(cz_1^0 + 1)] - \phi [n_1^0 A(1 - e^{-\lambda z_1^1}) - q] - \theta \left[\frac{e^{\lambda z_1^0}}{h} - n_1^0 \right] - \eta z_1^0 \quad (\text{E.8})$$

The Kuhn-Tucker conditions are:

$$\frac{\partial \mathcal{L}_1}{\partial n_1^0} = k(cz_1^0 + 1) - \phi A(1 - e^{-\lambda z_1^1}) + \theta = 0 \quad (\text{E.9})$$

$$\frac{\partial \mathcal{L}_1}{\partial z_1^0} = kcn_1^0 - \frac{\theta \lambda e^{\lambda z_1^0}}{h} - \eta = 0 \quad (\text{E.10})$$

$$\frac{\partial \mathcal{L}_1}{\partial z_1^1} = kc - \phi An_1^0 \lambda e^{-\lambda z_1^1} = 0 \quad (\text{E.11})$$

$$0 = \phi \left[n_1^0 A(1 - e^{-\lambda z_1^1}) - q \right] \quad (\text{E.12})$$

$$0 = \theta \left[\frac{e^{\lambda z_1^0}}{h} - n_1^0 \right] \quad (\text{E.13})$$

$$0 = \eta z_1^0 \quad (\text{E.14})$$

$$\theta \geq 0, \phi \geq 0, \eta \geq 0 \quad (\text{E.15})$$

The marginal cost is thus simply $\frac{\partial \mathcal{L}_1}{\partial q} = \phi$. Condition E.11 ensures that it is strictly positive, such that Equation E.3 is necessarily saturated at the optimum, as in the 0-layer case (this is due to Condition E.12).

In the case where $z_1^0 > 0$ (i.e. production workers at the bottom are able to deal with at least some of the most common problems), then Condition E.14 implies that $\eta = 0$. Plugging this into Condition E.10 implies $\theta > 0$ and therefore $n_1^0 = \frac{e^{\lambda z_1^0}}{h}$ because of Condition E.13. This case corresponds to the baseline Caliendo and Rossi-Hansberg (2012) specification.

However, in our setting, it is possible to get $z_1^0 = 0$ for some values of q .⁷ Because of Condition E.14, this relaxes the condition on η which is no longer necessarily 0 ($\eta \geq 0$). This in turn relaxes the constraint on θ , which is not necessarily strictly positive in Condition E.10, such that cases where $\theta = 0$ are possible in Conditions E.9, E.10 and E.13. This implies in particular that n_1^0 can become strictly smaller than $\frac{e^{\lambda z_1^0}}{h} = \frac{1}{h}$ (Condition E.13 for $z_1^0 = 0$).

Firms with 2 or more layers. If Equations 8 and 9 hold, all managers at all layers have enough time to deal with all the problems that are transferred to them and the expected output of a firm that is organized in $L \geq 2$ layers is:

$$q \leq n_L^0 A(1 - e^{-\lambda z_L^L}) \iff \frac{1}{\lambda} \ln \frac{n_L^0 A}{n_L^0 A - q} \leq z_L^L \quad (\text{E.16})$$

The cost function to be minimized is:

$$\forall q \in \mathbb{R}^+, \quad C_L((n_L^l)_l, (z_L^l)_l; q, k) = k \left(\underbrace{cz_L^L + 1}_{\text{Entrepreneur}} + \underbrace{\sum_{l=1}^{L-1} n_L^l (cz_L^l + 1)}_{\text{Intermediate managers}} + \underbrace{n_L^0 (cz_L^0 + 1)}_{\text{Production workers}} \right), \quad (\text{E.17})$$

$$= k \left(\sum_{l=0}^L n_L^l (cz_L^l + 1) \right) \quad (\text{E.18})$$

⁷ Caliendo and Rossi-Hansberg (2012) impose a parametric assumption in order to rule out this second case, strictly speaking for $L \geq 2$ (see their Appendix 2). With cumulative knowledge, Appendix E.2.2 (proof in the case where $L \geq 2$ and $z_L^0 = 0$) shows that this parametric assumption takes the following form: $c \leq \lambda h$. This condition ensures that $z_L^0 > 0$ because it becomes less costly to train production workers at the bottom than to transfer problems upwards, although managers have a productivity advantage as long as $h < 1$ since they can deal with $\frac{1}{h} > 1$ problems per unit of time instead of just one. However, it does not hold in our calibrations. In Section 5, we show that, in our main calibration, $h = 0.193$ and $c = 0.220$ (with λ normalized to 1). The Kuhn-Tucker conditions imply that the choices $z_1^0 = 0$, $n_1^0 < \frac{1}{h}$ are optimal if the target to attain in terms of output is relatively small: $q < A \frac{\lambda}{\lambda + ch} \frac{1}{h}$.

subject to the following constraints:

$$q \leq n_L^0 A(1 - e^{-\lambda z_L^L}) \quad (\text{E.19})$$

$$\forall l \in \{0, \dots, L-1\}, \quad n_L^0 \leq \frac{e^{\lambda z_L^l}}{h} n_L^{l+1} \quad (\text{E.20})$$

$$(n_L^L = 1) \quad (\text{E.21})$$

$$0 \leq z_L^0 \quad (\text{E.22})$$

The full Lagrangian associated with this program is thus:

$$\begin{aligned} \mathcal{L}_L \left((n_L^l)_l, (z_L^l)_l; q, k \right) &= k \left[\sum_{l=0}^L n_L^l (cz_L^l + 1) \right] - \phi \left[n_L^0 A(1 - e^{-\lambda z_L^L}) - q \right] \\ &\quad - \sum_{l=0}^{L-1} \theta_l \left[\frac{e^{\lambda z_L^l}}{h} n_L^{l+1} - n_L^0 \right] - \eta z_L^0 \end{aligned} \quad (\text{E.23})$$

The Kuhn-Tucker conditions are:

$$\frac{\partial \mathcal{L}_L}{\partial n_L^0} = k(cz_L^0 + 1) - \phi A(1 - e^{-\lambda z_L^L}) + \sum_{l=0}^{L-1} \theta_l = 0 \quad (\text{E.24})$$

$$\frac{\partial \mathcal{L}_L}{\partial z_L^0} = kcn_L^0 - \theta_0 \frac{\lambda e^{\lambda z_L^0}}{h} n_L^1 - \eta = 0 \quad (\text{E.25})$$

$$\frac{\partial \mathcal{L}_L}{\partial z_L^L} = kc - \phi A n_L^0 \lambda e^{-\lambda z_L^L} = 0 \quad (\text{E.26})$$

$$\forall l \in \{1, \dots, L-1\}, \quad \frac{\partial \mathcal{L}_L}{\partial n_L^l} = k(cz_L^l + 1) - \theta_{l-1} \frac{e^{\lambda z_L^{l-1}}}{h} = 0 \quad (\text{E.27})$$

$$\forall l \in \{1, \dots, L-1\}, \quad \frac{\partial \mathcal{L}_L}{\partial z_L^l} = kcn_L^l - \theta_l \frac{\lambda e^{\lambda z_L^l}}{h} n_L^{l+1} = 0 \quad (\text{E.28})$$

$$0 = \phi \left[n_L^0 A(1 - e^{-\lambda z_L^L}) - q \right] \quad (\text{E.29})$$

$$\forall l \in \{0, \dots, L-1\}, \quad 0 = \theta_l \left[\frac{e^{\lambda z_L^l}}{h} n_L^{l+1} - n_L^0 \right] \quad (\text{E.30})$$

$$0 = \eta z_L^0 \quad (\text{E.31})$$

$$\forall l \in \{0, \dots, L-1\}, \quad \theta_l \geq 0, \phi \geq 0, \eta \geq 0 \quad (\text{E.32})$$

Again, the marginal cost is simply $\frac{\partial \mathcal{L}_L}{\partial q} = \phi$. Equation E.26 ensures that this marginal cost is strictly positive such that Equation E.16 is saturated, as in the previous (0 and 1-layer) cases: this is due to Condition E.29. Furthermore, Equations E.27 and E.28 insure that all θ_l ($\forall l \in \{0, \dots, L-1\}$) are strictly positive, which implies that the set of constraints in Equation E.20 are always saturated, even when $z_L^0 = 0$. Interestingly, this contrasts with the 1-layer case.

For the results and proofs which follow, it is convenient to combine the previous set of conditions into the

following additional relations:

$$n_L^0 = \frac{e^{\lambda z_L^{L-1}}}{h} \quad (\text{E.33})$$

$$\forall l \in \{1, \dots, L-1\}, \quad n_L^l = e^{\lambda(z_L^{L-1} - z_L^{l-1})} \quad (\text{E.34})$$

$$0 = kcn_L^0 - \lambda n_L^1 k(cz_L^1 + 1) - \eta \quad (\text{E.35})$$

$$\forall l \in \{1, \dots, L-2\}, \quad 0 = kcn_L^l - \lambda n_L^{l+1} k(cz_L^{l+1} + 1) \quad (\text{E.36})$$

$$0 = kcn_L^{L-1} + \sum_{l=0}^{L-1} \lambda n_L^l k(cz_L^l + 1) - \phi A(1 - e^{-\lambda z_L^L}) \lambda n_L^0 \quad (\text{E.37})$$

$$0 = kc \underbrace{n_L^L}_1 - \phi A \lambda e^{-\lambda z_L^L} n_L^0 \quad (\text{E.38})$$

Equation E.33 comes from the bracket being saturated in Equation E.30 for $l = L-1$ (and observing that $n_L^L = 1$). Equation E.34 comes from the bracket being saturated in Equation E.30 and Equation E.33. Equation E.35 combines Equation E.25 with Equation E.27 for $l = 1$. Equation E.36 combines Equation E.28 with Equation E.27 (with a rank shift). Equation E.37 combines Equation E.28 for $l = L-1$ with Equation E.24 (to retrieve θ_{L-1}) and Equation E.27 (for all other θ_l). Equation E.38 simply repeats Equation E.26.

E.2 Additional Results from the Cost Minimization Programs

In their mainly theoretical contribution, Caliendo and Rossi-Hansberg (2012) rule out the case in which $z_L^0 = 0$ by imposing a convenient restriction on structural parameters (see footnote 7 in this Appendix), which also has the benefit of simplifying several of their proofs. Unfortunately, this parametric restriction is rejected in the French data in our main calibration (which implies that some French firms are likely to feature $\eta > 0$ and $z_L^0 = 0$ for at least some organizational forms L). As a consequence, many derivations in Caliendo and Rossi-Hansberg (2012) no longer hold. Our assumption of skill overlap (aka cumulative knowledge at the worker level) also alters some of their results. It is however possible to derive the set of results which follows. They apply to regions of the firms' optimization problem where they do not find it optimal to switch to a different organization (or to a different regime within a given organizational form). Formally, this implies that we consider marginal deviations in which the precise set of saturated constraints described in Section E.1 remains the same (or equivalently, in which the set of strictly positive Lagrange multipliers remains the same).

E.2.1 Characteristics of the Cost Function for a Given Number of Layers L

In terms of the cost function, our set-up features a result that is similar to part 2 of Proposition 1 from Caliendo and Rossi-Hansberg (2012).

Proposition 2. *Given a number of layers L , the marginal cost is always positive and so the cost curve is strictly increasing in q .*

Furthermore, the cost function remains homogeneous of degree one in k , which implies that the cost-minimizing firm organization given q is insensitive to k .

The first part of Proposition 2 results from the fact that, for $L \geq 1$, the first-order condition for z_L^L shows that ϕ must be positive; and for $L = 0$, the derivative of the cost function E.2 is positive. The second part of Proposition 2 results from the fact that all Lagrange multipliers are proportional to k , such that k drops from all first order conditions. In our model with cumulative knowledge, and in contrast with the specification proposed in Caliendo and Rossi-Hansberg (2012), however, the marginal cost is not necessarily increasing in q : see Proposition 3. This result of non-monotonicity when knowledge is cumulative is likely to render the management of production costs in organizations more “complex” for firms.

E.2.2 Sensitivity Analysis with Respect to q of Firms' Optimal Employment and Skill Decisions

As in part 1 of Proposition 1 from Caliendo and Rossi-Hansberg (2012), we can evaluate how the values of z_L^l , n_L^l , and ϕ vary with q . However the proof is somewhat different with cumulative knowledge such that the proposition below holds under a different set of sufficient conditions:

Proposition 3. *For a given number of layers L , we get:*

$$\begin{aligned}\frac{dz_L^l}{dq} &\geq 0 \quad \forall l \in \{0, 1, \dots, L\} \\ \frac{dn_L^l}{dq} &\geq 0 \quad \forall l \in \{0, 1, \dots, L-1\} \\ n_L^L &= 1 \text{ by design, such that } \frac{dn_L^L}{dq} = 0\end{aligned}$$

These results always hold for $L = 0$ (self-employed entrepreneurs), and under the following (sufficient) conditions,⁸ depending on firms' number of layers: $h < 1$ in firms with $L = 1$, or $h \leq 0.25$ in firms with $L \geq 2$ and $\eta = 0$ (thus $z_L^0 \geq 0$), or $h \leq \frac{c}{\lambda} \leq 0.25$ in firms with $L \geq 2$ and $\eta > 0$ (thus $z_L^0 = 0$).

Furthermore, the previous inequalities are strict, except for n_L^0 and z_L^0 when the latter is constrained at zero ($\eta > 0$).

Proposition 4. *Under the same parameter restrictions as Proposition 3, $\frac{d\phi}{dq}$ can be positive or negative, depending on the number of layers and the parameter values:*

- $\frac{d\phi}{dq} > 0$ for self-employed entrepreneurs ($L = 0$) as well as in 1-layer firms ($L = 1$) as long as $\theta > 0$, such that the employment of the bottom layer (n_L^0) reaches its upper bound (Equation E.6);
- $\frac{d\phi}{dq} < 0$ in 1-layer firms ($L = 1$) where $\theta > 0$, such that n_L^0 does not necessarily reach the upper bound in Equation E.6;
- $\frac{d\phi}{dq}$ can take any sign in firms with more than 2 layers ($L \geq 2$).

Proof for the case $L = 0$

Proof. In this class of firms, output is a monotonic function of z_0^0 : $q = A(1 - e^{-\lambda z_0^0})$, such that $z_0^0 = \frac{1}{\lambda} \ln \frac{A}{A-q}$ and $C_0(q) = k(cz_0^0 + 1)$. It is therefore immediate that $\frac{dz_0^0}{dq} > 0$ and $\frac{d\phi}{dq} > 0$. Note that in this case, z_0^0 can adjust, but not n_0^0 . $\square$

Proof for the case $L = 1$ In this class of firms, there are three different cases to consider:⁹

- (a) $\eta = 0$ and $\theta > 0$, which implies $z_1^0 \geq 0$ and $n_1^0 = \frac{e^{\lambda z_1^0}}{h} > \frac{1}{h}$
- (b) $\eta > 0$ and $\theta > 0$, which implies $z_1^0 = 0$ and $n_1^0 = \frac{e^{\lambda z_1^0}}{h} = \frac{1}{h}$
- (c) $\eta > 0$ and $\theta = 0$, which implies $z_1^0 = 0$ and $n_1^0 \leq \frac{1}{h}$

Proof for the case $L = 1$, $\eta > 0$ and $\theta > 0$, which implies $z_1^0 = 0$ and $n_1^0 = \frac{e^{\lambda z_1^0}}{h} = \frac{1}{h}$. The proof for case (b) is analogous to the proof for $L = 0$. Indeed, we get that output simply takes the following form: $q =$

⁸Notice that these conditions on the exogenous parameters are increasingly binding. Low values of h and $\frac{c}{\lambda}$ correspond to efficient communication or information technologies, respectively.

⁹Indeed, the case where $\eta = 0$ and $\theta = 0$ would imply $n_1^0 = 0$ from Equation E.10, which is impossible.

$\frac{A}{h}(1 - e^{-\lambda z_1^1})$, such that $z_1^1 = \frac{1}{\lambda} \ln \frac{A/h}{A/h-q}$ and $C_1(q) = k(cz_1^1 + 1 + \frac{1}{h})$. It is therefore immediate that $\frac{dz_1^1}{dq} > 0$ and $\frac{d\phi}{dq} > 0$. Note that in this case, z_1^1 can adjust, but neither z_1^0 nor n_1^0 .¹⁰ $\square$

Proof for the case $L = 1$, $\eta = 0$ and $\theta > 0$, which implies $z_1^0 \geq 0$ and $n_1^0 = \frac{e^{\lambda z_1^0}}{h}$. The proof for case (a) is somewhat more complex. The Kuhn-Tucker conditions can be used to deliver the following set of relations:¹¹

$$n_1^0 kc + \lambda n_1^0 k(cz_1^0 + 1) = \phi \lambda q \quad (\text{E.39})$$

$$\phi A \lambda e^{-\lambda z_1^1} n_1^0 = kc \quad (\text{E.40})$$

$$A(1 - e^{-\lambda z_1^1}) n_1^0 = q \quad (\text{E.41})$$

Equation E.39 combines Equations E.9 and E.10 and recognizes the values of q and n_1^0 in the (saturated) brackets of Equations E.12 and E.13, respectively. Equation E.40 corresponds to the saturated constraint in Equation E.13, while Equation E.41 corresponds to the saturated constraint in Equation E.12.

Differentiating each with respect to q , we obtain:

$$\phi + q \frac{d\phi}{dq} = (2n_1^0 kc + \lambda n_1^0 k(cz_1^0 + 1)) \frac{dz_1^0}{dq} \quad (\text{E.42})$$

$$\frac{d\phi}{dq} = \phi \lambda \left(\frac{dz_1^1}{dq} - \frac{dz_1^0}{dq} \right) \quad (\text{E.43})$$

$$1 = A \lambda n_1^0 \frac{dz_1^0}{dq} + A \lambda e^{-\lambda z_1^1} n_1^0 \left(\frac{dz_1^1}{dq} - \frac{dz_1^0}{dq} \right) \quad (\text{E.44})$$

Combining all three of these, we find:

$$(kc + q\phi\lambda) \left(\frac{dz_1^1}{dq} - \frac{dz_1^0}{dq} \right) = (2n_1^0 kc + \lambda n_1^0 k(cz_1^0 + 1) - \phi A \lambda n_1^0) \frac{dz_1^0}{dq} \quad (\text{E.45})$$

Multiplying Equation E.40 by Equation E.41 and subtracting kc , we get that $\phi A \lambda n_1^0 = \phi \lambda q + kc$, which can be plugged into the previous equation to deliver:

$$(kc + q\phi\lambda) \left(\frac{dz_1^1}{dq} - \frac{dz_1^0}{dq} \right) = (n_1^0 - 1)kc \frac{dz_1^0}{dq}. \quad (\text{E.46})$$

Whenever $h < 1$, $n_1^0 > 1$ such that $\left(\frac{dz_1^1}{dq} - \frac{dz_1^0}{dq} \right)$ takes the same sign as $\frac{dz_1^0}{dq}$. Given Equation E.44, that sign must be positive, which implies $\frac{dz_1^1}{dq} > \frac{dz_1^0}{dq} > 0$, and therefore $\frac{d\phi}{dq} > 0$ and $\frac{dn_1^0}{dq} > 0$. $\square$

Proof for the case $L = 1$, $\eta > 0$ and $\theta = 0$, which implies $z_1^0 = 0$ and $n_1^0 < \frac{1}{h}$. Lastly, consider case (c). Again,

¹⁰Working on the restrictions implied by the two constraints $\eta > 0$ and $\theta > 0$, it is possible to characterize the domain under which this regime is likely to be optimal:

$$\frac{A}{h} \cdot \left[\frac{\frac{kc}{\theta\lambda} + k}{\frac{kch}{\lambda} + \frac{kc}{\theta\lambda} + k} \right] > q > \frac{\lambda A}{h(ch + \lambda)}$$

¹¹Solving this set of equations in z_1^0 , we obtain the following non-linear equation, which can't be solved in closed form but delivers a unique solution:

$$k \left(cz_1^0 + \frac{c}{\lambda} + 1 \right) = \frac{k^2 hcq}{\lambda e^{\lambda z_1^0} (A e^{\lambda z_1^0} - hq)}$$

the Kuhn-Tucker conditions can be used to deliver the following set of relations:¹²

$$\phi A(1 - e^{-\lambda z_1^1}) = k \quad (\text{E.47})$$

$$\phi A \lambda e^{-\lambda z_1^1} n_1^0 = kc \quad (\text{E.48})$$

$$A(1 - e^{-\lambda z_1^1}) n_1^0 = q \quad (\text{E.49})$$

Equation E.47 simply corresponds to Equation E.9 with $\theta = 0$ and $z_1^0 = 0$. Equation E.48 corresponds to the saturated constraint in Equation E.13, while Equation E.49 corresponds to the saturated constraint in Equation E.12. Differentiating each of the previous relations with respect to q , we obtain:

$$(1 - e^{-\lambda z_1^1}) \frac{d\phi}{dq} = -\phi \lambda e^{-\lambda z_1^1} \frac{dz_1^1}{dq} \quad (\text{E.50})$$

$$\phi \lambda n_1^0 \frac{dz_1^1}{dq} - \phi \frac{dn_1^0}{dq} = n_1^0 \frac{d\phi}{dq} \quad (\text{E.51})$$

$$A(1 - e^{-\lambda z_1^1}) \frac{dn_1^0}{dq} + A \lambda e^{-\lambda z_1^1} n_1^0 \frac{dz_1^1}{dq} = 1 \quad (\text{E.52})$$

Combining Equations E.50 and E.51, we obtain:

$$\lambda n_1^0 \frac{dz_1^1}{dq} = (1 - e^{-\lambda z_1^1}) \frac{dn_1^0}{dq} \quad (\text{E.53})$$

Therefore, $\frac{dz_1^1}{dq}$ and $\frac{dn_1^0}{dq}$ must take the same sign, and Equation E.52 implies that that sign must be positive: $\frac{dz_1^1}{dq} > 0$ and $\frac{dn_1^0}{dq} > 0$. Finally, Equation E.50 implies that $\frac{d\phi}{dq} < 0$. $\square$

Proof for the case $L \geq 2$ In this class of firms, there are two different cases to consider:

- (a) $\eta = 0$, which implies $z_L^0 \geq 0$
- (b) $\eta > 0$, which implies $z_L^0 = 0$

Proof for the case $L \geq 2$ and $\eta = 0$, which implies $z_L^0 \geq 0$. In this regime, we have $\eta = 0$ in Equation E.32. Plugging Equations E.33 and E.34 into Equations E.35 to E.38, the latter simplify to:¹³

$$\lambda h(cz_L^1 + 1) = ce^{\lambda z_L^0} \quad (\text{E.54})$$

$$\forall l \in \{1, \dots, L-2\}, \quad \lambda(cz_L^{l+1} + 1) = ce^{\lambda(z_L^l - z_L^{l-1})} \quad (\text{E.55})$$

$$\sum_{l=0}^{L-1} n_L^l kc + \lambda n_L^0 k(cz_L^0 + 1) = \phi \lambda q \quad (\text{E.56})$$

$$\phi A \lambda e^{-\lambda z_L^L} n_L^0 = kc \quad (\text{E.57})$$

$$A(1 - e^{-\lambda z_L^L}) n_L^0 = q \quad (\text{E.58})$$

¹²In this regime, the firm's optimization problem can be solved for in closed form:

$$\begin{aligned} e^{-\lambda z_1^1} &= 1 + \frac{\lambda q + \sqrt{\lambda^2 q^2 + 4ac\lambda q}}{2Ac} \\ \phi A &= \frac{k}{1 - e^{-\lambda z_1^1}} \\ n_1^0 &= \frac{q}{A(1 - e^{-\lambda z_1^1})} \end{aligned}$$

¹³It should be noticed that Equation E.55 does not operate when $L = 2$. However, in this regime, Equations E.54, E.56, E.57 and E.58 are sufficient to derive the same results as in the case where $L > 2$ and Equation E.55 holds.

Differentiating each of the previous equations with respect to q , we obtain:

$$n_L^0 \frac{dz_L^0}{dq} = n_L^1 \frac{dz_L^1}{dq} \quad (\text{E.59})$$

$$\forall l \in \{1, \dots, L-2\}, \quad n_L^l \left(\frac{dz_L^l}{dq} - \frac{dz_L^{l+1}}{dq} \right) = n_L^{l+1} \frac{dz_L^{l+1}}{dq} \quad (\text{E.60})$$

$$\phi \lambda q \frac{dz_L^{L-1}}{dq} + k c n_L^{L-1} \left(\frac{dz_L^{L-1}}{dq} - \frac{dz_L^{L-2}}{dq} \right) = \phi + q \frac{d\phi}{dq} \quad (\text{E.61})$$

$$\phi \lambda \left(\frac{dz_L^L}{dq} - \frac{dz_L^{L-1}}{dq} \right) = \frac{d\phi}{dq} \quad (\text{E.62})$$

$$A \lambda n_L^0 \frac{dz_L^{L-1}}{dq} + A \lambda n_L^0 e^{-\lambda z_L^L} \left(\frac{dz_L^L}{dq} - \frac{dz_L^{L-1}}{dq} \right) = 1 \quad (\text{E.63})$$

Notice that the derivation of Equation E.61 above is somewhat indirect and involves the following steps:

$$\begin{aligned} k c \sum_{l=0}^{L-1} \frac{dn_L^l}{dq} + \lambda k (c z_L^0 + 1) \frac{dn_L^0}{dq} + \lambda n_L^0 k c \frac{dz_L^0}{dq} &= \phi \lambda + \lambda q \frac{d\phi}{dq} \\ \therefore k c \sum_{l=1}^{L-1} \lambda n_L^l \left(\frac{dz_L^{L-1}}{dq} - \frac{dz_L^{l-1}}{dq} \right) + k c \lambda n_L^0 \frac{dz_L^{L-1}}{dq} + \lambda^2 k (c z_L^0 + 1) n_L^0 \frac{dz_L^{L-1}}{dq} + \lambda n_L^0 k c \frac{dz_L^0}{dq} &= \phi \lambda + \lambda q \frac{d\phi}{dq} \\ \therefore \phi \lambda q \frac{dz_L^{L-1}}{dq} - k c \sum_{l=1}^{L-1} n_L^l \frac{dz_L^{l-1}}{dq} + n_L^0 k c \frac{dz_L^0}{dq} &= \phi + q \frac{d\phi}{dq} \\ \therefore \phi \lambda q \frac{dz_L^{L-1}}{dq} - k c \sum_{l=1}^{L-2} \left(n_L^l \frac{dz_L^l}{dq} - n_L^{l+1} \frac{dz_L^{l+1}}{dq} \right) - k c n_L^{L-1} \frac{dz_L^{L-2}}{dq} + n_L^0 k c \frac{dz_L^0}{dq} &= \phi + q \frac{d\phi}{dq} \\ \therefore \phi \lambda q \frac{dz_L^{L-1}}{dq} - k c \left(n_L^1 \frac{dz_L^1}{dq} - n_L^{L-1} \frac{dz_L^{L-1}}{dq} \right) - k c n_L^{L-1} \frac{dz_L^{L-2}}{dq} + n_L^0 k c \frac{dz_L^0}{dq} &= \phi + q \frac{d\phi}{dq} \\ \therefore \phi \lambda q \frac{dz_L^{L-1}}{dq} + k c n_L^{L-1} \left(\frac{dz_L^{L-1}}{dq} - \frac{dz_L^{L-2}}{dq} \right) &= \phi + q \frac{d\phi}{dq} \end{aligned}$$

In the previous set of equations, the third line uses the fact that $\sum_{l=0}^{L-1} n_L^l k c + \lambda n_L^0 k (c z_L^0 + 1) = \phi \lambda q$ from Equation E.56. The fourth line uses the fact that $n_L^l \frac{dz_L^{l-1}}{dq} = n_L^l \frac{dz_L^l}{dq} - n_L^{l+1} \frac{dz_L^{l+1}}{dq}$ for $l \in \{1, \dots, L-2\}$ from Equation E.60, and the fifth line uses the fact that $n_L^0 \frac{dz_L^0}{dq} = n_L^1 \frac{dz_L^1}{dq}$ from Equation E.59.

Returning to Equations E.61 to E.63 and combining them, we obtain:

$$\phi \lambda q \frac{dz_L^{L-1}}{dq} + k c n_L^{L-1} \left(\frac{dz_L^{L-1}}{dq} - \frac{dz_L^{L-2}}{dq} \right) = \phi A \lambda n_L^0 \frac{dz_L^{L-1}}{dq} + k c \left(\frac{dz_L^L}{dq} - \frac{dz_L^{L-1}}{dq} \right) + \phi \lambda q \left(\frac{dz_L^L}{dq} - \frac{dz_L^{L-1}}{dq} \right) \quad (\text{E.64})$$

and using the fact that $\phi \lambda q = \phi A \lambda n_L^0 - k c$ (as implied by Equation E.38 where q is replaced with its value in Equation E.29), this simplifies to:

$$k c n_L^{L-1} \left(\frac{dz_L^{L-1}}{dq} - \frac{dz_L^{L-2}}{dq} \right) = k c \frac{dz_L^L}{dq} + \phi \lambda q \left(\frac{dz_L^L}{dq} - \frac{dz_L^{L-1}}{dq} \right) \quad (\text{E.65})$$

This last result will be useful at a later stage of the proof. We now have to introduce the (sufficient) condition that $h \leq 0.25$, which secures monotone rankings in terms of employment across layers. To show this, we first recognize that Equation E.30 as evaluated for $l = 0$ implies that $\frac{n_L^0}{n_L^1} = \frac{e^{\lambda z_L^0}}{h} \geq \frac{1}{h} \geq 4$. Second, our assumption of cumulative knowledge implies that z_L^l must increase across layers (i.e. with respect to l for a given L). Equations E.35 (where $\eta = 0$) and E.36 thus imply that the ratio $\frac{n_L^l}{n_L^{l+1}}$ increases with l for $l \in \{0, \dots, L-2\}$, and each such ratio is therefore greater than the first one, $\frac{n_L^0}{n_L^1} \geq \frac{1}{h} \geq 4$. The fact that $\frac{n_L^0}{n_L^1} \geq \frac{1}{h} \geq 4$ together with Equation E.59 also imply the following rankings between $\frac{dz_L^0}{dq}$ and $\frac{dz_L^1}{dq}$, depending

on the sign of $\frac{dz_L^0}{dq}$:

- if $\frac{dz_L^0}{dq} > 0$, then $\frac{dz_L^1}{dq} \geq \frac{1}{h} \frac{dz_L^0}{dq} \geq 4 \frac{dz_L^0}{dq} (> 0)$
- if $\frac{dz_L^0}{dq} \leq 0$, then $\frac{dz_L^1}{dq} \leq \frac{1}{h} \frac{dz_L^0}{dq} \leq 4 \frac{dz_L^0}{dq} (\leq 0)$.

The challenge of the proof at this stage is to determine which of these two regimes is valid. To that end, we rely on recurrence reasoning in each regime:¹⁴

- *Recurrence equation, positive case* ($\frac{dz_L^0}{dq} > 0$): Suppose that $\frac{dz_L^l}{dq} > 2 \frac{dz_L^{l-1}}{dq} > 0$ for some $l \in \{1, \dots, L-2\}$. Notice that in particular, this holds true for $l = 1$ whenever $\frac{dz_L^0}{dq} > 0$. Equation E.60 implies that:

$$\frac{dz_L^{l+1}}{dq} = \frac{n_L^l}{n_L^{l+1}} \left(\frac{dz_L^l}{dq} - \frac{dz_L^{l-1}}{dq} \right) > \frac{n_L^l}{n_L^{l+1}} \left(\frac{dz_L^l}{dq} - \frac{1}{2} \frac{dz_L^l}{dq} \right) = \frac{n_L^l}{2n_L^{l+1}} \frac{dz_L^l}{dq} > 2 \frac{dz_L^l}{dq} > 0$$

where the last inequality relies on the result $\frac{n_L^l}{n_L^{l+1}} \geq \frac{1}{h} \geq 4$ which was previously proved for $l \in \{0, \dots, L-2\}$. By recurrence, this implies that whenever $\frac{dz_L^0}{dq} > 0$, all ratios $\frac{dz_L^l}{dq}$ for $l \in \{0, \dots, L-1\}$ take the same sign (are strictly positive) and increase in absolute value.

- *Recurrence equation, negative case* ($\frac{dz_L^0}{dq} \leq 0$): a similar reasoning holds with large inequalities and implies that all ratios $\frac{dz_L^l}{dq}$ for $l \in \{0, \dots, L-1\}$ take the same sign (are negative or zero) and increase in absolute value.

However, if $\frac{dz_L^{L-1}}{dq} \leq \frac{dz_L^{L-2}}{dq} \leq 0$, then Equation E.65 implies that $\frac{dz_L^L}{dq}$ must be negative or zero, since in that case:

$$\underbrace{(kc + \phi\lambda q)}_{>0} \frac{dz_L^L}{dq} = kcn_L^{L-1} \underbrace{\left(\frac{dz_L^{L-1}}{dq} - \frac{dz_L^{L-2}}{dq} \right)}_{\leq 0} + \phi\lambda q \underbrace{\frac{dz_L^{L-1}}{dq}}_{\leq 0}$$

This contradicts Equation E.63, since the latter would imply:

$$A\lambda n_L^0 e^{\lambda z_L^L} \underbrace{\frac{dz_L^L}{dq}}_{\leq 0} + \lambda q \underbrace{\frac{dz_L^{L-1}}{dq}}_{\leq 0} = 1$$

which is impossible.

We can conclude from the previous developments that for $l \in \{0, \dots, L-1\}$, all ratios $\frac{dz_L^l}{dq}$ must be strictly positive and increase with l .

Given Equation E.62, the sign of $\frac{d\phi}{dq}$ is the same as the sign of $\left(\frac{dz_L^L}{dq} - \frac{dz_L^{L-1}}{dq} \right)$, but it can be either positive

¹⁴Solving these recurrence equations actually motivates the value of the upper bound for h (0.25) in the necessary condition that is required in this proof. Focusing on the first regime ($\frac{dz_L^0}{dq} > 0$), let us define $\alpha > 0$ as a candidate value such that

$\frac{dz_L^l}{dq} > \alpha \frac{dz_L^{l-1}}{dq} \implies \frac{dz_L^{l+1}}{dq} > \alpha \frac{dz_L^l}{dq}$. Our proof is based on the fact that Equation E.60 implies that:

$$\frac{dz_L^{l+1}}{dq} = \frac{n_L^l}{n_L^{l+1}} \left(\frac{dz_L^l}{dq} - \frac{dz_L^{l-1}}{dq} \right) \geq \frac{1}{h} \left(\frac{dz_L^l}{dq} - \frac{dz_L^{l-1}}{dq} \right) \geq \frac{1}{h} \left(1 - \frac{1}{\alpha} \right) \frac{dz_L^l}{dq}$$

Thus, the following relation between parameters has to hold: $\frac{1}{h} \left(1 - \frac{1}{\alpha} \right) \geq \alpha \iff \alpha^2 h - \alpha + 1 \leq 0$. There are admissible real values for α if and only if this polynomial has real solutions. This happens if and only if $h \leq 0.25$. For $h = 0.25$, the only admissible candidate value is $\alpha = 2$. For $h \in [0; \frac{1}{4}]$, there is a entire range of admissible values for α : $\left[\frac{1-\sqrt{1-4h}}{2h}; \frac{1+\sqrt{1-4h}}{2h} \right]$. For $h \in [0; \frac{1}{4}]$, this interval always contains the value $\alpha = 2$ which is retained in the main proof.

or negative in Equations E.61 to E.63.¹⁵

Lastly, differentiating Equations E.33 and E.34 with respect to q , we obtain that $\frac{dn_L^0}{dq} = \lambda n_L^0 \frac{dz_L^{L-1}}{dq} \geq \frac{dn_L^l}{dq} = \lambda n_L^l \left(\frac{dz_L^{L-1}}{dq} - \frac{dz_L^{l-1}}{dq} \right)$ for $l \in \{1, \dots, L-1\}$, such that:

$$\frac{dn_L^0}{dq} > \frac{dn_L^1}{dq} > \dots > \frac{dn_L^{L-1}}{dq} > 0 = \frac{dn_L^L}{dq}$$

□

Proof for the case $L \geq 2$ and $\eta > 0$, which implies $z_L^0 = 0$. We first notice that $\eta \geq 0$ in Equation E.35 implies:

$$\begin{aligned} \eta \geq 0 &\iff n_L^0 k [c - \lambda h(cz_L^1 + 1)] \geq 0 \\ &\implies c \geq \lambda h \end{aligned}$$

This contradicts the parametric assumption of the kind that is introduced in Caliendo and Rossi-Hansberg (2012), which would precisely rule out the regime where $z_L^0 = 0$.

The proof in this regime is slightly altered and introduces the additional sufficient condition that $h \leq \frac{c}{\lambda} \leq 0.25$. It proceeds as follows. Equations E.55 to E.58 are replaced with the following set of constraints (building on the fact that at the optimum, $n_L^0 = \frac{e^{\lambda z_L^{L-1}}}{h}$ and $n_L^1 = e^{\lambda(z_L^{L-1}-0)} = h n_L^0$):¹⁶

$$\forall l \in \{1, \dots, L-2\}, \quad \lambda(cz_L^{l+1} + 1) = ce^{\lambda(z_L^l - z_L^{l-1})} \quad (\text{E.66})$$

$$\sum_{l=2}^{L-1} n_L^l k c + n_L^0 k [hc + \lambda h(cz_L^1 + 1) + \lambda] = \phi \lambda q \quad (\text{E.67})$$

$$\phi A \lambda e^{-\lambda z_L^L} n_L^0 = k c \quad (\text{E.68})$$

$$A(1 - e^{-\lambda z_L^L}) n_L^0 = q \quad (\text{E.69})$$

Differentiating each of the previous equations with respect to q we obtain:

$$n_L^1 \frac{dz_L^1}{dq} = n_L^2 \frac{dz_L^2}{dq} \quad (\text{E.70})$$

$$\forall l \in \{1, \dots, L-2\}, \quad n_L^l \left(\frac{dz_L^l}{dq} - \frac{dz_L^{l-1}}{dq} \right) = n_L^{l+1} \frac{dz_L^{l+1}}{dq} \quad (\text{E.71})$$

$$\phi \lambda q \frac{dz_L^{L-1}}{dq} + k c n_L^{L-1} \left(\frac{dz_L^{L-1}}{dq} - \frac{dz_L^{L-2}}{dq} \right) = \phi + q \frac{d\phi}{dq} \quad (\text{E.72})$$

$$\phi \lambda \left(\frac{dz_L^L}{dq} - \frac{dz_L^{L-1}}{dq} \right) = \frac{d\phi}{dq} \quad (\text{E.73})$$

$$A \lambda n_L^0 \frac{dz_L^{L-1}}{dq} + A \lambda n_L^0 e^{-\lambda z_L^L} \left(\frac{dz_L^L}{dq} - \frac{dz_L^{L-1}}{dq} \right) = 1 \quad (\text{E.74})$$

The first relation corresponds to Equation E.71 as evaluated at $l = 1$ and recognizing that $\frac{dz_L^0}{dq} = 0$. Interestingly, Equation E.72 coincides with Equation E.61 despite the fact that Equation E.67 does not

¹⁵Yet, it is possible to derive regimes where $\frac{d\phi}{dq} \geq 0$. Combining Equation E.65 together with the result from the recurrence reasoning that $\frac{dz_L^{L-1}}{dq} > 2 \frac{dz_L^{L-2}}{dq}$ implies that $\frac{d\phi}{dq} \geq 0$ whenever $n_L^{L-1} \geq 2$.

¹⁶Again, it should be noticed that Equation E.66 does not operate when $L = 2$. However, in this regime, Equations E.67, E.68 and E.69 are sufficient to derive the same results as in the case where $L > 2$ and Equation E.66 holds.

coincide with Equation E.56. This result is derived in the same way as previously:

$$\begin{aligned}
kc \sum_{l=2}^{L-1} \lambda n_L^l \left(\frac{dz_L^{L-1}}{dq} - \frac{dz_L^{l-1}}{dq} \right) + k\lambda n_L^0 \frac{dz_L^{L-1}}{dq} [hc + \lambda h(cz_L^1 + 1) + \lambda] + \lambda n_L^0 kch \frac{dz_L^1}{dq} &= \phi\lambda + \lambda q \frac{d\phi}{dq} \\
\therefore \phi\lambda q \frac{dz_L^{L-1}}{dq} - kc \sum_{l=2}^{L-1} n_L^l \frac{dz_L^{l-1}}{dq} + n_L^0 kch \frac{dz_L^1}{dq} &= \phi + q \frac{d\phi}{dq} \\
\therefore \phi\lambda q \frac{dz_L^{L-1}}{dq} - kc \sum_{l=2}^{L-2} \left(n_L^l \frac{dz_L^l}{dq} - n_L^{l+1} \frac{dz_L^{l+1}}{dq} \right) - kcn_L^{L-1} \frac{dz_L^{L-2}}{dq} + n_L^0 kch \frac{dz_L^1}{dq} &= \phi + q \frac{d\phi}{dq} \\
\therefore \phi\lambda q \frac{dz_L^{L-1}}{dq} - kc \left(n_L^2 \frac{dz_L^2}{dq} - n_L^{L-1} \frac{dz_L^{L-1}}{dq} \right) - kcn_L^{L-1} \frac{dz_L^{L-2}}{dq} + kcn_L^1 \frac{dz_L^1}{dq} &= \phi + q \frac{d\phi}{dq} \\
\therefore \phi\lambda q \frac{dz_L^{L-1}}{dq} + kcn_L^{L-1} \left(\frac{dz_L^{L-1}}{dq} - \frac{dz_L^{L-2}}{dq} \right) &= \phi + q \frac{d\phi}{dq}
\end{aligned}$$

In the previous set of equations, the second line uses the fact that $\sum_{l=2}^{L-1} n_L^l kc + n_L^0 k [hc + \lambda h(cz_L^1 + 1) + \lambda] = \phi\lambda q$ from Equation E.67. The third line uses the fact that $n_L^l \frac{dz_L^{l-1}}{dq} = n_L^l \frac{dz_L^l}{dq} - n_L^{l+1} \frac{dz_L^{l+1}}{dq}$ for $l \in \{1, \dots, L-2\}$ from Equation E.71, and the fourth line incorporates Equation E.70.

Since Equations E.72 to E.74 coincide with Equations E.61 to E.63, Equation E.65 still holds. Furthermore, Equation E.36 implies:

$$\forall l \in \{1, \dots, L-2\}, \quad \frac{n_L^l}{n_L^{l+1}} = \frac{\lambda}{c} (z_L^{l+1} + 1) \quad (\text{E.75})$$

In our framework of cumulative knowledge, as long as $h \leq \frac{c}{\lambda} \leq 0.25 \iff \frac{1}{h} \geq \frac{\lambda}{c} \geq 4$, this implies that the ratio $\frac{n_L^l}{n_L^{l+1}}$ increases with l for $l \in \{1, \dots, L-2\}$ and that each such ratio is greater than $\frac{\lambda}{c} \geq 4$.

The fact that $\frac{n_L^1}{n_L^2} \geq \frac{\lambda}{c} \geq 4$ together with Equation E.70 also imply the following rankings between $\frac{dz_L^1}{dq}$ and $\frac{dz_L^2}{dq}$, depending on the sign of $\frac{dz_L^1}{dq}$:

- if $\frac{dz_L^1}{dq} > 0$, then $\frac{dz_L^2}{dq} \geq \frac{\lambda}{c} \frac{dz_L^1}{dq} \geq 4 \frac{dz_L^1}{dq} (> 0)$
- if $\frac{dz_L^1}{dq} \leq 0$, then $\frac{dz_L^2}{dq} \leq \frac{\lambda}{c} \frac{dz_L^1}{dq} \leq 4 \frac{dz_L^1}{dq} (\leq 0)$.

As previously, the challenge of the proof at this stage is to determine which of these two regimes is valid.

To that end, we rely on recurrence reasoning in each regime:

- *Recurrence equation, positive case* ($\frac{dz_L^1}{dq} > 0$): Suppose that $\frac{dz_L^l}{dq} > 2 \frac{dz_L^{l-1}}{dq} > 0$ for some $l \in \{1, \dots, L-2\}$.

Notice that in particular, this holds true for $l = 2$ whenever $\frac{dz_L^1}{dq} > 0$. Equation E.71 implies that:

$$\frac{dz_L^{l+1}}{dq} = \frac{n_L^l}{n_L^{l+1}} \left(\frac{dz_L^l}{dq} - \frac{dz_L^{l-1}}{dq} \right) > \frac{n_L^l}{n_L^{l+1}} \left(\frac{dz_L^l}{dq} - \frac{1}{2} \frac{dz_L^l}{dq} \right) = \frac{n_L^l}{2n_L^{l+1}} \frac{dz_L^l}{dq} > 2 \frac{dz_L^l}{dq} > 0$$

where the last inequality relies on the result $\frac{n_L^l}{n_L^{l+1}} \geq \frac{\lambda}{c} \geq 4$ which was previously proved for $l \in \{1, \dots, L-2\}$. By recurrence, this implies that whenever $\frac{dz_L^0}{dq} > 0$, all ratios $\frac{dz_L^l}{dq}$ for $l \in \{1, \dots, L-1\}$ take the same sign (are strictly positive) and increase in absolute value.

- *Recurrence equation, negative case* ($\frac{dz_L^1}{dq} \leq 0$): a similar reasoning holds with large inequalities and implies that all ratios $\frac{dz_L^l}{dq}$ for $l \in \{1, \dots, L-1\}$ take the same sign (are negative or zero) and increase in absolute value.

However, if $\frac{dz_L^{L-1}}{dq} \leq \frac{dz_L^{L-2}}{dq} \leq 0$, then Equation E.65 implies that $\frac{dz_L^L}{dq}$ must be negative or zero, since in

that case:

$$\underbrace{(kc + \phi\lambda q)}_{>0} \frac{dz_L^L}{dq} = kn_L^{L-1} \underbrace{\left(\frac{dz_L^{L-1}}{dq} - \frac{dz_L^{L-2}}{dq} \right)}_{\leq 0} + \phi\lambda q \underbrace{\frac{dz_L^{L-1}}{dq}}_{\leq 0}$$

This contradicts Equation E.74, since the latter would imply:

$$A\lambda n_L^0 e^{\lambda z_L^L} \underbrace{\frac{dz_L^L}{dq}}_{\leq 0} + \lambda q \underbrace{\frac{dz_L^{L-1}}{dq}}_{\leq 0} = 1$$

which is impossible.

We can conclude from the previous developments that for $l \in \{1, \dots, L-1\}$, all ratios $\frac{dz_L^l}{dq}$ must be strictly positive and increase with l . Furthermore, $\frac{dz_L^0}{dq}$ is trivially 0. The treatment of $\frac{dz_L^L}{dq}$ is slightly more indirect, but Equation E.65 implies that:

$$\underbrace{(kc + \phi\lambda q)}_{>0} \frac{dz_L^L}{dq} = kn_L^{L-1} \underbrace{\left(\frac{dz_L^{L-1}}{dq} - \frac{dz_L^{L-2}}{dq} \right)}_{>0} + \phi\lambda q \underbrace{\frac{dz_L^{L-1}}{dq}}_{>0}$$

such that $\frac{dz_L^L}{dq}$ is necessarily strictly positive.

As in the previous case, given Equation E.73, the sign of $\frac{d\phi}{dq}$ is the same as the sign of $\left(\frac{dz_L^L}{dq} - \frac{dz_L^{L-1}}{dq} \right)$, but it can be either positive or negative in Equations E.72 to E.74.

Lastly, differentiating Equations E.33 and E.34 with respect to q , we obtain that $\frac{dn_L^0}{dq} = \lambda n_L^0 \frac{dz_L^{L-1}}{dq} \geq \frac{dn_L^1}{dq} = \lambda n_L^1 \frac{dz_L^{L-1}}{dq} \geq \frac{dn_L^l}{dq} = \lambda n_L^l \left(\frac{dz_L^{L-1}}{dq} - \frac{dz_L^{l-1}}{dq} \right)$ for $l \in \{2, \dots, L-1\}$, such that:

$$\frac{dn_L^0}{dq} > \frac{dn_L^1}{dq} > \dots > \frac{dn_L^{L-1}}{dq} > 0 = \frac{dn_L^L}{dq}$$

□

E.3 Introducing the Minimum Wage

E.3.1 Supply: Cost Minimization with a Minimum Wage Constraint

Our framework mimics the French institutions and assumes that the minimum wage is only legally binding in actual firms, i.e. organizations featuring $L \geq 1$. In our framework, this constraint hits the bottom (production) layer as follows:

$$k(1 + cz_L^0) \geq \bar{w},$$

which is more conveniently written in terms of z_L^0 (see Section 4.2) as:

$$z_L^0 \geq \frac{\bar{w} - k}{ck} \equiv \bar{z}_L^0.$$

As shown below, this modified constraint only marginally alters the previous proofs, such that all previous results are preserved.

Firms with 1 layer. The full Lagrangian associated with the new set of constraints is altered in the following way:

$$\mathcal{L}_1(n_1^0, z_1^0) = k [cz_1^1 + 1 + n_1^0(cz_1^0 + 1)] - \phi [n_1^0 A(1 - e^{-\lambda z_1^1}) - q] - \theta \left[\frac{e^{\lambda z_1^0}}{h} - n_1^0 \right] - \eta [z_1^0 - \bar{z}_1^0] \quad (\text{E.76})$$

The only Kuhn-Tucker condition that is altered is equation E.14. It is replaced with:

$$0 = \eta [z_1^0 - \bar{z}_1^0] \quad (\text{E.77})$$

Our previous developments only have to be slightly adjusted. In cases where the minimum wage is not binding, ie whenever $\eta = 0$ in Condition E.77 (such that $z_1^0 \geq \bar{z}_1^0$), then θ has to be strictly positive in Condition E.10. Combined with Condition E.13, this implies that $n_1^0 = \frac{e^{\lambda \bar{z}_1^0}}{h}$.

Alternatively, if the minimum wage binds such that the Lagrange multiplier $\eta \geq 0$ (and $z_1^0 = \bar{z}_1^0$), then θ is no longer constrained to be strictly positive in Conditions E.9, E.10 and E.13. More precisely, if $\theta = 0$, then n_1^0 can become strictly smaller than $\frac{e^{\lambda \bar{z}_1^0}}{h}$ (Condition E.13). Notice in particular that since the bound is higher than previously ($\bar{z}_1^0 \geq 0$ whenever the institutional minimum wage is not degenerate), all else equal, the organizational distortion is likely to occur more often. However, here, and in contrast with the baseline case without minimum wage, production workers are always “productive” (in the sense that they are always able to solve problems) since $z_1^0 = \bar{z}_1^0 > 0$.

It is possible to investigate more generally how the optimal choices for n_1^0 , z_1^1 and ϕ evolve with $\bar{z}_1^0$ as solutions to firms’ cost minimization programs:¹⁷

– In the regime where $\eta \geq 0$ and $\theta > 0$, we get that:

$$\frac{dz_1^0}{d\bar{z}_1^0} = 1, \quad \frac{dn_1^0}{d\bar{z}_1^0} = \lambda n_1^0 > 0, \quad \frac{dz_1^1}{d\bar{z}_1^0} = -\frac{hq}{Ae^{\lambda \bar{z}_1^0} - hq} < 0, \quad \frac{d\phi}{d\bar{z}_1^0} = -\frac{Akche^{\lambda \bar{z}_1^0}}{(Ae^{\lambda \bar{z}_1^0} - hq)^2} < 0$$

– In the regime where $\eta \geq 0$ and $\theta = 0$, we get that:

$$\frac{dz_1^1}{d\bar{z}_1^0} = -\frac{q}{Ae^{-\lambda \bar{z}_1^0}} \left[1 + \frac{\lambda q[\lambda q(c\bar{z}_1^0 + 1) + 2Ac]}{\sqrt{\lambda^2 q^2 (c\bar{z}_1^0 + 1)^2 + 4\bar{z}_1^0 c \lambda q (c\bar{z}_1^0 + 1)}} \right] < 0,$$

$$\frac{dz_1^0}{d\bar{z}_1^0} = 1, \quad \frac{dn_1^0}{d\bar{z}_1^0} = -\frac{\lambda q e^{-\lambda \bar{z}_1^0}}{A(1 - e^{-\lambda \bar{z}_1^0})^2} \cdot \frac{dz_1^1}{d\bar{z}_1^0} > 0, \quad \frac{d\phi}{d\bar{z}_1^0} = -\frac{\lambda k(c\bar{z}_1^0 + 1)e^{-\lambda \bar{z}_1^0}}{A(1 - e^{-\lambda \bar{z}_1^0})^2} \cdot \frac{dz_1^1}{d\bar{z}_1^0} > 0$$

Firms with 2 or more layers. The Lagrangian is altered as:

$$\begin{aligned} \mathcal{L}_L \left((n_L^0)_l, (z_L^l)_l \right) &= k \left[\sum_{l=0}^L n_L^l (cz_L^l + 1) \right] - \phi \left[n_L^1 A(1 - e^{-\lambda z_L^L}) - q \right] \\ &\quad - \sum_{l=0}^{L-1} \theta_l \left[\frac{e^{\lambda z_L^l}}{h} n_L^{l+1} - n_L^0 \right] - \eta [z_L^0 - \bar{z}_L^0] \end{aligned} \quad (\text{E.78})$$

This means that condition E.31 is simply altered as:

$$0 = \eta [z_L^0 - \bar{z}_L^0]$$

The transposition of our previous developments and proofs is almost straightforward, and all previous results

¹⁷Recall that $\frac{dn_1^1}{d\bar{z}_1^0} = 0$ since $n_1^1 = 1$.

remain valid. To be precise, the regime where $\eta = 0$ is unaltered, while the analysis for the regime where $\eta > 0$ is altered in the following way:

- We still have $n_L^0 = \frac{e^{\lambda z_L^{L-1}}}{h}$ but $n_L^1 = e^{\lambda(z_L^{L-1} - \bar{z}_L^0)} = hn_L^0 e^{-\lambda \bar{z}_L^0}$;
- Equation E.67 is altered as:

$$\sum_{l=2}^{L-1} n_L^l kc + n_L^0 k \left[hce^{-\lambda \bar{z}_L^0} + \lambda h(cz_L^1 + 1) + \lambda(c\bar{z}_L^0 + 1) \right] = \phi \lambda q$$

- Equations E.70 to E.74 still hold such that the remainder of the proof also holds.

The comparative statics with respect to $\bar{z}_L^0$ are more complex than in the case where $L = 1$:

- For $L = 2$, differentiating Equations E.67 to E.69 leads to the following results:

$$\begin{aligned} \frac{dz_2^2}{d\bar{z}_2^0} - \frac{dz_2^1}{d\bar{z}_2^0} &= -e^{\lambda z_2^2} \frac{dz_2^1}{d\bar{z}_2^0} \\ \frac{d\phi}{d\bar{z}_2^0} &= -\phi \lambda e^{\lambda z_2^2} \frac{dz_2^1}{d\bar{z}_2^0} \\ kcn_2^0(1 - he^{-\lambda \bar{z}_2^0}) &= - \left[\phi \lambda q(1 + e^{\lambda z_2^2}) + kchn_2^0 \right] \frac{dz_2^1}{d\bar{z}_2^0} \end{aligned}$$

The last equation implies $\frac{dz_2^1}{d\bar{z}_2^0} < 0$ such that $\frac{dz_2^2}{d\bar{z}_2^0} > 0$ and $\frac{d\phi}{d\bar{z}_2^0} > 0$

- For $L > 2$, no general results hold: see Figure 5.

E.3.2 Partial Equilibrium Effects of Minimum Wage: Proofs

This Section of the appendix contains the proof of Proposition 1 from Section 4, as well as the proposition and the proof for Proposition 5, which is mentioned in the same section of the paper. In contrast to the previous analyses in Section E.3.1, the optimal level of production q is endogenized (while k remains exogenously determined in partial equilibrium).

Proof of Proposition 1 : Suppose that a firm is indifferent between producing with M and L layers, and that the minimum wage is binding on z_M^0 but not on z_L^0 . If so, then an increase in the minimum wage reduces profits for a firm with M layers but not for a firm with L layers. We start with:

$$\frac{d\pi_x}{d\bar{w}} = \frac{d\pi_x}{dz_x^0} = \frac{\partial \pi_x}{\partial z_x^0} + \frac{\partial \pi_x}{\partial n_x^0} \frac{dn_x^0}{dz_x^0} + \sum_{l=1}^x \frac{\partial \pi_x}{\partial z_x^l} \frac{dz_x^l}{dz_x^0} + \frac{\partial \pi_x}{\partial q_x} \frac{dq_x}{dz_x^0}, \quad \forall x \in \{L, M\}$$

However, an envelope condition tells us that $\frac{\partial \pi_x}{\partial n_x^0} = \frac{\partial \pi_x}{\partial z_x^l} = \frac{\partial \pi_x}{\partial q_x} = 0$. Therefore, we have:

$$\frac{d\pi_x}{d\bar{w}} = \frac{\partial \pi_x}{\partial z_x^0} = -\frac{\partial C_x}{\partial z_x^0} = -(n_x^0 kc - \lambda n_x^1 k(cz_x^1 + 1)) = -\eta_x$$

which is zero for L but negative for M . As a result, we know that $\frac{d\pi_M}{d\bar{w}} < 0$ and $\frac{d\pi_L}{d\bar{w}} = 0$, and so a firm that is on the margin between the two will shift to production with L layers.

If, on the other hand the minimum wage is binding on both L and M , the result is ambiguous, because $\eta_M > 0$ and $\eta_L > 0$. Our intuition is that the multiplier η should be larger for an organization with more layers, where the minimum wage should be more binding, but at this stage, we are unable to prove that intuition.

Proposition 5. *Given a number L of layers, a (marginal) increase in a binding minimum wage has the following impact on firms, depending on their optimal hierarchical organization:*

- For $L = 1$, if n_1^0 is below the level which would saturate the time constraint of the manager,¹⁸ z_1^1 and n_1^0 decrease and the marginal cost of production, which we denote by ϕ , increases with the minimum wage if and only if $z_1^1 > \frac{\sigma-1}{\lambda}$. Intuitively, firms in this regime need to charge higher prices and contract production to limit the costs generated by higher minimum wages.
- For $L = 1$, if n_1^0 saturates the time constraint of the manager, then z_1^1 and n_1^0 increase and ϕ decreases with the minimum wage. This means that the optimal strategy in this regime is the opposite of the previous: firms need to lower prices and increase their volume of production in order to absorb the minimum wage constraint.
- For $L \geq 2$, if $h \leq 0.25$ and $z_L^L > \frac{\sigma-1}{\lambda}$,¹⁹ then n_L^0 and all z_L^l increase while ϕ decreases with a marginally-binding minimum wage.

However, as the minimum wage continues to increase and becomes more than marginally binding on z_L^0 , the results become ambiguous. Indeed, the increase in production costs tends to lower the optimal output target q by an amount that becomes sufficient to offset the initial tendency to increase n_L^0 . This means that the optimal strategy in this regime is first to increase volumes in order to achieve lower marginal costs and decrease prices, up to a point where the converse holds true (until it becomes optimal to remove a hierarchical layer to save on fixed production costs, see Proposition 1).

In addition, for $L \geq 2$, the effect on the number of workers in higher layers of the firm is always ambiguous, as is the effect on total firm size.

Proof of Proposition 5: There are 3 different regimes to consider: they correspond to those where the minimum wage constraint is binding.

Proof for the case where $L = 1$, $\eta > 0$ and $\theta = 0$ such that $z_1^0 = \bar{z}_1^0$ and $n_1^0 \leq \frac{e^{\lambda \bar{z}_1^0}}{h}$. In this case, we have 3 first-order conditions which can be combined with the profit-maximization Condition 17, $q = \alpha k \left(\frac{\sigma}{\sigma-1} \phi \right)^{-\sigma}$, to deliver:

$$k(c\bar{z}_1^0 + 1) = \phi A(1 - e^{-\lambda z_1^1}) \quad (\text{E.79})$$

$$kc = \phi A \lambda e^{-\lambda z_1^1} n_1^0 \quad (\text{E.80})$$

$$A(1 - e^{-\lambda z_1^1}) n_1^0 = \alpha k \left(\frac{\sigma}{\sigma-1} \phi \right)^{-\sigma} \quad (\text{E.81})$$

Differentiating each condition with respect to $\bar{z}_1^0$, i.e. the parameter corresponding to the lower bound in the minimum wage constraint, we obtain, first, $\frac{dz_1^0}{d\bar{z}_1^0} = 1$, and the following set of equations:

$$kc = A(1 - e^{-\lambda z_1^1}) \frac{d\phi}{d\bar{z}_1^0} + \phi A \lambda e^{-\lambda z_1^1} \frac{dz_1^1}{d\bar{z}_1^0} \quad (\text{E.82})$$

$$\phi \lambda n_1^0 \frac{dz_1^1}{d\bar{z}_1^0} = n_1^0 \frac{d\phi}{d\bar{z}_1^0} + \phi \frac{dn_1^0}{d\bar{z}_1^0} \quad (\text{E.83})$$

$$A \lambda e^{-\lambda z_1^1} n_1^0 \frac{dz_1^1}{d\bar{z}_1^0} + A(1 - e^{-\lambda z_1^1}) \frac{dn_1^0}{d\bar{z}_1^0} = -\sigma \alpha k \left(\frac{\sigma}{\sigma-1} \right)^{-\sigma} \phi^{-\sigma-1} \frac{d\phi}{d\bar{z}_1^0} \quad (\text{E.84})$$

¹⁸Technically, this implies that the Lagrange multiplier θ introduced in Appendix E.3.2 is constrained to 0.

¹⁹While this condition depends on an endogenous outcome of the model (z_L^L), we should note that, in our simulations, this corresponds to a weak assumption. As a consequence, many firms seem to fall into these cases in our later simulations.

Equation E.82 can be solved for:

$$\frac{d\phi}{d\bar{z}_1^0} = \frac{kc}{A(1 - e^{-\lambda z_1^1})} - \frac{\phi \lambda e^{-\lambda z_1^1}}{1 - e^{-\lambda z_1^1}} \frac{dz_1^1}{d\bar{z}_1^0}.$$

Equation E.83 then rearranges as:

$$A(1 - e^{-\lambda z_1^1}) \frac{dn_1^0}{d\bar{z}_1^0} = A\lambda n_1^0 \frac{dz_1^1}{d\bar{z}_1^0} - \frac{kc n_1^0}{\phi}$$

and Equation E.84 rearranges to deliver:

$$\phi A \lambda \left(1 + (1 - \sigma)e^{-\lambda z_1^1}\right) \frac{dz_1^1}{d\bar{z}_1^0} = (1 - \sigma)kc \iff \frac{\phi A \lambda}{kc} \frac{dz_1^1}{d\bar{z}_1^0} = \frac{\sigma - 1}{(\sigma - 1)e^{-\lambda z_1^1} - 1}$$

Since σ is typically calibrated at values that exceed 1, we obtain that $\frac{dz_1^1}{d\bar{z}_1^0}$ is of the same sign as $(\sigma - 1)e^{-\lambda z_1^1} - 1$. This implies:

$$\frac{dz_1^1}{d\bar{z}_1^0} < 0 \iff z_1^1 > \frac{\ln(\sigma - 1)}{\lambda},$$

which corresponds to our calibration to the French data.

Next, substituting the rearranged form of Equation E.84 into Equation E.83, we have:

$$A(1 - e^{-\lambda z_1^1}) \frac{dn_1^0}{d\bar{z}_1^0} = \frac{kc n_1^0}{\phi} \left[\frac{\sigma + (1 - \sigma)e^{-\lambda z_1^1}}{(\sigma - 1)e^{-\lambda z_1^1} - 1} \right].$$

The numerator in the final square brackets is always positive, such that $\frac{dn_1^0}{d\bar{z}_1^0}$ is of the same sign as $(\sigma - 1)e^{-\lambda z_1^1} - 1$. This implies:

$$\frac{dn_1^0}{d\bar{z}_1^0} < 0 \iff z_1^1 > \frac{\ln(\sigma - 1)}{\lambda},$$

which, again, corresponds to our calibration to the French data.

Finally, Equation E.82 rearranges to deliver $A(1 - e^{-\lambda z_1^1}) \frac{d\phi}{d\bar{z}_1^0} = -\frac{kc}{(\sigma - 1)e^{-\lambda z_1^1} - 1}$ such that:

$$\frac{d\phi}{d\bar{z}_1^0} > 0 \iff z_1^1 > \frac{\ln(\sigma - 1)}{\lambda}.$$

□

Proof for the case where $L = 1$, $\eta > 0$ and $\theta > 0$ such that $z_1^0 = \bar{z}_1^0$ and $n_1^0 = \frac{e^{\lambda \bar{z}_1^0}}{h}$. There are two relevant first-order conditions (for z_1^1 and ϕ , since z_1^0 is fixed by the minimum wage), which can be combined with profit maximization to deliver:

$$kch = \phi A \lambda e^{\lambda(z_1^0 - z_1^1)} \tag{E.85}$$

$$A(1 - e^{-\lambda z_1^1}) e^{\lambda z_1^0} = \alpha h k \left(\frac{\sigma}{\sigma - 1} \phi \right)^{-\sigma} \tag{E.86}$$

Differentiating each equation with respect to z_1^0 , we find:

$$\frac{d\phi}{d\bar{z}_1^0} = \phi\lambda \left(\frac{dz_1^1}{d\bar{z}_1^0} - 1 \right) \quad (\text{E.87})$$

$$\phi\lambda \left[e^{-\lambda z_1^1} \frac{dz_1^1}{d\bar{z}_1^0} + (1 - e^{-\lambda z_1^1}) \right] = -\sigma(1 - e^{-\lambda z_1^1}) \frac{d\phi}{d\bar{z}_1^0} \quad (\text{E.88})$$

Combining and rearranging, we get:

$$\frac{dz_1^1}{d\bar{z}_1^0} = \frac{(\sigma - 1)(1 - e^{-\lambda z_1^1})}{e^{-\lambda z_1^1} + \sigma(1 - e^{-\lambda z_1^1})} > 0$$

and substituting back into Equation E.87:

$$\frac{d\phi}{d\bar{z}_1^0} = \phi\lambda \left(\frac{(\sigma - 1)(1 - e^{-\lambda z_1^1}) - e^{-\lambda z_1^1} - \sigma(1 - e^{-\lambda z_1^1})}{e^{-\lambda z_1^1} + \sigma(1 - e^{-\lambda z_1^1})} \right) = \phi\lambda \left(\frac{-1}{e^{-\lambda z_1^1} + \sigma(1 - e^{-\lambda z_1^1})} \right) < 0.$$

□

Proof for the case where $L \geq 2$ and $\eta \geq 0$, such that $z_L^0 = \bar{z}_L^0$ (and $n_L^0 = \frac{e^{\lambda z_L^{L-1}}}{h}$). General results can only be derived in the case where the minimum wage is only marginally binding, i.e. whenever $\eta = 0$ and $kcn_L^0 = \lambda n_L^1 k(cz_L^1 + 1)$.

There are four relevant first-order conditions, to be combined with profit maximization:

$$ce^{\lambda(z_L^l - z_L^{l-1})} = \lambda(cz_L^{l+1} + 1) \quad (\text{E.89})$$

$$\sum_{l=0}^{L-1} n_L^l kc + \lambda n_L^0 k(cz_L^0 + 1) = \lambda \alpha k \left(\frac{\sigma}{\sigma - 1} \right)^{-\sigma} \phi^{1-\sigma} \quad (\text{E.90})$$

$$kc = \phi A \lambda e^{-\lambda z_L^L} n_L^0 \quad (\text{E.91})$$

$$A(1 - e^{-\lambda z_L^L}) n_L^0 = \alpha k \left(\frac{\sigma}{\sigma - 1} \right)^{-\sigma} \phi^{-\sigma} \quad (\text{E.92})$$

Differentiating each equation with respect to $\bar{z}_L^0$, we obtain:

$$\forall l \in \{1, \dots, L-2\}, \quad \frac{dz_L^{l+1}}{d\bar{z}_L^0} = \frac{n_L^l}{n_L^{l+1}} \left(\frac{dz_L^l}{d\bar{z}_L^0} - \frac{dz_L^{l-1}}{d\bar{z}_L^0} \right) \quad (\text{E.93})$$

$$\phi \lambda q \frac{dz_L^{L-1}}{d\bar{z}_L^0} + kcn_L^{L-1} \left(\frac{dz_L^{L-1}}{d\bar{z}_L^0} - \frac{dz_L^{L-2}}{d\bar{z}_L^0} \right) = (1 - \sigma) q \frac{d\phi}{d\bar{z}_L^0} \quad (\text{E.94})$$

$$\frac{d\phi}{d\bar{z}_L^0} = \phi \lambda \left(\frac{dz_L^L}{d\bar{z}_L^0} - \frac{dz_L^{L-1}}{d\bar{z}_L^0} \right) \quad (\text{E.95})$$

$$A\lambda(1 - e^{-\lambda z_L^L}) n_L^0 \frac{dz_L^{L-1}}{d\bar{z}_L^0} + A\lambda n_L^0 e^{-\lambda z_L^L} \frac{dz_L^L}{d\bar{z}_L^0} = \frac{-\sigma q}{\phi} \frac{d\phi}{d\bar{z}_L^0} \quad (\text{E.96})$$

Combining Equation E.95 and E.96, we find:

$$A\lambda n_L^0 \left[\frac{dz_L^{L-1}}{d\bar{z}_L^0} + \frac{e^{-\lambda z_L^L}}{\phi \lambda} \frac{d\phi}{d\bar{z}_L^0} \right] = \frac{-\sigma q}{\phi} \frac{d\phi}{d\bar{z}_L^0}$$

and therefore $A\lambda n_L^0 \frac{dz_L^{L-1}}{d\bar{z}_L^0} = -\frac{\sigma q + A n_L^0 e^{-\lambda z_L^L}}{\phi} \frac{d\phi}{d\bar{z}_L^0}$, so $\frac{dz_L^{L-1}}{d\bar{z}_L^0}$ and $\frac{d\phi}{d\bar{z}_L^0}$ take opposite signs. Plugging this into Equation E.96 and rearranging, we obtain $\frac{dz_L^L}{d\bar{z}_L^0} = \frac{(1-\sigma)(1-e^{-\lambda z_L^L})}{\phi \lambda} \frac{d\phi}{d\bar{z}_L^0}$, so the sign of $\frac{dz_L^L}{d\bar{z}_L^0}$ is the same as the

sign of $\frac{dz_L^{L-1}}{d\bar{z}_L^0}$ and opposite to $\frac{d\phi}{d\bar{z}_L^0}$.

Next, we can plug the solution for $\frac{dz_L^{L-1}}{d\bar{z}_L^0}$ into Equation E.94:

$$kcn_L^{L-1} \left(\frac{dz_L^{L-2}}{d\bar{z}_L^0} - \frac{dz_L^{L-1}}{d\bar{z}_L^0} \right) = \left[-1 + (\sigma - 1)e^{-\lambda z_L^L} \right] q \frac{d\phi}{d\bar{z}_L^0}.$$

Therefore, $\left(\frac{dz_L^{L-2}}{d\bar{z}_L^0} - \frac{dz_L^{L-1}}{d\bar{z}_L^0} \right)$ takes the same sign as $\frac{dz_L^{L-1}}{d\bar{z}_L^0}$ if and only if $1 + (1 - \sigma)e^{-\lambda z_L^L} > 0$. If this condition is satisfied, we have $\frac{dz_L^{L-2}}{d\bar{z}_L^0} = X_{L-2} \frac{dz_L^{L-1}}{d\bar{z}_L^0}$ where $X_{L-2} > 1$. We also know that $\frac{n_L^l}{n_1^{l+1}}$ is increasing in l for $l \in \{1, \dots, L-2\}$, and as before we have $\frac{n_L^1}{n_2^1} \geq 4$, so that $\frac{n_L^l}{n_1^{l+1}} \geq 4$ for all $l \leq L-2$. Then we can rely on a descending chain of inductive reasoning:

- if $\frac{dz_L^l}{d\bar{z}_L^0} = X_l \frac{dz_L^{l+1}}{d\bar{z}_L^0}$, where $X_l \geq \frac{1}{2}$, for $l \in \{2, \dots, L-2\}$, then $\frac{dz_L^{l-1}}{d\bar{z}_L^0} = \frac{dz_L^l}{d\bar{z}_L^0} - \frac{n_L^{l+1}}{n_L^l} \frac{dz_L^{l+1}}{d\bar{z}_L^0} = \left(1 - \frac{n_L^{l+1}}{n_L^l X_l} \right) \frac{dz_L^l}{d\bar{z}_L^0}$
- then $X_{l-1} \equiv 1 - \frac{n_L^{l+1}}{n_L^l X_l} \geq \frac{1}{2}$, so $\frac{dz_L^l}{d\bar{z}_L^0}$ takes same sign for all $l \in 1, \dots, L$.

Finally, we have $\frac{n_L^1}{n_2^1} \left(\frac{dz_L^1}{d\bar{z}_L^0} - \frac{dz_L^0}{d\bar{z}_L^0} \right) = \frac{dz_L^2}{d\bar{z}_L^0} = \frac{1}{X_1} \frac{dz_L^1}{d\bar{z}_L^0}$ where $\frac{dz_L^0}{d\bar{z}_L^0} > 0$, and therefore $\frac{dz_L^1}{d\bar{z}_L^0} = \frac{\frac{n_L^1}{n_2^1}}{\frac{n_L^1}{n_2^1} - \frac{1}{X_1}} \frac{dz_L^0}{d\bar{z}_L^0} > 0$.

Therefore, $\frac{dz_L^l}{d\bar{z}_L^0} > 0$ for all $l \in 1, \dots, L$, and $\frac{d\phi}{d\bar{z}_L^0} < 0$. However, $\frac{dn_L^l}{d\bar{z}_L^0}$ is of ambiguous sign except for $\frac{dn_L^0}{d\bar{z}_L^0} = \lambda n_L^0 \frac{dz_L^{L-1}}{d\bar{z}_L^0} > 0$.

It is important to note that, in the case of $L \geq 2$, we need to assume that the minimum wage is only marginally binding for the proof to work; in that case, $\eta = 0$, and we can use $kcn_L^0 = \lambda n_L^1 k(cz_L^1 + 1)$ to simplify Equation E.90 and Equation E.94 in the proof. Once the minimum wage is strictly above the value of z_L^0 that a firm would choose in the absence of the minimum wage, that condition no longer holds, and there will be a tendency for firms to decrease q by more than the above analysis suggests, leading to lower values of z_L^l and n_L^l .

□

F Additional Simulation Results

F.1 Detailed Calibration Results Across Organizations

This appendix presents additional simulation results from Section 5.2. Specifically, Table F1 reports average worker wages and average firm size, both overall and broken down by the number of hierarchical layers within firms, as well as by layer within each firm type. The structure of the table mirrors that of regression Table 1 and Panel (A) of Table 2;²⁰ it is worth noting, however, that compositional effects may differ slightly between the regression and simulation results: in the regression tables, firms are classified by their number of layers as observed in the pre-regression period (2002), so the estimates incorporate the effects of possible subsequent delayering. In contrast, in the simulation results, firms with 1, 2, or 3 layers are grouped based on their status in the simulated equilibrium, meaning the composition of these groups may differ across scenarios. Despite this distinction, the comparisons remain rich and informative.

In Panel (A) of Table F1, the average wage per worker - which stands in for overall labor costs - generally increases with the tightness of the minimum wage constraint in all types of firms, although the difference is not noticeable up to the second digit between the “no MW” and “baseline” scenarios among 3-layer firms, and barely noticeable in the full population. Furthermore, we obtain that at the bottom layer (l_0), this relationship is increasing in the minimum wage in all categories of firms, while it is non-monotonic in layers l_1 and l_2 . These results are strikingly similar to those obtained with the DDD estimator in Table 1: insignificant when estimated across all layers, positive and significant at layer l_0 and non-monotonic across populations of firms in layers l_1 and l_2 .

In Panel (B) of Table F1, we find that the relationship between the tightness of the minimum wage constraint and firm size is negative in the pooled sample (except for the 24% increase), but non-monotonic for two-layer firms: initially negative, then turning positive. In other firm categories, the relationship is positive. A similar pattern emerges for the bottom layer, l_0 , except that the relationship is uniformly positive for two-layer firms. While the fit is not perfect, these results align fairly well with the triple-difference regression estimates shown in Table 2: the overall effect on firm size in the pooled sample is significantly negative, but the layer-by-layer effects are insignificant and sometimes positive.

F.2 Simulated Productivity Adjustments as a Function of α

To better understand the firm-level adjustments that underlie the aggregate Q - and R - productivity patterns discussed in Section 5.2, it is interesting to analyze the relationships between α , the idiosyncratic demand shifter which (fully) captures firm level heterogeneity, and the different firm-level indicators of performance and productivity. Figure F1 plots profits, sales per worker as well as the Q - and R -productivity indices as a function of α .²¹

Panel (A) shows that minimum-wage increases generally reduce profits, consistent with Machin et al. (2003), except for low- α firms facing intermediate increases, for which the equilibrium response of k can generate (extremely) modest profit gains. More broadly, the simulations indicate that moderate minimum-wage constraints have only limited adverse effects on profits. Furthermore, the firms facing the worst market conditions in terms of α only start being significantly selected out of the market for extremely stringent minimum wage constraints (“+24%”, scenario (6)). Panel (C) shows that minimum-wage increases uniformly reduce the measure of pure technical efficiency, the Q -productivity index. Moreover, this indicator deteriorates

²⁰See Appendix F.3 for a broader assessment of the model’s fit.

²¹The revenue-productivity index simply corresponds to the Q - productivity index multiplied by firm-level prices (Caliendo and Rossi-Hansberg, 2012).

Table F1: General Equilibrium Implications of Minimum Wages Across Firms & Layers

(A) Focus on Wages [Labor Costs]

	(1) Baseline	(2) No MW (-4%)	(3) + 4%	(4) + 8%	(5) + 16%	(6) + 24%
	<i>Average wage (in €/hour)</i>					
All firms	20.29	20.28	20.52	21.06	21.96	23.01
1-layer firms	20.95	20.94	20.99	21.03	21.55	23.00
2-layer firms	20.42	20.40	20.58	21.06	22.01	23.01
3-layer firms	20.14	20.14	20.40	20.96	-	-
	<i>Average wage in Layer l_0 (in €/hour)</i>					
All firms	19.16	19.15	19.43	20.10	21.52	23.00
1-layer firms	20.95	20.94	20.99	21.03	21.55	23.00
2-layer firms	19.25	19.23	19.39	20.03	21.51	23.00
3-layer firms	18.92	18.92	19.29	20.03	-	-
	<i>Average wage in Layer l_1 (in €/hour)</i>					
All firms	28.02	28.00	28.77	30.67	30.06	24.64
1-layer firms (entrepreneurs)	34.66	34.65	34.70	34.70	34.72	30.24
2-layer firms	29.22	29.16	29.85	30.69	30.06	24.64
3-layer firms	27.12	27.12	27.49	28.28	-	-
	<i>Average wage in Layer l_2 (in €/hour)</i>					
All firms with $L \geq 2$	31.78	31.78	31.84	32.37	-	-
2-layer firms (entrepreneurs)	40.37	40.33	41.15	42.71	42.80	36.06
3-layer firms	31.78	31.78	31.84	32.37	-	-
	<i>Average wage in Layer l_3 (in €/hour)</i>					
3-layer firms (entrepreneurs)	45.22	45.22	45.28	45.79	-	-

(B) Focus on Firms' Sizes and Organizations

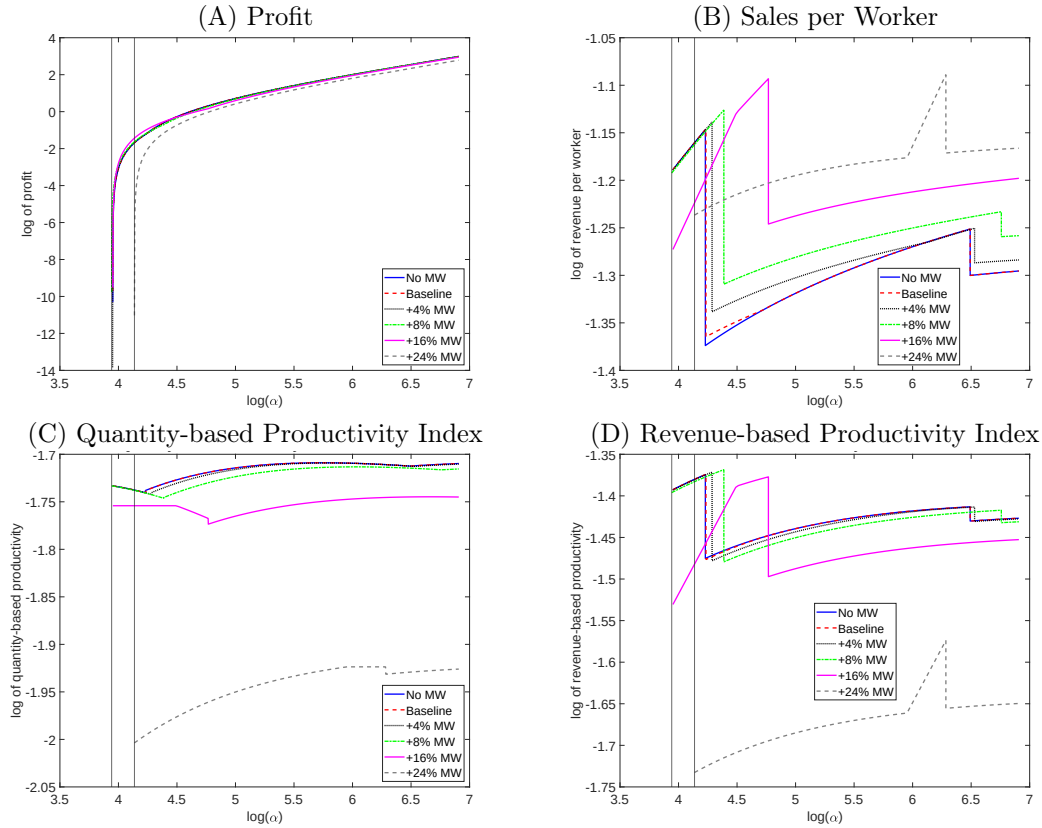
	(1) Baseline	(2) No MW (-4%)	(3) + 4%	(4) + 8%	(5) + 16%	(6) + 24%
<i>(as shares of total number of agents)</i>	<i>More details on the structure of firm population</i>					
Agents in 1-layer firms	0.03	0.03	0.03	0.04	0.06	0.07
Agents in 2-layer firms	0.27	0.27	0.33	0.57	0.51	0.38
Agents in 3-layer firms	0.34	0.34	0.27	0.00	0.00	0.00
	<i>Average firm size (in jobs)</i>					
All firms	92.38	92.43	90.39	84.98	82.99	86.11
1-layer firms	12.61	12.59	12.61	12.64	17.10	15.04
2-layer firms	87.11	87.14	92.81	103.07	129.66	128.97
3-layer firms	215.52	215.51	217.15	248.39	-	-
	<i>Average size of Layer l_0 (in jobs)</i>					
All firms	80.00	80.01	79.16	76.36	77.70	84.27
1-layer firms	11.61	11.59	11.61	11.64	16.10	14.04
2-layer firms	75.99	75.95	81.37	92.21	121.13	126.46
3-layer firms	184.58	184.58	188.40	220.93	-	-
	<i>Average size of Layer l_1 (in jobs)</i>					
All firms	10.95	10.99	10.04	8.04	4.82	1.34
1-layer firms	1	1	1	1	1	1
2-layer firms	10.13	10.19	10.43	9.86	7.52	1.50
3-layer firms	26.65	26.65	24.72	23.62	-	-
	<i>Average size of Layer l_2 (in jobs)</i>					
All firms with $L \geq 2$	1.77	1.77	1.58	1.01	1	1
2-layer firms	1	1	1	1	1	1
3-layer firms	3.28	3.28	3.04	2.85	-	-
	<i>Average size of Layer l_3 (in jobs)</i>					
3-layer firms	1	1	1	1	-	-

Notes: This table presents simulations of alternative scenarios where the minimum wage is either removed (Column (2)), or increased by 4%, 8%, 16% or 24%, respectively, in Columns (3) to (6), relative to the baseline calibration presented in Column (1). The baseline scenario is described in detail in Section 5.1.

much earlier than the other productivity measures, with sizeable declines already appearing in scenario (5) (the “+16%” scenario).

The effects of minimum-wage increases on sales per worker (Panel B) or *Revenue* productivity (Panel D) differ markedly from those on technical efficiency. For low values of α — which correspond to the most populated section of the support of the distribution in our calibration results — minimum-wage increases often raise revenue productivity, although the response is highly heterogeneous and non-monotonic. As shown in Table 6, these heterogeneous firm-level responses translate into positive aggregate effects in all scenarios calibrated to the French economy in 2006, but the complexity of the underlying patterns may help explain why the empirical literature has reached mixed conclusions regarding the effects of minimum wages on productivity, with estimates varying across institutional settings and the magnitude of the policy shock (Rizov et al., 2016; Riley and Rosazza Bondibene, 2017). These heterogeneous results are driven by firms’ endogenous organizational responses, which tend to counteract unfavorable demand conditions (low α). When demand conditions are unfavorable, firms reorganize by adopting smaller and flatter hierarchies, thereby increasing labor productivity and partially offsetting the adverse effects of low α . Consequently, revenue productivity inherits the non-monotonicities and kinks of the endogenous cost function shown in Panel (B) of Figure 5. In particular, and in contrast to Melitz (2003), firms facing the least favorable demand conditions are not necessarily the least productive, because they try and counterbalance them by remaining small and endogenously adjusting their organizations in order to achieve higher productivity, although the cost of this strategy (in terms of skill upgrading) ultimately becomes too large and affects profits heavily.

Figure F1: Firms’ Selection and Performance as a Function of α



Notes: This figure is derived from simulations of the calibrated model, in different scenarios in terms of the minimum wage. The parameters are calibrated to the following values: $\lambda = 1$, $c = 0.220$, $h = 0.193$, $A = 0.240$. In Panel (C), the index of “Quantity-based Productivity” corresponds to $q/C_L(q, 1)$, as in Equation 14. In Panel (D), the index of “Revenue-based Productivity” corresponds to $pq/C_L(q, 1)$.

F.3 Assessment of Model Performance on Non-targeted Data Moments

Table F2 combines the regression estimates reported in Section 3.4 with the simulation results presented in Table 6. It evaluates the model’s ability to reproduce a set of non-targeted dynamic moments, thereby providing a powerful assessment of its empirical performance. By contrast, all moments used to identify the model parameters are static, as reported in Table 5. This table is discussed in Section 5.2.

Table F2: Model Performance on Non-targeted Data Moments

	(1)	(2)	(3)	(4)	(5)	(6)
		Simulation results			DDD impact	
	Baseline	Baseline -4% (no MW)	Baseline +4%	Av. impact 4% increase	Exposure +4% x SH^{MW}	Total
Av. firm size (jobs)	92.38	92.43	90.39	-1.1%	-0.3%	-0.9%
Layer l_0 (jobs)	80.00	80.01	79.16	-0.5%	NS	-1.8%
Av. nb layers in firms	1.91	1.91	1.82	-0.05	-0.01	-0.01
Av. wage [labor cost] in firms	20.29	20.28	20.52	+0.6%	NS	+1.4%
Av. wage [labor cost] at l_0	19.16	19.15	19.43	+0.8%	+0.2%	+1.2%
Av. revenue / worker (€/hour)	28.42	28.37	28.90	+0.9%	+0.4%	+0.9%

Notes: This table compares the simulation results of scenarios where the minimum wage is either removed, i.e. decreased by 4% relative to the baseline scenario, or increased by 4% (Table 6), with regressions results associated with a 4% shock on the MW (similar to that experienced by GMR2 firms), for the average sample firm in terms of share of workers paid at the MW (see Section 3.4).

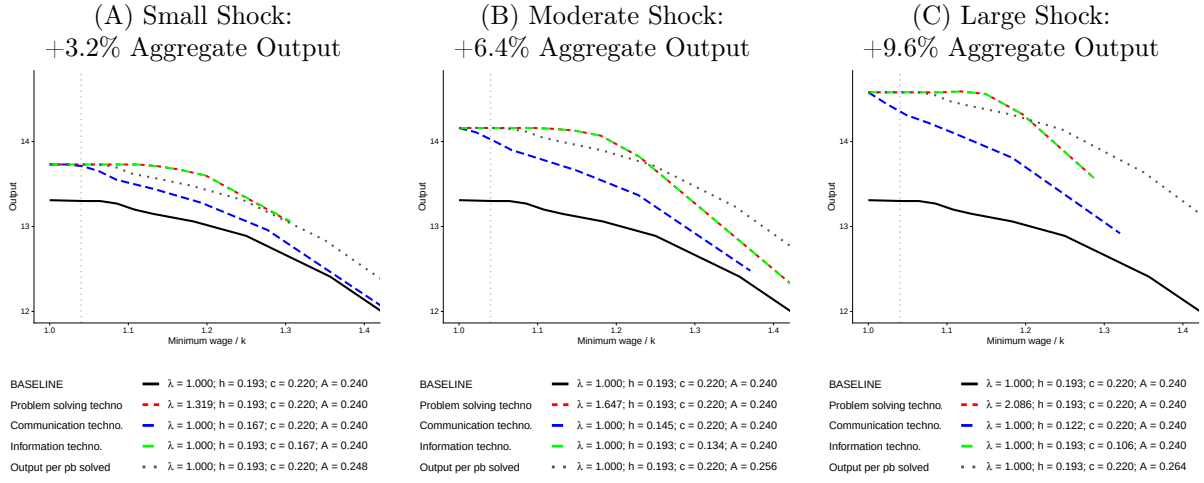
F.4 Varying the Magnitude of Technological Shocks

Figure F2 reports the detailed simulation results for technological shocks of varying magnitudes. The main findings, focusing on their effects on aggregate output, are summarized in the last paragraph of Section 5.3. Panel (A) simply replicates the results from Panel (A) in Figure 8 for comparison. In Panel (B), the technological shock is calibrated to generate a productivity premium twice as large—6.4% instead of 3.2%—in the “no minimum wage” scenario. Panel (C) considers a tripled shock, yielding a 9.6% productivity premium. The precise values of the underlying technological parameters used to achieve these targets are provided in the legend of Figure F2. As in Figure 8, when the aggregate productivity parameter A is increased, the resulting curves shift upward but remain homothetic to those in the baseline scenario across all panels. Consequently, regardless of the magnitude of the technological shock affecting A , the relative cost of the minimum wage constraint remains unchanged compared to the baseline scenario described in Section 5.1. Economies whose curves are steeper than this benchmark should be interpreted as experiencing a higher relative cost of the minimum wage constraint, while flatter curves indicate a lower relative cost.

With this benchmark in mind, as discussed in the main text, we find that the relative cost of minimum wage constraints tends to be higher in economies characterized by efficient communication technologies (i.e., low h) than in those characterized by higher overall productivity (i.e., higher A). Moreover, this disadvantage is amplified as the magnitude of the technological shock increases. For example, a minimum wage set 20% above the baseline labor cost leads to an output drop of just over 2% in economies with higher A in all panels. By contrast, in economies with improved communication efficiency (lower h), the same minimum wage constraint results in an output drop of 3.5% under the small technological shock in Panel (A), 4.9% under the medium shock in Panel (B), and 6.0% under the large shock in Panel (C). This amplification effect is intuitive: minimum wages reduce the advantages for firms of organizing themselves into hierarchies. The benefits lost are particularly significant when hierarchies are “technologically” efficient, i.e. when h is lowest.

In economies characterized by low values of c or high values of λ , the relative cost of moderate minimum wage constraints remains lower than in the A benchmark scenario. However, this advantage diminishes as

Figure F2: Impact of Minimum Wage For Higher Technological Shocks



Notes: This figure replicates Panel (A) of Figure 8 and further explores the role of technological shocks by varying their magnitudes: in Panel (B), the shocks are calibrated to generate an economy-wide productivity premium of 6.4% in the absence of a minimum wage, while in Panel (C), the calibration yields a productivity premium of 9.6%. See legend for parameter values.

the magnitude of the technological shocks increases. For instance, when the minimum wage is set 20% above the baseline price of labor (k), output declines by 1.0% in the small-shock scenario of Panel (A), 1.3% in the medium-shock scenario of Panel (B), and 2.0% in the large-shock scenario of Panel (C) – all relative to the corresponding no-minimum-wage economy. Importantly, this ranking reverses when minimum wage constraints become more stringent: in the upper range, large shocks to c or λ tend to amplify the aggregate output losses from minimum wage regulations, to the point that the resulting performance is worse than under the A benchmark. For example, when the minimum wage reaches $1.3k$, output falls by 4.6% in the small-shock scenario of Panel (A), by 6.3% in Panel (B), and by 7.2% in Panel (C). By contrast, in the benchmark scenario where the shock affects parameter A , output declines by just under 5% across all three cases.

G Sensitivity Analyses of Simulation Results

This Appendix contains a series of sensitivity analyses of the calibration and simulation results from Section 5. Our main goal is to assess whether our results can be safely interpreted as representative of the French economy. We present the moments used for each scenario in Panel (A) of Table G1, and the values actually attained in each simulated model in Panel (B) of Table G1. The results of the calibrations give us the parameter values found in Panel (C) of Table G1. Lastly, Figure G1 contains the simulation results associated with each calibration.

In a first set of calibrations, in Columns (i) to (iv) of Table G1, we consider four alternative calibration strategies, based on increasingly comprehensive datasets, in order to check that our main results are not driven by some of the selections imposed by our identification strategy in our main estimation sample (see Appendix B):²² (i) one in which we include all firms in GMRs 1 through 5 with at least one (instead of five) paid employee; (ii) one in which we further include firms with at least 1 employee with a well defined but late RWT agreement (between 2002 and 2006) which were subject to SMIC rather than a GMR,²³ as well as the firms from (i); (iii) one in which we use sample (i) but also incorporate a moment capturing the percentage of firms that consist of just one self-employed individual. There is no dataset available which measures this share for 2006. Rather, we measure it from Insee’s FARE datasets for 2010 to 2016 (Insee, 2010–2016) and assume in this robustness check that the average value is approximately relevant for 2006. Lastly, specification (iv) relies on sample (ii) but also inserts the same estimate of self-employment as in specification (iii).²⁴ None of the other aspects of the calibration strategy are altered, with the exception of the firm death rate (δ), which is calculated to be consistent with the different scenarios: $\delta = 0.327$ in scenarios (i) and (iii), and $\delta = 0.370$ in scenarios (ii) and (iv). The values of parameters obtained in Panel (C) show that the fixed cost parameters f_E and f absorb most of the variations that are introduced, including the one corresponding to the insertion of an additional moment for self-employed entrepreneurs. In contrast, parameters c , h and A are only weakly affected, and the same holds true for the distribution of α (which only features thinner tails), such that the general equilibrium and simulation results are qualitatively preserved, as shown in Figure G1.

In a second set of calibrations, presented in Columns (v) to (vii) of Table G1, we return to the original calibration sample but modify the set of empirical moments used as targets. Column (v) relies on moments expressed in hours (or more accurately, in full-time equivalents based on 1,607 hours per year), rather than in jobs as in all previous calibrations. Column (vi) replaces the average wage moment with the Kaitz index (see Appendix A), while Column (vii) uses the ratio of the minimum to the average wage. In these last two cases, the model generates insufficient wage dispersion and fails to match the alternative empirical targets. Improving the model’s performance on wage outcomes would likely require introducing worker-level heterogeneity, as in Garicano and Rossi-Hansberg (2006). We leave this challenging extension for future research. In this series of experiments, the general equilibrium and simulation results are fully preserved, as shown in Figure G1.

²²In the main text, we use calibration targets drawn directly from our regression sample in order to be able to benchmark the model’s simulation results against the regression estimates — see Section 5.2 and Appendix F.3.

²³Recall that these firms are excluded from the regression analysis because, over the 2002–2006 period, their adjustments confound their responses to the reduction in working time and to changes in the minimum wage.

²⁴Due to the fact that the sum of the shares across all possible organizational forms has to sum to 1, the first two specifications actually contain an implicit moment which pools all firms with 0 and 1 hierarchical layers, while in specifications (iii) and (iv), the implicit moment only concerns 1-layer firms.

Table G1: Calibration Sensitivity Analyses

(A) Target Values of Moments

Calibration strategy	(i)	(ii)	(iii)	(iv)	(v)	(vi)	(vii)
Share of agents who are teachers	0.098	0.098	0.098	0.098	0.098	0.098	0.098
Share of workers in new firms	0.209	0.241	0.209	0.241	-	0.203	0.203
Share of hours in new firms	-	-	-	-	0.204	-	-
Share of 0-layer firms	-	-	0.170	0.170	-	-	-
Share of 2-layer firms	0.352	0.267	0.292	0.222	0.446	0.446	0.446
Share of 3-layer firms	0.161	0.099	0.134	0.082	0.228	0.228	0.228
Average firm size, jobs	66.17	28.10	66.17	28.10	-	93.48	93.48
Average size of 2-layer firms, jobs	77.60	42.53	77.60	42.53	-	87.23	87.23
Average size of 3-layer firms, jobs	210.30	122.90	210.30	122.90	-	214.00	214.00
Average firm size, hours/1,607	-	-	-	-	78.71	-	-
Average size of 2-layer firms, hours/1,607	-	-	-	-	74.78	-	-
Average size of 3-layer firms, hours/1,607	-	-	-	-	178.70	-	-
Average (hourly) wage (€)	19.70	20.04	19.70	20.04	20.21	-	-
Kaitz index (hourly wage)	-	-	-	-	-	0.540	-
MW costs/ Mean costs (per hour)	-	-	-	-	-	-	0.422
Share of firms' workers bound by MW	0.048	0.055	0.048	0.055	-	0.047	0.047
Share of firms' hours bound by MW	-	-	-	-	0.032	-	-

(B) Simulated Values of Moments

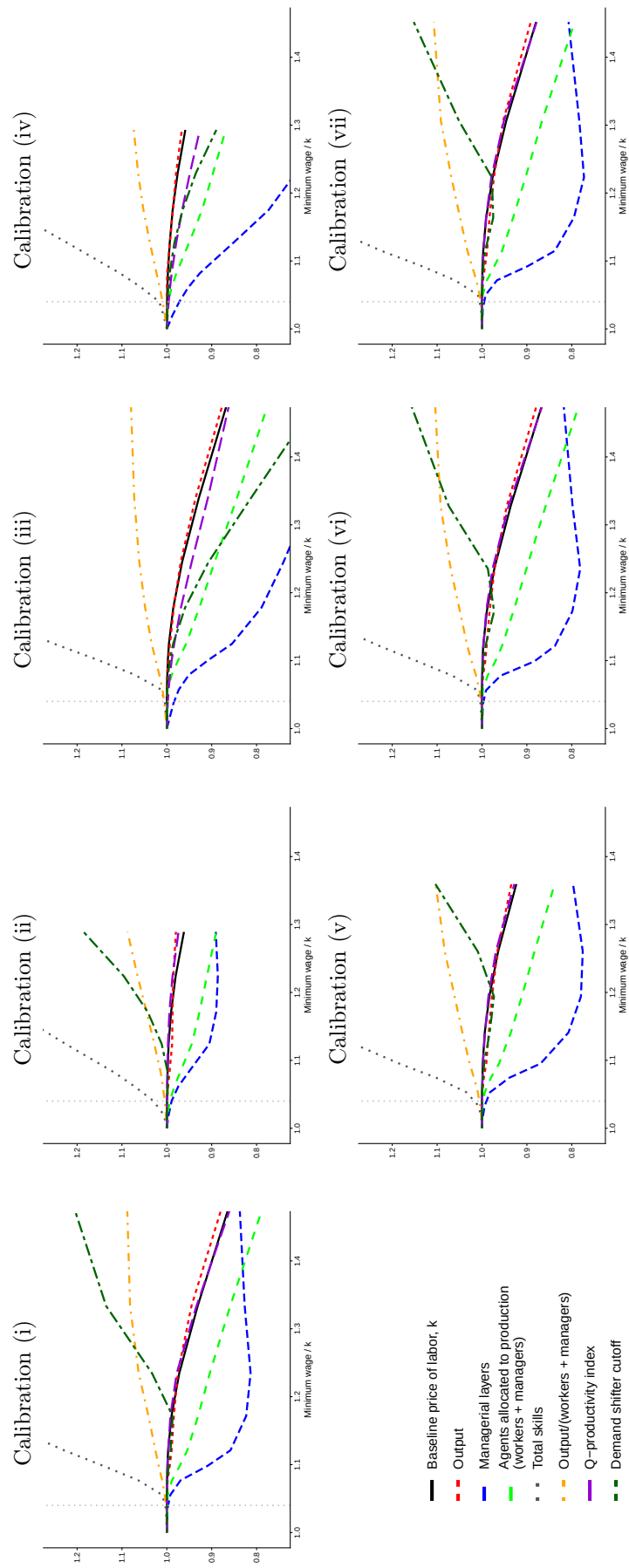
Calibration strategy	(i)	(ii)	(iii)	(iv)	(v)	(vi)	(vii)
Share of agents who are teachers	0.098	0.122	0.099	0.124	0.098	0.099	0.098
Share of workers in new firms	0.216	0.208	0.223	0.217	-	0.223	0.222
Share of hours in new firms	-	-	-	-	0.220	-	-
Share of 0-layer firms	-	-	0.172	0.166	-	-	-
Share of 2-layer firms	0.351	0.261	0.294	0.215	0.448	0.478	0.471
Share of 3-layer firms	0.160	0.097	0.134	0.082	0.228	0.215	0.229
Average firm size, jobs	66.48	29.99	66.00	30.23	-	90.25	87.48
Average size of 2-layer firms, jobs	77.14	47.12	78.11	46.60	-	89.60	82.82
Average size of 3-layer firms, jobs	210.28	132.30	211.79	142.94	-	202.65	195.89
Average firm size, hours/1,607	-	-	-	-	78.08	-	-
Average size of 2-layer firms, hours/1,607	-	-	-	-	74.23	-	-
Average size of 3-layer firms, hours/1,607	-	-	-	-	180.74	-	-
Average (hourly) wage (€)	19.71	20.60	19.70	19.69	20.22	-	-
Kaitz index (hourly wage)	-	-	-	-	-	0.959	-
MW costs/ Mean costs (per hour)	-	-	-	-	-	-	0.903
Share of firms' workers bound by MW	0.047	0.057	0.047	0.058	-	0.045	0.044
Share of firms' hours bound by MW	-	-	-	-	0.032	-	-

(C) Calibrated Values of Parameters

Calibration strategy	(i)	(ii)	(iii)	(iv)	(v)	(vi)	(vii)
Cost of knowledge acquisition, c	0.225	0.332	0.236	0.329	0.250	0.228	0.235
Time cost of problem communication, h	0.192	0.222	0.196	0.227	0.194	0.193	0.193
Output per problem solved, A (~ 100 s of €/hour)	0.221	0.234	0.283	0.260	0.230	0.247	0.240
Fixed cost of entry, f_E	4.576	4.566	63.357	21.918	4.383	5.159	4.986
Fixed cost of production, f	4.796	2.471	0.116	0.218	5.032	5.537	5.435
Minimum wage $\bar{w}$, relative to average wage	0.910	0.864	0.911	0.862	0.890	0.907	0.903
2nd-node of α distribution ($\alpha = 50$)	-11.784	-11.721	-11.511	-11.470	-12.108	-12.155	-12.132
3rd-node of α distribution ($\alpha = 500$)	-16.335	-13.880	-12.410	-12.720	-15.414	-15.458	-15.413
4th-node of α distribution ($\alpha = 1000$)	-16.754	-32.249	-79.444	-66.467	-25.187	-25.053	-25.145

Notes: Each column presents the values of moments or parameters computed on alternative datasets or data moments: Column (i) includes all firms in GMRs 1 through 5 with at least 1 paid employee; Column (ii) further includes firms with at least 1 employee and late RWT agreements (subject to SMIC), as well as the firms from (i); Column (iii) uses sample (i) but adds a moment from Insee's FARE dataset for 2010-16 which measures the percentage of firms that consist of one self-employed individual; and Column (iv) uses sample (ii) but with the same self-employment moment added. Due to the fact that the sum of the shares across all possible organizational forms has to sum to 1, Columns (i) and (ii) contain an implicit moment which pools all firms with 0 and 1 hierarchical layers, while in Columns (iii) and (iv), the implicit moment only concerns 1-layer firms. Lastly, Column (v) return to the main sample but relies on data moments that are specified in hours (or more accurately, in full time equivalents when normalized by 1,607 hours per year), while Column (vi) replaces the moment specified in terms of average wage with the Kaitz index (see Appendix A) and Column (vii) with the ratio of minimum wage to average wage.

Figure G1: Calibration Sensitivity Analyses: Results & Simulations



Notes: This figure presents simulation results for each of the calibrations presented in Table G1.

H Framework with Unemployment

In this Appendix, we examine whether a rigid labor market – a feature often attributed to the French economy – is likely to affect our results. To this end, we extend the model by augmenting it with an endogenous unemployment rate, determined by workers’ job-search effort. Consequently, the calibration exercise targets an additional empirical moment, which corresponds to the empirically measured unemployment rate.

To be precise, we assume that, in each period, non-entrepreneurs (including workers, managers, fixed-cost workers, and teachers) must search for a job. If they find a job, they receive net wage k (i.e. the total wage $k(1 + cz)$, minus the training costs $kc z$), while they receive an unemployment benefit b if they don’t. The probability that they find a job is e , which they choose optimally subject to a job search cost $c(e) = \frac{\ln(1-e)+e}{-x}$. As a result, non-entrepreneurs select the optimal amount of effort: $e = \frac{\chi(k-b)}{1+\chi(k-b)}$. The free-entry and production conditions for entrepreneurs require that they compare their (expected or known) profits as entrepreneurs to a worker’s expected utility $V = ek + (1 - e)b - c(e)$ rather than just k . In the simulations which follow, we assume that unemployment benefits b are set according to a replacement rate that is applied to the net wage k . This replacement rate is set to 0.57, a value that is in line with the rate applied in France. Lastly, the labor-market clearing condition also changes: if u represents the unemployment rate (within the population “at risk”, i.e. $1 - M$), Equation 20 is replaced with:

$$1 - u = M \left(\frac{\left[\delta f_E + \int_{\bar{\alpha}}^{+\infty} (C(q(\alpha); 1) + f) g(\alpha) d\alpha \right]}{1 - G(\bar{\alpha})} - u \right). \quad (\text{H.1})$$

Table H1 reports the calibrated moments and parameters. Introducing the possibility of being unemployed for workers alters mainly the values that are obtained for the parameters driving occupational choices, namely f_E and f , but only marginally. The distribution of α is also estimated to have slightly thinner tails than in the main calibration.

Table H2 replicates the general equilibrium simulation scenarios presented in the main text. The simulations confirm that our core results hold: the unemployment margin does not play a significant adjustment role when firms are subjected to increasingly binding minimum wage constraints. First, the unemployment rate remains unchanged to the third decimal place when the minimum wage constraint is introduced (columns (1) and (2)). It only adjusts noticeably in scenarios with large increases, relative to the baseline calibration based on French data. In contrast, skill upgrading remains the main adjustment channel: as production workers become more expensive, the demand for skills rises in the economy (it nearly doubles in the “+16%” scenario) to enhance their productivity. Ultimately, the global “cost” of introducing a minimum wage constraint in terms of total output is negligible in the baseline calibration, as in the Section 5.2, and reaches 21% in the highest scenario, as compared with 19% in the specification of the main text. The global cost is thus somewhat amplified, but not by much. The calibrated values obtained for the production threshold $\bar{\alpha}$ and for the demand conditions across classes of firms show that selection is somewhat reduced in the sense that firms with lower values of α remain active in equilibrium. Meanwhile, average firm size and the Q -productivity index are both very similar to their baseline values across all scenarios.

Table H1: Baseline Calibration to French Data

(A) Empirical Targets

Moment	Data	Model
Share of agents who are teachers	0.098	0.098
Share of workers in new firms	0.203	0.222
Share of 2-layer firms	0.446	0.448
Share of 3-layer firms	0.228	0.228
Average firm size	93.48	92.31
Average size of 2-layer firms	87.23	87.06
Average size of 3-layer firms	214.00	215.48
Average (hourly) wage (€)	20.21	20.26
Share of workers (in firms) bound by minimum wage	0.047	0.047
Unemployment rate (France, 2006)	0.088	0.088

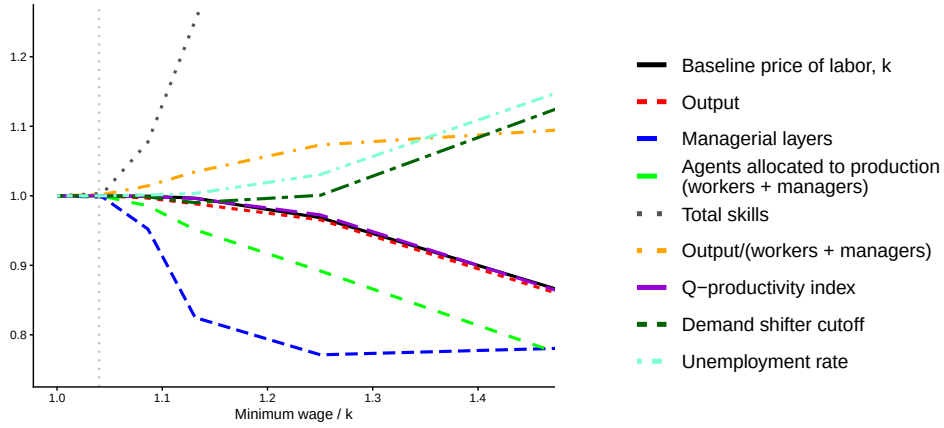
(B) Calibrated Values of Parameters

Parameter	Value
Cost of knowledge acquisition, c	0.220
Time cost of problem communication, h	0.194
Output per problem solved, A (~ 100 s of €/hour)	0.240
Fixed cost of entry, f_E	5.921
Fixed cost of production, f	5.784
Minimum wage $\bar{w}$, relative to average wage	0.914
Job search parameter χ	135.79
2nd-node of α distribution ($\alpha = 50$)	-12.13
3rd-node of α distribution ($\alpha = 500$)	-15.43
4th-node of α distribution ($\alpha = 1000$)	-24.55

Notes: Panel (A) shows the empirical moments to which the model detailed in Appendix H is calibrated. They are all computed from the DADS, except the share of teachers, which is sourced from INSEE's Tableaux de l'Economie Française (2020 edition, data for 2018), as in the main text (see Section 5). Panel (B) displays the values of the structural parameters that are obtained in the calibration.

Table H2: Minimum Wages with Unemployment in General Equilibrium

	(1)	(2)	(3)	(4)	(5)	(6)
	Baseline	No MW	+ 4%	+ 8%	+ 16%	+ 24%
Minimum wage, relative to k	1.04	1.00	1.09	1.13	1.25	1.61
Net (real) wage, k (€/hour)	17.74	17.74	17.73	17.68	17.18	14.23
Average wage per worker, $k(cz + 1)$	20.26	20.25	20.49	21.02	21.93	22.97
Share of firms bound by MW	0.19	0.00	0.41	0.57	0.94	1.00
Share of workers bound by MW	0.05	0.00	0.46	0.85	0.93	0.99
<i>(as shares of total number of agents)</i>		<i>Occupational choices and total skills:</i>				
Fixed-cost workers	0.24	0.24	0.24	0.24	0.24	0.23
Workers/managers	0.58	0.58	0.57	0.55	0.52	0.41
Firms (entrepreneurs), M	0.01	0.01	0.01	0.01	0.01	0.00
Teachers ($\propto$ total skills in economy)	0.09	0.09	0.10	0.11	0.15	0.26
Unemployed	0.09	0.09	0.09	0.09	0.09	0.11
<i>(In real terms when relevant)</i>		<i>Production and productivity:</i>				
Total output ($\sim$ €/hour)	12.13	12.13	12.09	11.99	11.71	9.64
Profits per firm ($\sim$ €/hour)	5.66	5.66	5.65	5.56	5.49	5.93
Average firm size (in jobs)	92.31	92.36	90.32	84.92	82.99	85.91
Average revenue per worker (€/hour)	28.38	28.32	28.85	29.63	29.91	30.42
Q -productivity index, $q/C_L(q, 1)$ (€/hour)	17.84	17.84	17.82	17.77	17.35	14.21
Production cutoff, $\bar{\alpha}$	51.66	51.66	51.58	51.15	51.70	62.10



Notes: This table presents the results from simulations computing the general equilibrium of the version of our model that incorporates job search and unemployment benefits (as described in Appendix H). Total output can be interpreted as per agent and per hour, in €: $P \int M\mu(\alpha)q(\alpha)d\alpha$, since P is normalized to 1. Average output per worker (and per hour) corresponds to $\frac{P \int M\mu(\alpha)q(\alpha)d\alpha}{\# \text{ Workers} + \# \text{ Managers}}$. Average revenue per worker (and per hour) corresponds to $\frac{p(\alpha) \int M\mu(\alpha)q(\alpha)d\alpha}{\# \text{ Workers} + \# \text{ Managers}}$.

I Framework without Organizations: Unproductive Managers

This Appendix aims to compare the quantification of our main model specification with a comparable set-up where the endogenous organizational (productivity) channel is essentially shut down. In this framework however, we continue to assume that knowledge (skills) remains both relevant for production ($\lambda < \infty$), and costly to acquire ($c \geq 0$). We also continue to assume that managers are essential to “bundling” production workers in firms, which enables them to reach their optimal size given their idiosyncratic demand shifter, α . However, in contrast to the specification in the main text, we now assume that they are otherwise irrelevant for production such that parameter h becomes irrelevant.²⁵

In such a setting, no firm finds it optimal to increase the number of managerial layers beyond $L = 1$, and managers do not need skills, but all firms still have to optimize n_1^0 , the number of production workers, as well as their level of skills z_1^0 . The firms’ cost minimization program thus takes the following (simplified) form:

$$C(q, k) = \min_{n_1^0, z_1^0} C_1(q, k) = kn_1^0(cz_1^0 + 1) + k \quad (\text{I.1})$$

subject to:

$$\bar{z}_1^0 \leq z_1^0 \quad (\text{I.2})$$

$$q \leq n_1^0 A(1 - e^{-\lambda z_1^0}), \quad (\text{I.3})$$

where $\bar{z}_1^0 = 0$ in the economy without a minimum wage, and $\bar{z}_1^0 = \frac{\bar{w}-k}{ck}$ (as in Equation 21) when a minimum wage is introduced.

Solving this problem shows that the optimal level of knowledge of production workers, z_1^0 , verifies the following relationship:

$$\frac{c}{\lambda} e^{\lambda z_1^0} (1 - e^{-\lambda z_1^0}) = cz_1^0 + 1 \quad \text{if} \quad z_1^0 > \bar{z}_1^0 \quad (\text{I.4})$$

$$z_1^0 = \bar{z}_1^0 \quad \text{otherwise} \quad (\text{I.5})$$

In particular, Equations I.4 and I.5 do not depend on α , the only parameter that is heterogeneous across firms, which implies that all firms select the same level of skills for any of their production workers. Therefore, in such an economy, there is no wage heterogeneity, neither within firms, nor across firms. In addition, Equations I.4 and I.5 do not depend on k either, which implies that the level of skills is unaffected by the

²⁵This implies that $h = 0$ is an admissible value and that the condition $c > \lambda h$ still holds formally.

Notice that the set-up of this Appendix is a little less extreme than the limit case considered in Caliendo and Rossi-Hansberg (2012), where knowledge, and not only managers, becomes irrelevant for production: $\lambda \rightarrow \infty$, and $c \geq 0$ (which implies that $c \leq \lambda h$, i.e. the parametric condition imposed by the French data no longer holds). In such a setting, the distribution of production problems becomes a mass point located at $d = 0$ and there is no heterogeneity in terms of the knowledge that is required to solve them all. Both workers’ and managers’ knowledge become irrelevant. In such a setting, organizations still emerge as soon as their production is higher than A , but only $L = 1$ managerial layer is required to “bundle” the optimal number of production workers (as in this Appendix). In addition, the optimal level of skills is trivially $z_1^0 = 0$ whatever the idiosyncratic demand conditions, α , which implies that all workers earn the same wage, k .

In such a setting, minimum wages start to be binding as soon as their value exceeds k . As in the model of the present Appendix, firms have to train all workers up to the level $\bar{z}_1^0 = \frac{\bar{w}-k}{ck}$, although this strategy is useless to them. Introducing such a binding minimum wage in this environment however appears to have drastic consequences: indeed, no complex firm survives, because the free entry condition (Equation 19) and the zero-profit cutoff equation (Equation 18) become generically incompatible. This result is due to the fact that, as soon as a minimum wage is introduced, the profit function becomes homogeneous of degree 1 in k , which implies that the general equilibrium price adjustments in k become entirely incapable of balancing either the free entry condition or the zero-profit cutoff equation. The cutoff $\bar{\alpha}$ is the only endogenous variable that is left to balance these two generically incompatible equations, which implies that no equilibrium exists where both minimum wages are binding and complex firms (with $L = 1$ managerial layer) are active. Rather, the cost of introducing any minimum wage is immediately huge: it forces the economy to turn into a state where only $L = 0$ firms (i.e. self-employed agents, who are not bound by the minimum wage) operate, thus reducing drastically the production of all varieties (to A), even those which are highly valued by consumers.

general equilibrium price adjustments.

The profit maximization program in this set-up remains identical to Equation 15. It implies that in equilibrium, both $n_1^0(\alpha, k)$ and $q(\alpha, k)$ are linear in α :

$$n_1^0(\alpha, k) = \alpha k \left(\frac{\sigma - 1}{\sigma} \frac{\lambda A}{c k} \right)^\sigma \frac{e^{-\sigma \lambda z_1^0}}{A(1 - e^{-\lambda z_1^0})} \quad (\text{I.6})$$

$$q(\alpha, k) = \alpha k \left(\frac{\sigma - 1}{\sigma} \frac{\lambda A}{c k} \right)^\sigma e^{-\sigma \lambda z_1^0}, \quad (\text{I.7})$$

where, again, z_1^0 is an endogenous object but appears to be determined independently of α and k (see Equations I.4 and I.5).

Minimum wages. When minimum wages are introduced in such an economy and become binding, skills increase mechanically following Equation I.5. As there is no wage heterogeneity between workers, the latter are all paid exactly the minimum wage, as soon as the latter begins to be binding. In addition, firms' sizes adjust according to Equations I.6 and I.7, where $z_1^0 = \bar{z}_1^0$: the increase in production workers' skills has a negative impact on both n_1^0 and q , but the latter is dampened by the general equilibrium adjustment of prices, k .²⁶

To compare the qualitative and quantitative predictions arising from this set-up with our main results, we calibrate this model to the French data. As shown in Table I1, we rely on the same data moments as in the main text, except those relating to the number of layers in firms (since the number of layers is at most one in the present set-up) and to the share of workers who are bound by the minimum wage (since this share can only be 100% in the present set-up).²⁷ As a consequence, in order to achieve (just) identification, we have to reintroduce two additional targets, mainly to identify the compensation of teachers relative to workers and the distribution of α .²⁸ In the absence of public data on the former aspect and to ease the comparison with our baseline calibration, we simply introduce the value that is attained in the main text for the average wage divided by k , and use it as a calibration target. In terms of the distribution of α , we abandon the flexible specification of our main calibration and simply reintroduce the Pareto assumption that was used in Caliendo and Rossi-Hansberg (2012), so that $G(\alpha) = 1 - \alpha^{-\gamma}$. We also follow their calibration procedure in this respect, and thus introduce a calibration target corresponding to the coefficient obtained from a log-log

²⁶Lastly, both the increase in skills and the decrease in firm sizes alter the free entry condition (Equation 19) and the zero-profit cutoff equation (Equation 18), which in turn determine the production cut-off $\bar{\alpha}$.

A specific feature of this set-up with "unproductive managers" is that the production cut-off $\bar{\alpha}$ is unaffected by minimum wages when the distribution of α is a Pareto, $G(\alpha) = 1 - \alpha^{-\gamma}$. This is essentially due to the fact that marginal costs are constant (in contrast to the set-up in the main text). To see this, start from the fact that, evaluated at the optimal q , the profit function can be written as:

$$\pi(\alpha, k) = \frac{\alpha k}{\sigma - 1} \left(\frac{\sigma}{\sigma - 1} \right)^{-\sigma} \phi^{1-\sigma} - k(1 + f)$$

where $\phi = \frac{kc}{A\lambda e^{-\lambda z_1^0}}$ is the (constant) marginal cost. Plugging this into the zero-profit condition on entry and simplifying gives us the following expression for the production cut-off:

$$\bar{\alpha} = \left(\frac{1 + f}{(\gamma - 1)\delta f_E} \right)^{\frac{1}{\gamma}}$$

which varies only with fixed parameters γ , δ , f , and f_E - but not z_1^0 . Notice that this result does not hold when managers are productive (i.e. in the model of the main text if the distribution of α is constrained to a Pareto).

²⁷In other terms, the set-up without productive managers is not able to rationalize the empirical share of workers who are bound by the minimum wage (which is only 5% in Table 5) and the existence of firms with several hierarchical layers (68% in Table 5).

²⁸Additionally, we present the calibrated minimum wage in absolute terms rather than as a percentage of the average wage (since the latter is 1 by definition).

regression of the cumulative distribution of firm sizes (-1.095 in their Figure VII), which ultimately identifies the Pareto coefficient, γ .

Calibration results. The calibration results are displayed in Panel (C) of Table II. The model abstracts from hierarchical organizations, thereby ruling out both wage and skill heterogeneity. Relative to the baseline specification, this leads to a much lower estimated cost of knowledge acquisition, c —roughly one-sixth of its baseline value — and a substantially larger aggregate productivity parameter, A , estimated at about three times its baseline level. The fixed costs of entry are also estimated to be more than ten times higher than in the main specification, while the fixed costs of production are in contrast in the same order of magnitude.

Table II: Calibration of Model with Unproductive Managers

(A) Empirical Targets			
Moment		Data	Model
1. Share of agents who are teachers		0.098	0.098
2. Share of workers in new firms		0.203	0.213
3. Average firm size		93.48	93.40
4. Average (hourly) wage (€)		20.21	20.19

(B) Additional Targets			
Moment		Target	Model
1. Average wage, relative to k (from main calibration)		1.141	1.154
2. Coefficient for log-log regression of cumulative distribution of firm size from Caliendo and Rossi-Hansberg (2012)		-1.095	-1.131

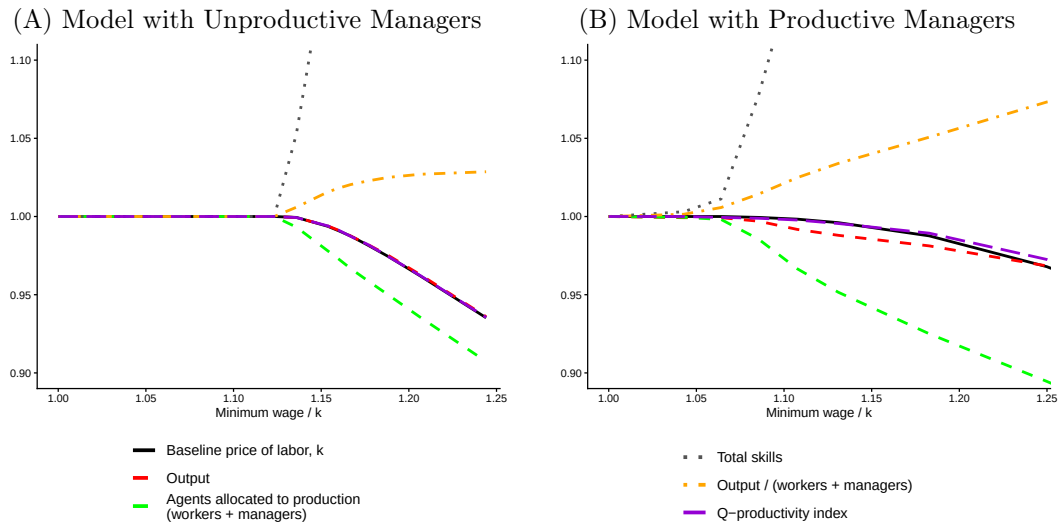
(C) Calibrated Values of Parameters	
Parameter	Value
Cost of knowledge acquisition, c	0.037
Output per problem solved, A ($\sim 10^2$ €/hour)	0.739
Fixed cost of entry, f_E	61.557
Fixed cost of production, f	6.236
Minimum wage $\bar{w}$	0.202
Pareto parameter for α, γ	1.090

Notes: In Panel (A), all moments are computed from the DADS, except the share of teachers, which is sourced from INSEE's Tableaux de l'Economie Française (2020 edition, data for 2018), as in the main text (see Section 5).

Lastly, Figure II replicates Figure 6 and simulates the impact of increasingly stringent minimum wages in such an economy on the main aggregate outcomes. Given the lack of heterogeneity in wages across firms in this setting, firms find it optimal to train their workers (and pay them accordingly) even in absence of a minimum wage. As a consequence, we obtain that the minimum wage only starts to be binding for values above $1.12k$, while it started to be binding for at least some workers for all values exceeding k in the main calibration. At this point, skills adjust even more quickly than in the baseline set-up of the main text. Most importantly, the absence of the organizational channel²⁹ implies that output per worker increases only moderately, thanks to skills only. As a consequence, the Q -productivity index and aggregate output both decrease four times faster than in the baseline set-up, when measuring this slope between the point where the minimum wage starts to be binding and the maximal value that is considered in Figure II ($1.24k$).

²⁹Hierarchical layers range between $L = 0$ and $L = 1$, such that the scope for adjustment is extremely limited.

Figure I1: Comparison of Models with *vs.* w/o Organizational Productivity Channel



Notes: This figure presents simulations results for alternative scenarios where the minimum wage is either removed or set at increasingly high levels (relative to k , the baseline price of labor in general equilibrium). Panel (A) considers the economy which is described in this Appendix, where managers do not contribute directly to production (but remain essential to “bundle” workers into firms in cases where it is optimal to achieve large output targets given the value of the demand shifter). Panel (B) replicates Figure 6 for comparison. It describes the economy where managers contribute directly to production such that complex firms’ organizations emerge in equilibrium, as in the main text.

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